MASTER AZURE – BECOME CLOUD READY • STUDENT NOTES

Session 05 – Live

Introduction to Databricks

Big data • Types of data • Spark vs PySpark vs Databricks • Databricks account • Read CSV • Create a table • Genie

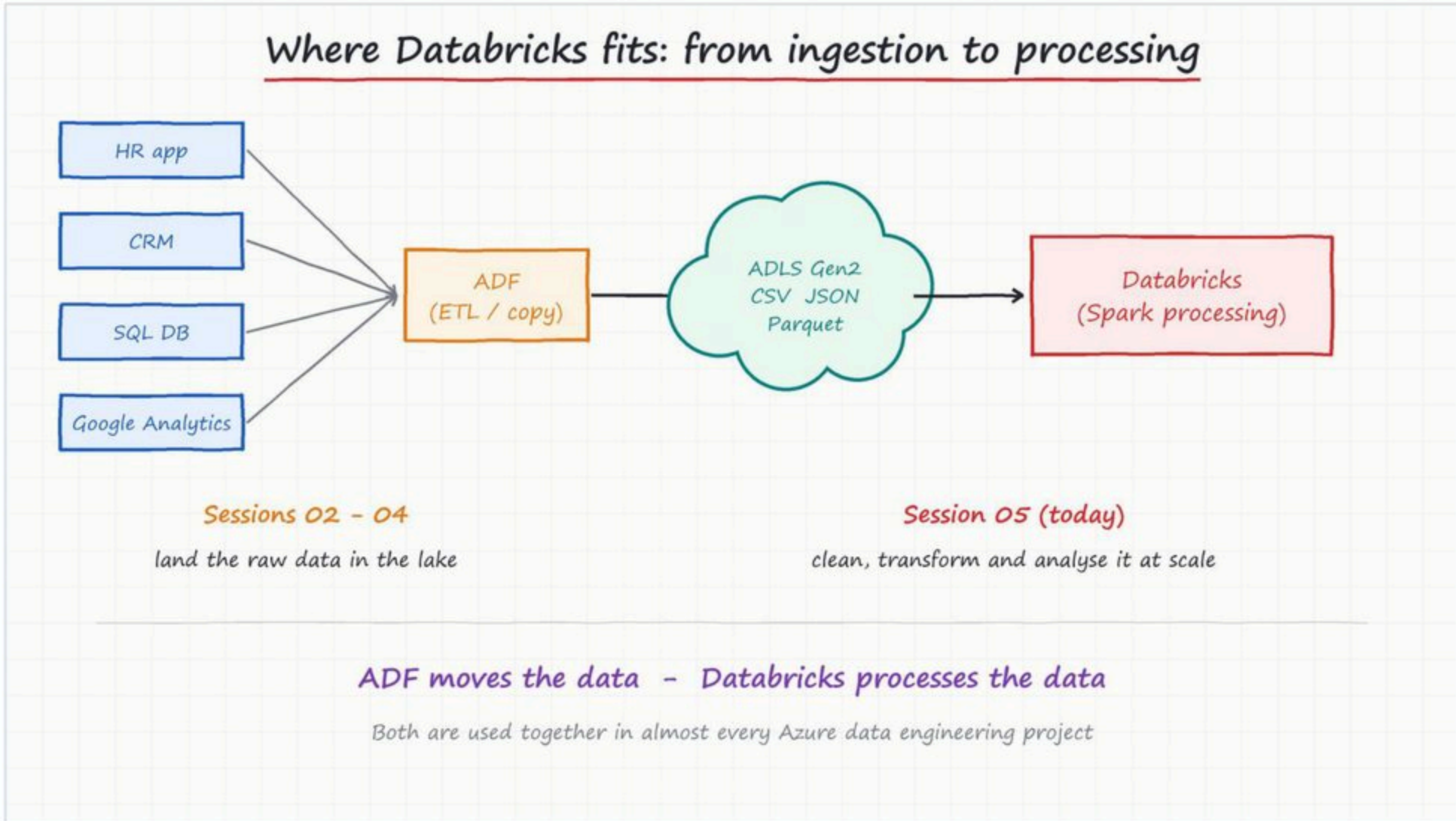
Hands-on: Databricks Free Edition • Unity Catalog volume • PySpark • Spark SQL • Genie

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: **linkedin.com/in/uwayo**

Introduction – why this session matters

- ADF moves data – something must PROCESS it at scale
- Databricks is the most requested processing tool in Azure data jobs
- Spark / PySpark skills are asked in almost every data engineering interview
- Today you get your own free workspace to practise every day
- From a CSV file to a SQL table to AI answers – in one session

Recap – where Databricks fits



Learning objectives

- Explain big data with the 5 Vs and why one computer is not enough
- Recognise structured, semi-structured and unstructured data
- Explain the difference between Spark, PySpark and Databricks
- Create a Databricks workspace (Free Edition) and know the Azure option
- Upload a CSV file to a Unity Catalog volume and read it with PySpark
- Query data with Spark SQL and save it as a permanent table
- Ask questions about a table in plain English with Genie

Session agenda

#	Topic	Type
1	Big data concept and types of data	Theory
2	Spark vs PySpark vs Databricks	Theory
3	Databricks account: Azure Databricks and Free Edition	Demo + hands-on
4	Compute: clusters, runtimes, workers and SQL warehouses	Demo
5	Lab 1 – Upload Order.csv to a Unity Catalog volume	Hands-on
6	Lab 2 – Read the CSV with PySpark (header, inferSchema)	Hands-on
7	Lab 3 – Spark SQL, AI assistant and fixing errors	Hands-on
8	Lab 4 – Create a table + Lab 5 – Genie	Hands-on
9	Troubleshooting, Q&A, quiz and homework	Discussion

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

Before we start – what you need

Item	Example used in these notes	Notes
A web browser	Chrome / Edge	An incognito window is fine
An email address	user@contoso.com	For the Databricks Free Edition sign-up
Sample file	Order.csv (1,000 sales orders, 12 columns)	Shared in the course group
Optional: Azure subscription	rg-adf-training	Only to see the Azure Databricks create page
Catalog / schema / volume	workspace / default / sales_data_files	Created in Lab 1
Notebook	Practice-Notebook	Created in Lab 2

⚠ **Free Edition needs no credit card. Do not create a paid Azure Databricks workspace just for practice – clusters cost money.**

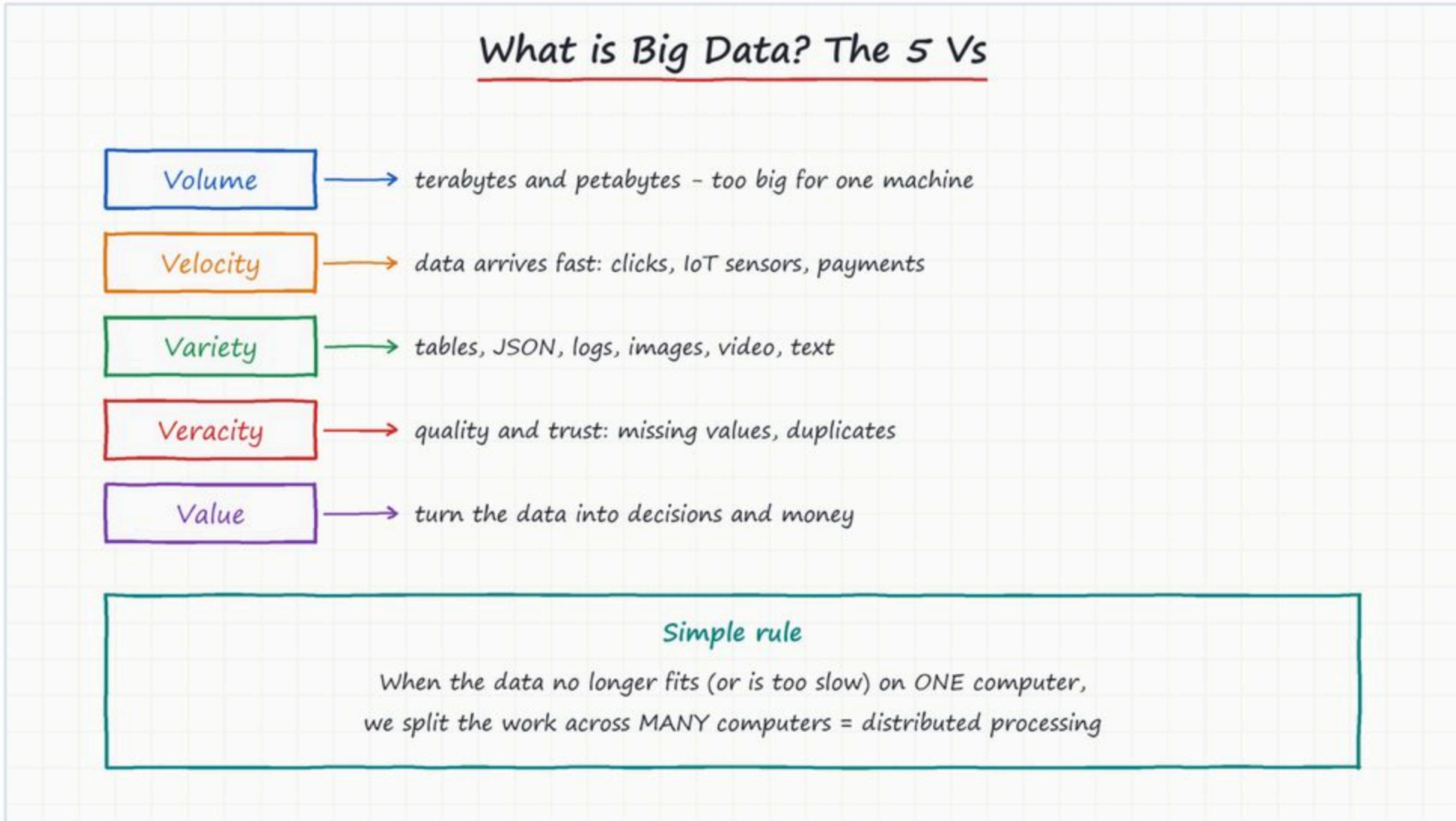
Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

PART 1

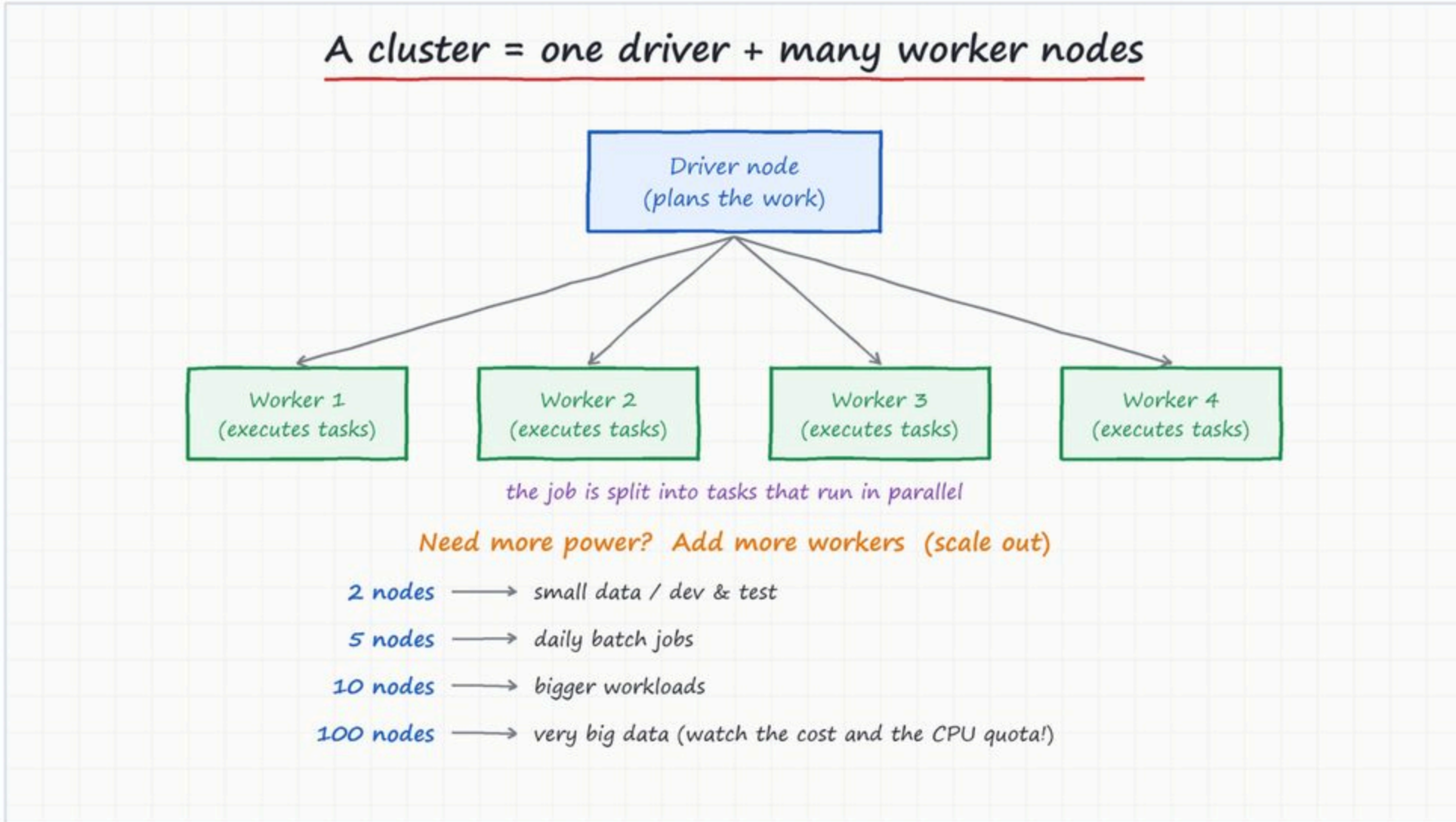
Big data and types of data

The 5 Vs • Distributed processing • Structured, semi-structured, unstructured

What is big data? The 5 Vs

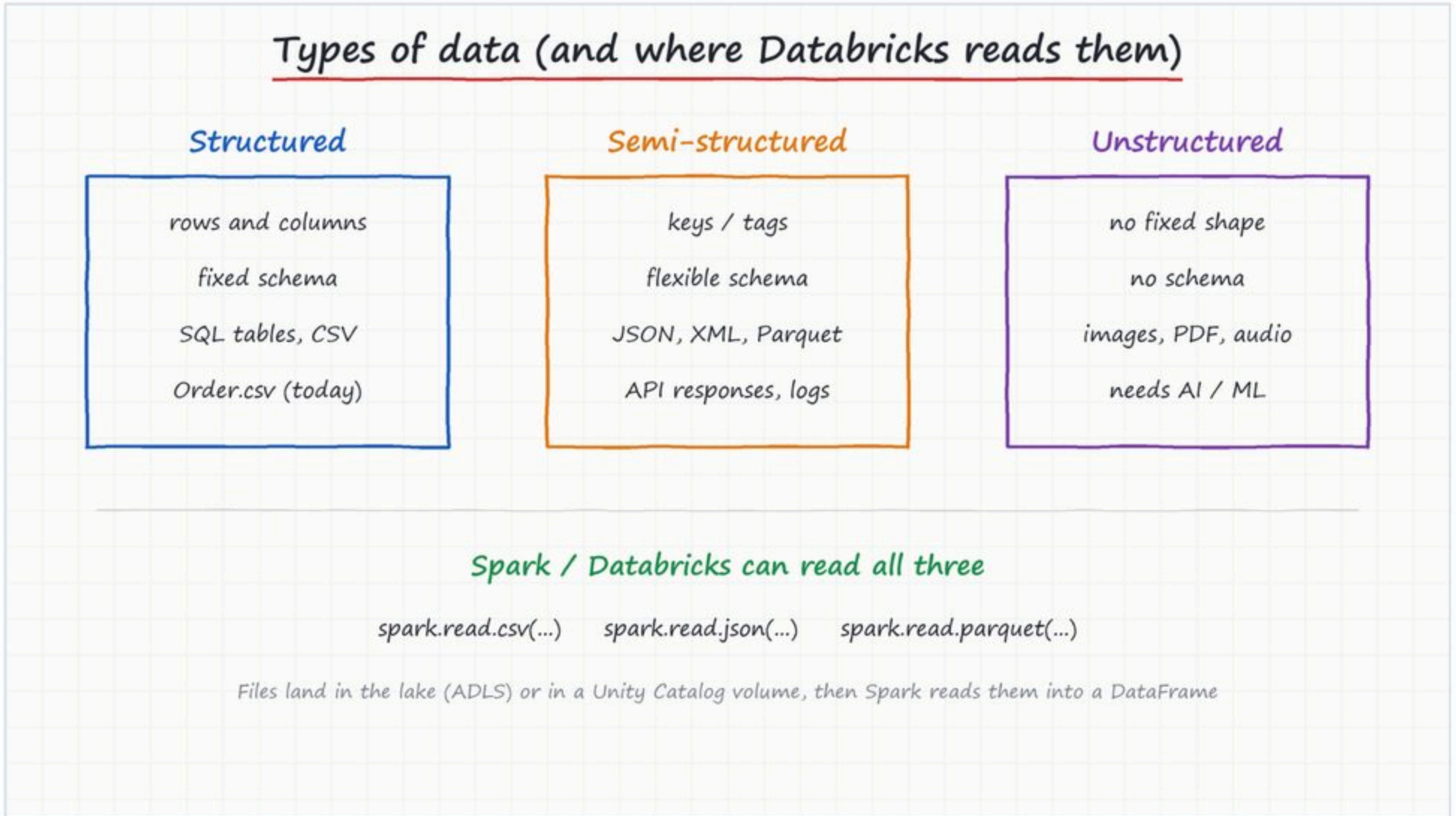


Scaling out: one driver, many workers



Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

Types of data



Big data vocabulary you will hear

Term	Meaning
Cluster	A group of machines (nodes) that work together
Driver / worker	The node that plans the work / the nodes that execute it
Partition	A piece of the data processed by one task
DataFrame	A distributed table with named, typed columns
Batch vs streaming	Process data in chunks on a schedule / continuously as it arrives
Data lake	Cheap storage for files of any type (ADLS Gen2)
Lakehouse	Data lake + table features (Delta Lake): ACID, SQL, governance

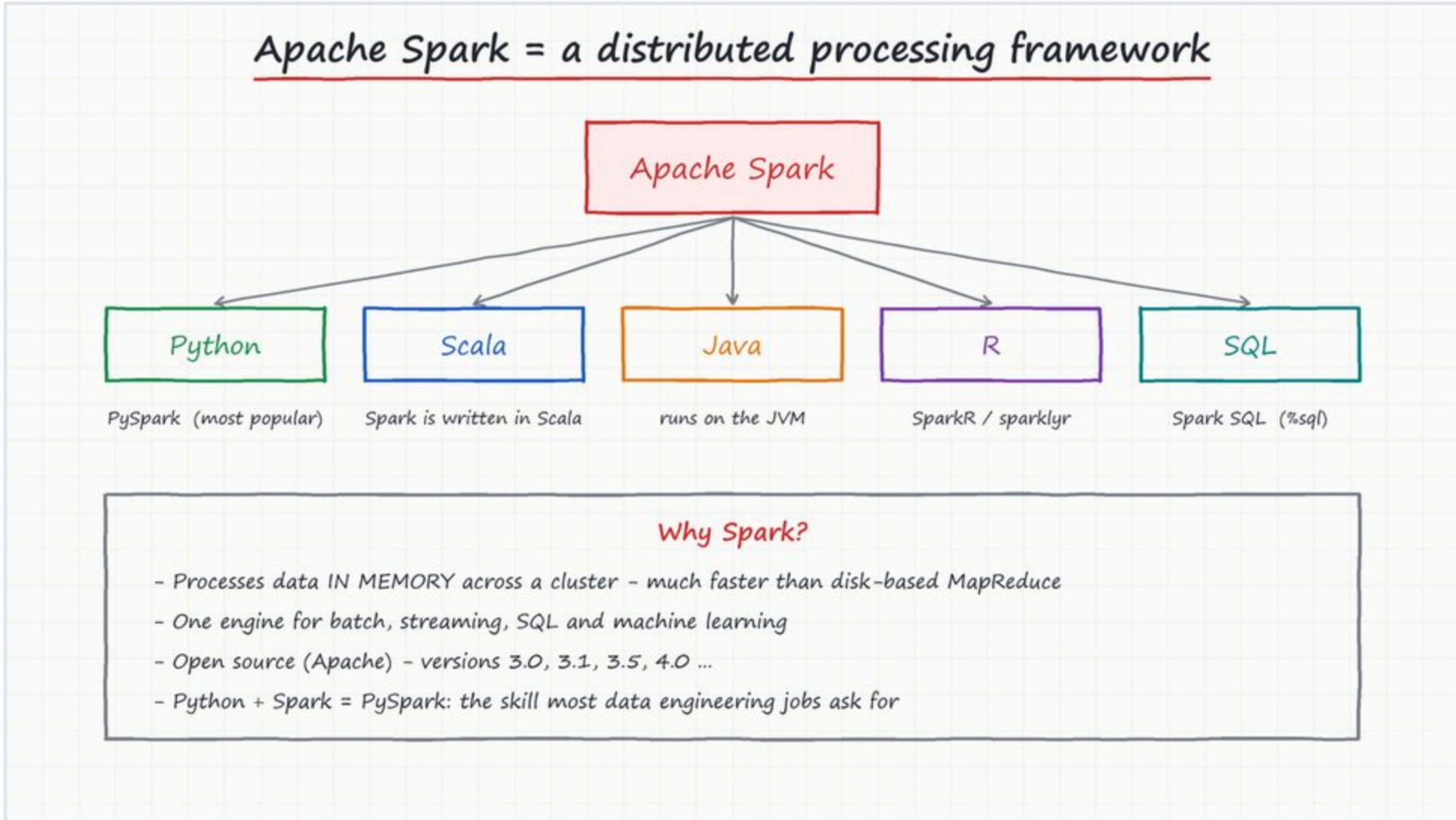
Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

PART 2

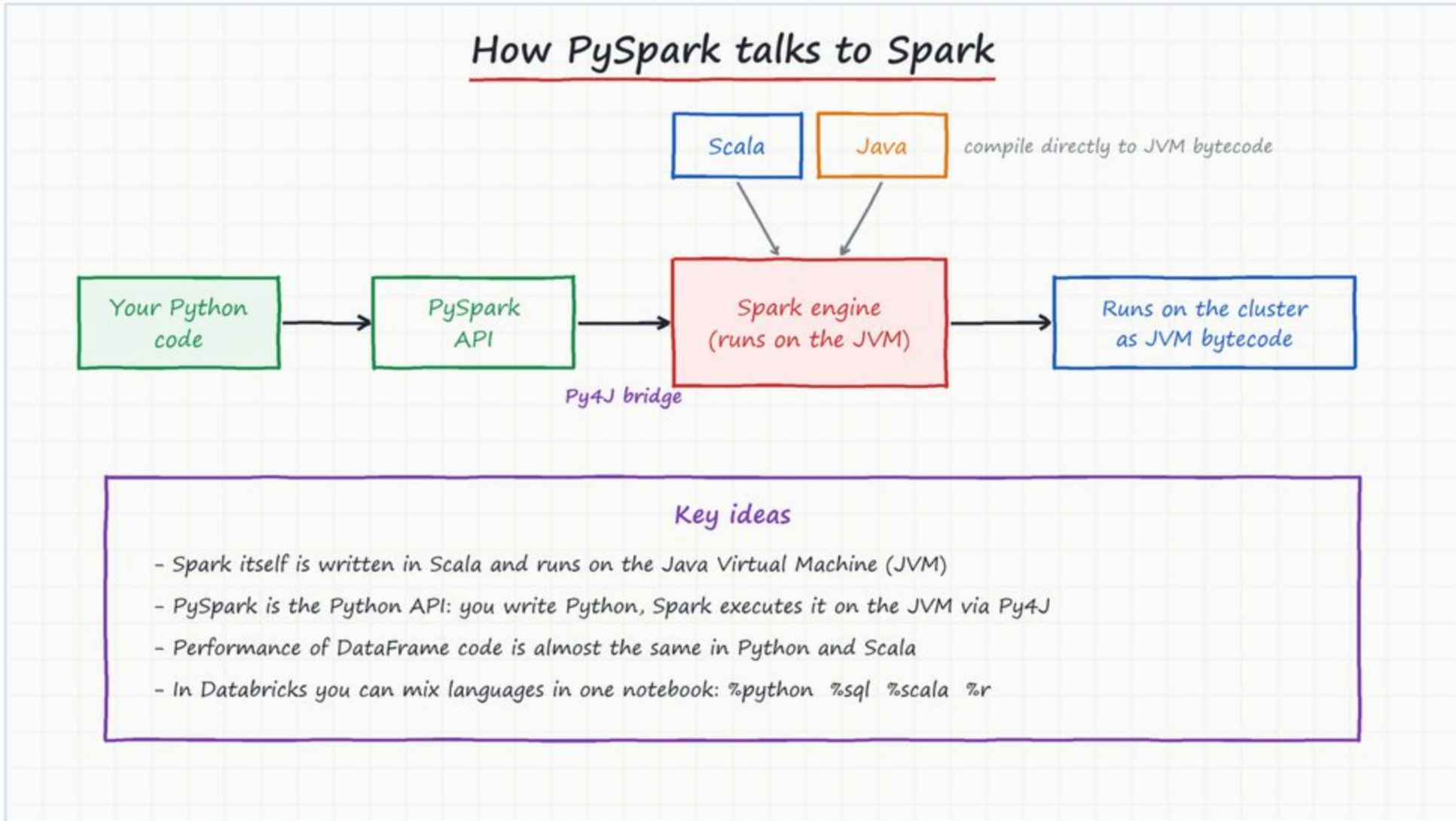
Spark vs PySpark vs Databricks

The engine • The Python API • The platform

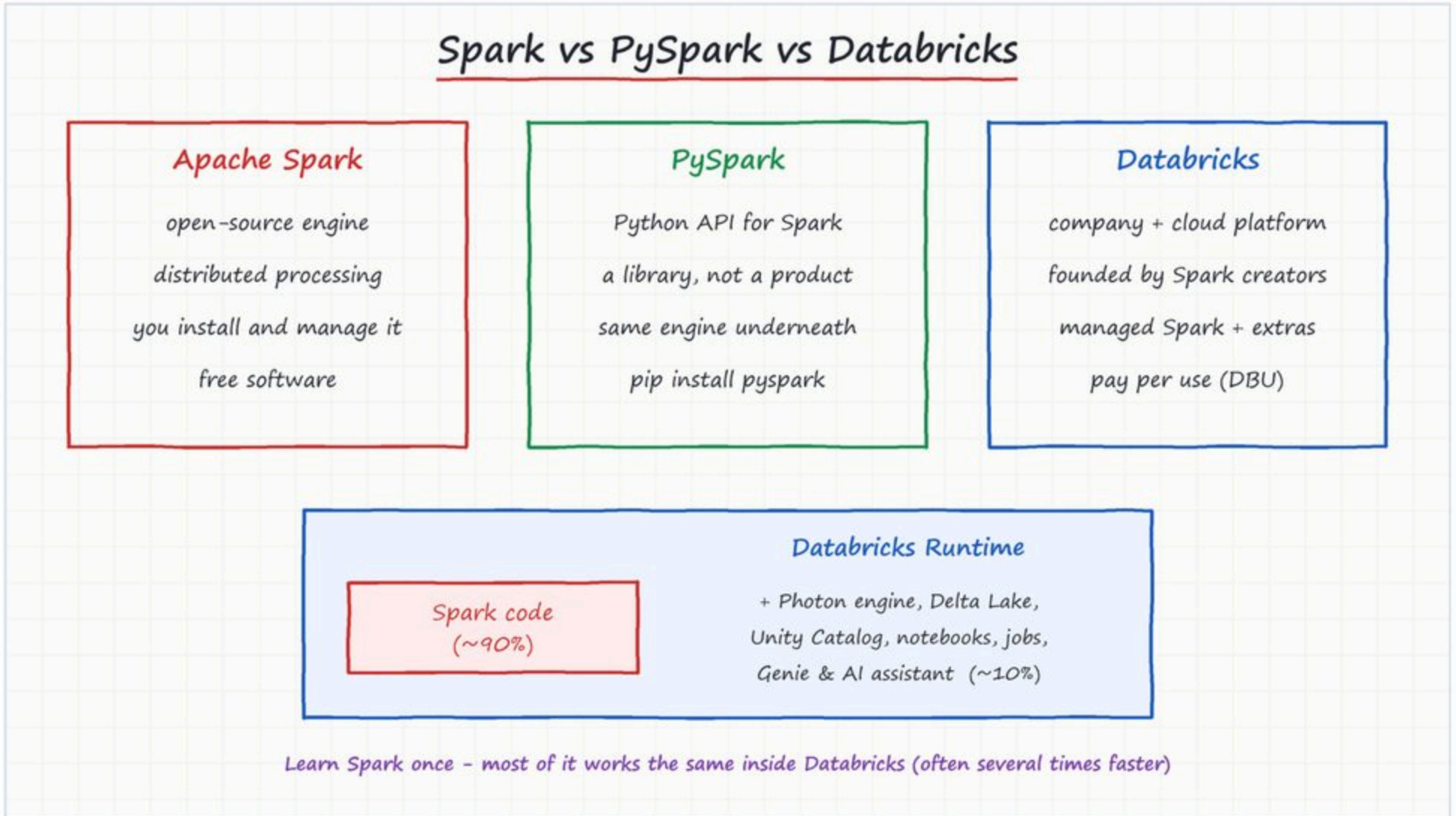
Apache Spark – a distributed processing framework



How PySpark talks to the Spark engine



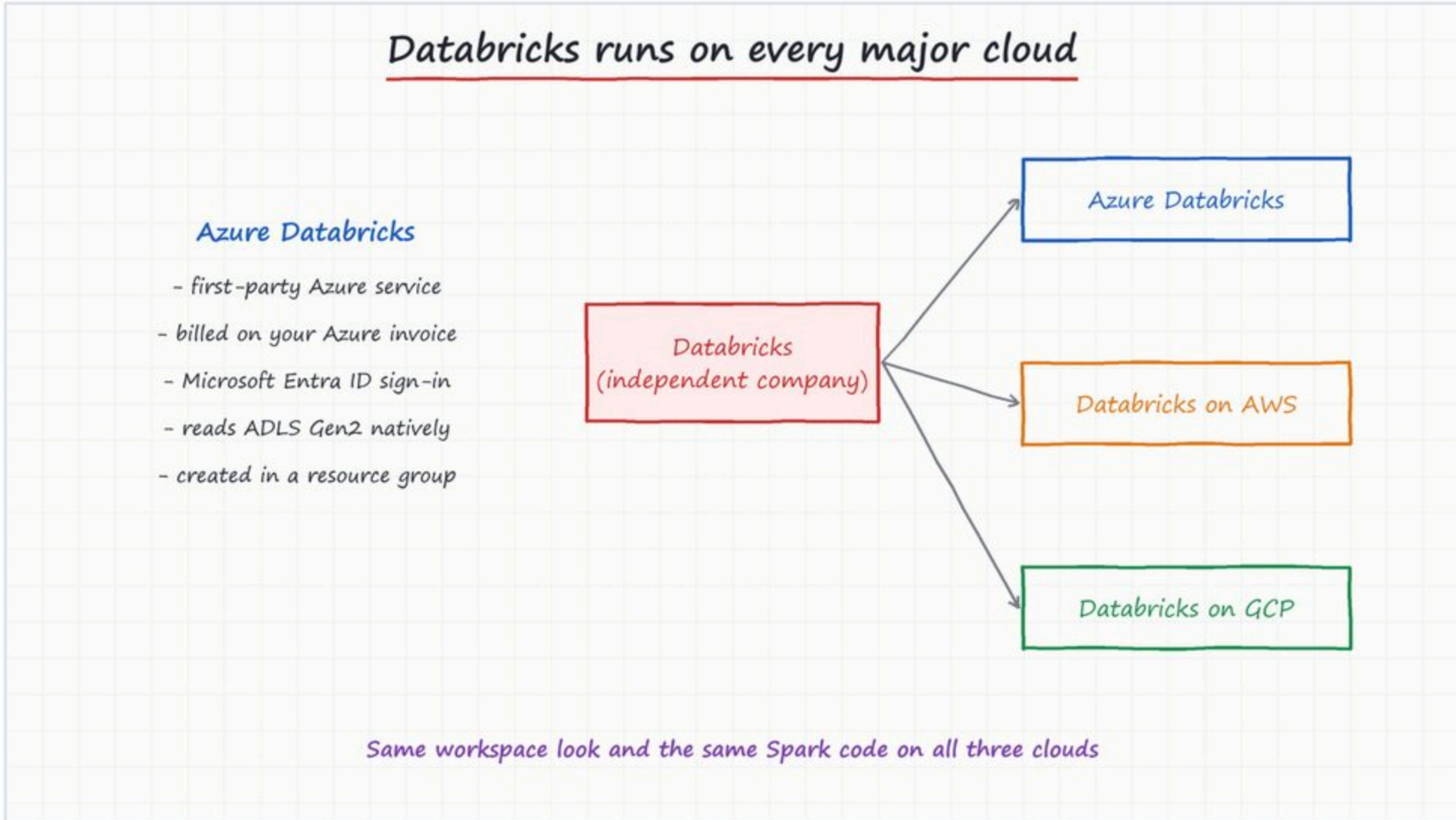
Spark vs PySpark vs Databricks



Spark vs PySpark vs Databricks – summary

	What it is
Apache Spark	Open-source distributed processing engine (Scala, runs on the JVM)
PySpark	Python API for Spark – write Python, Spark executes it
Spark SQL	SQL API for Spark – the %sql cells in a notebook
Databricks (company)	Founded by the creators of Spark; sells a data + AI platform
Databricks (platform)	Managed Spark + Runtime/Photon + Delta Lake + Unity Catalog + Genie
Azure Databricks	Databricks offered as a first-party Azure service

Databricks runs on every major cloud

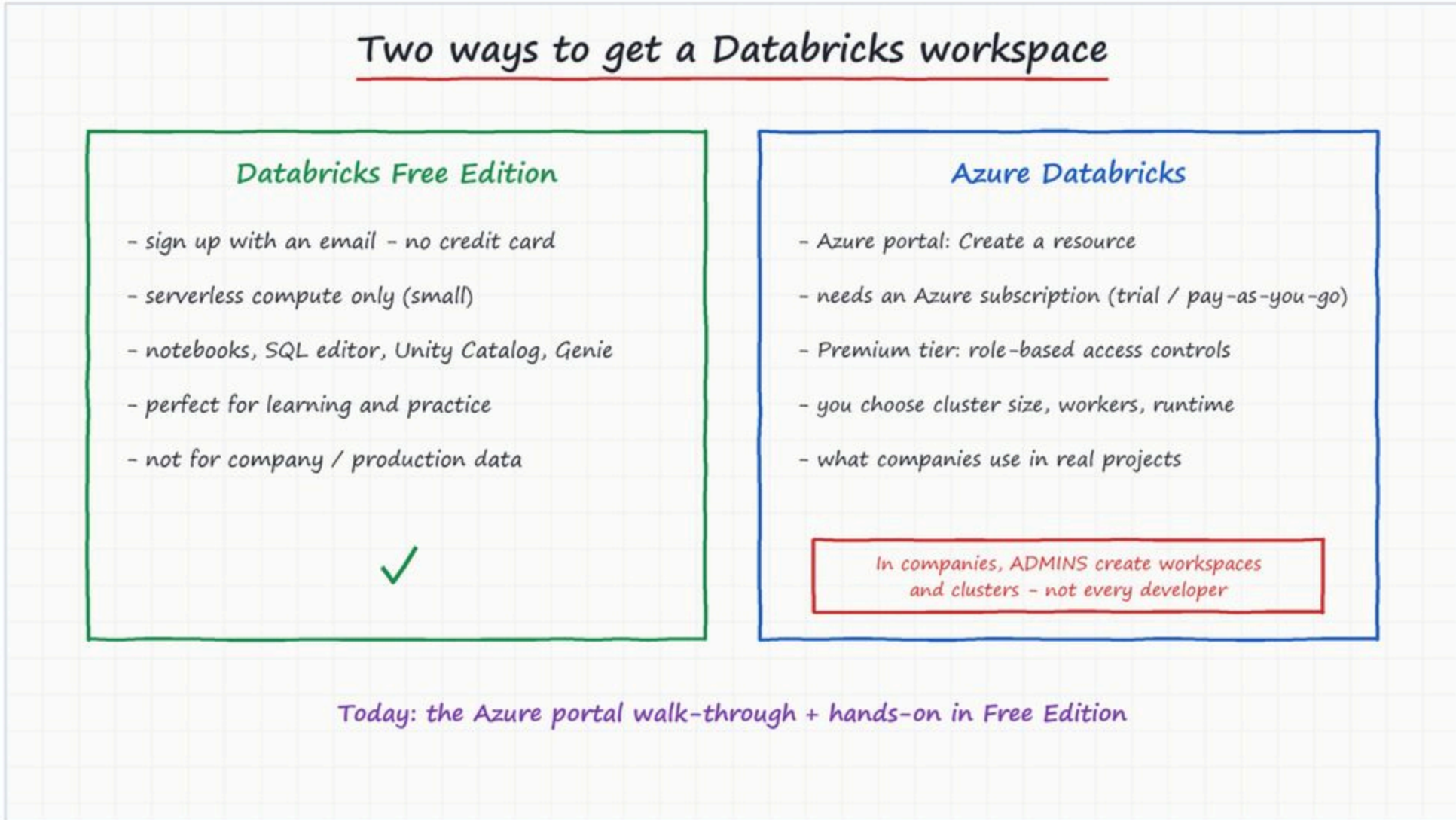


PART 3

How to create a Databricks account

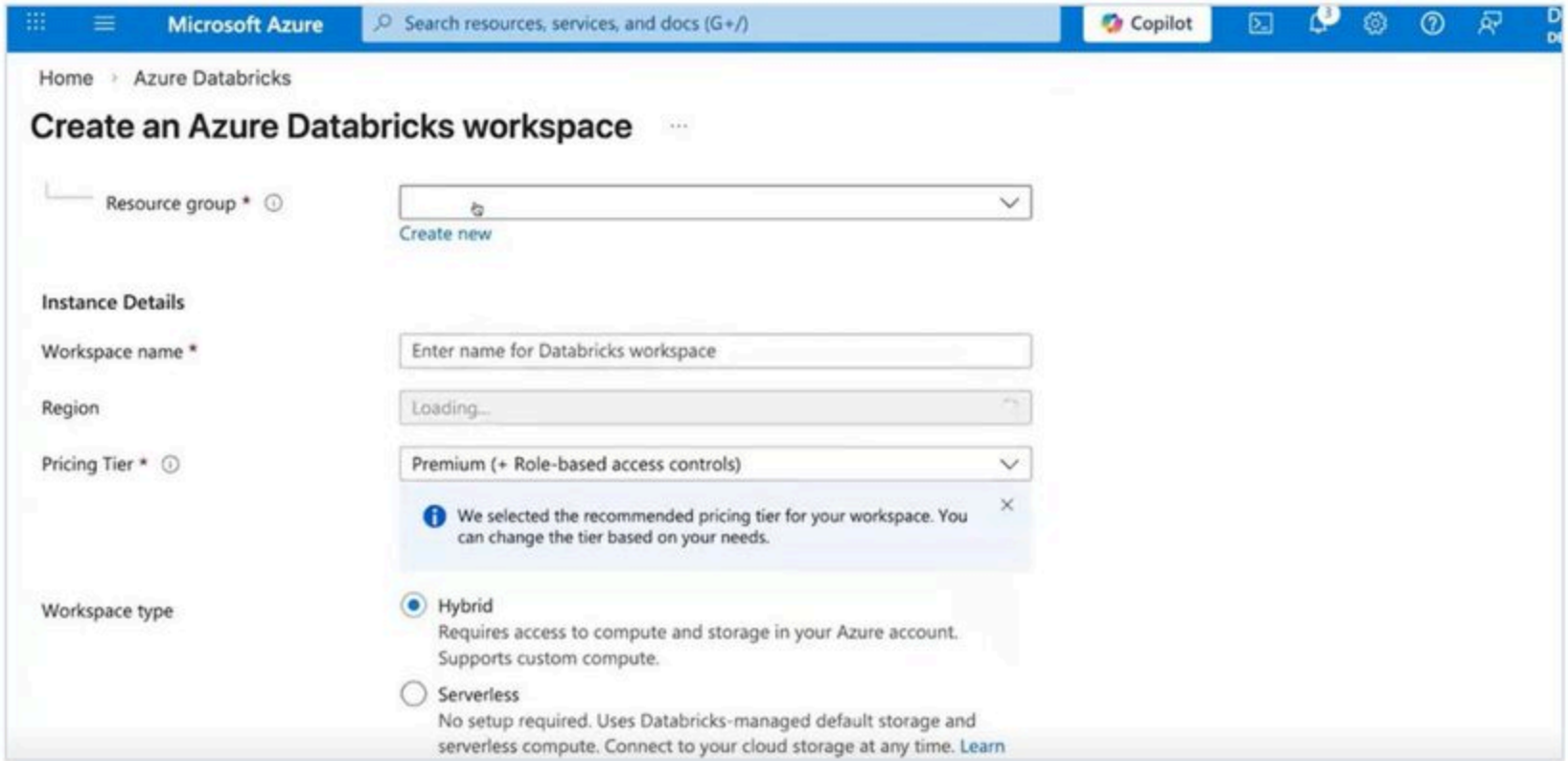
Azure Databricks (portal) • Databricks Free Edition (hands-on)

Two ways to get a Databricks workspace



PART 3

Azure Databricks – Create a workspace (Basics)

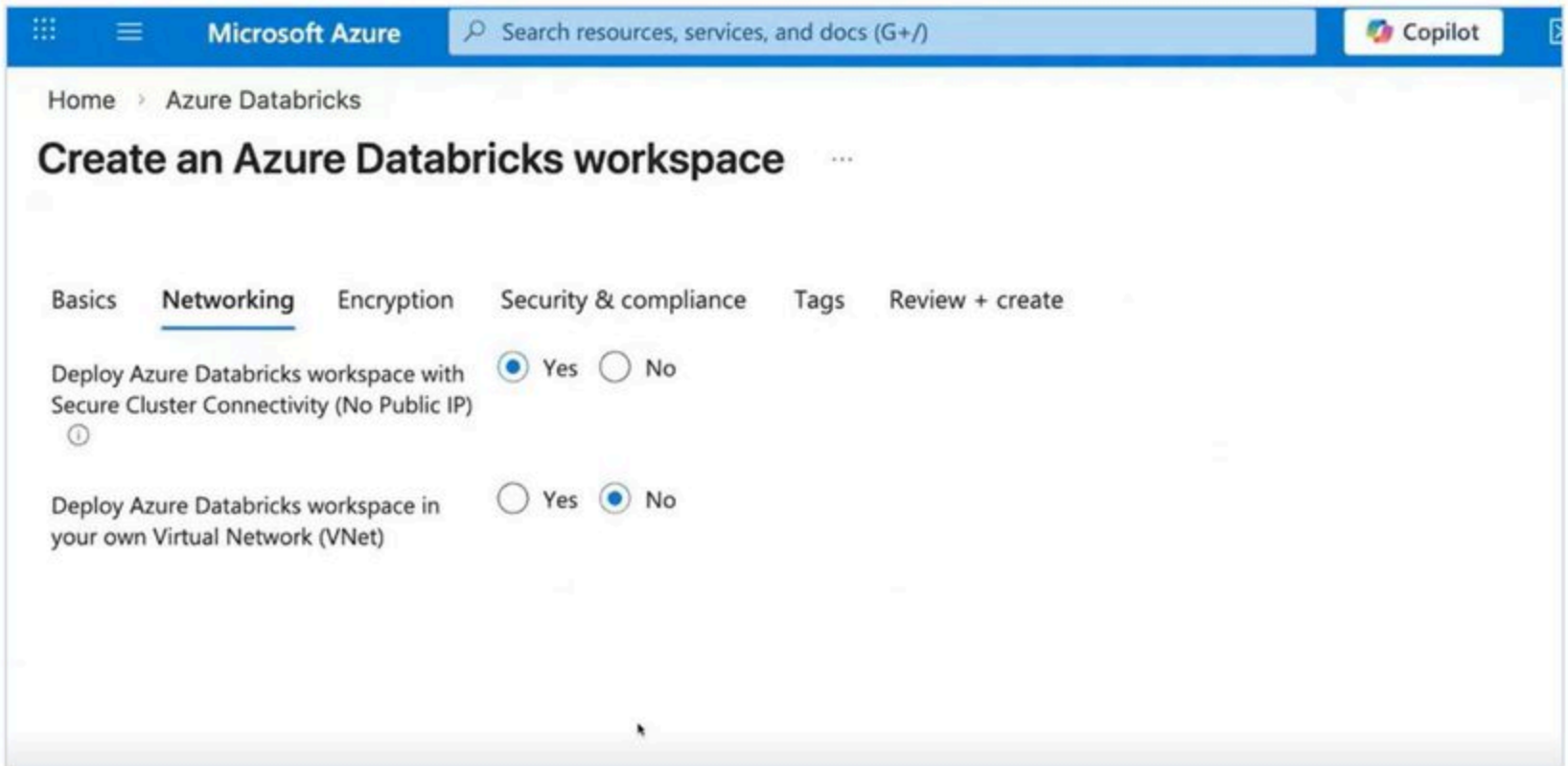


- Portal → Create a resource → Azure Databricks
- Resource group + workspace name + region
- Pricing tier: Premium (+ role-based access controls)
- Workspace type: Hybrid or Serverless

⚠ A workspace itself costs almost nothing – running compute is what you pay for.

PART 3

Azure Databricks – Networking tab



- Secure Cluster Connectivity (No Public IP): Yes
- Deploy in your own VNet: No (default)
- Then Review + create

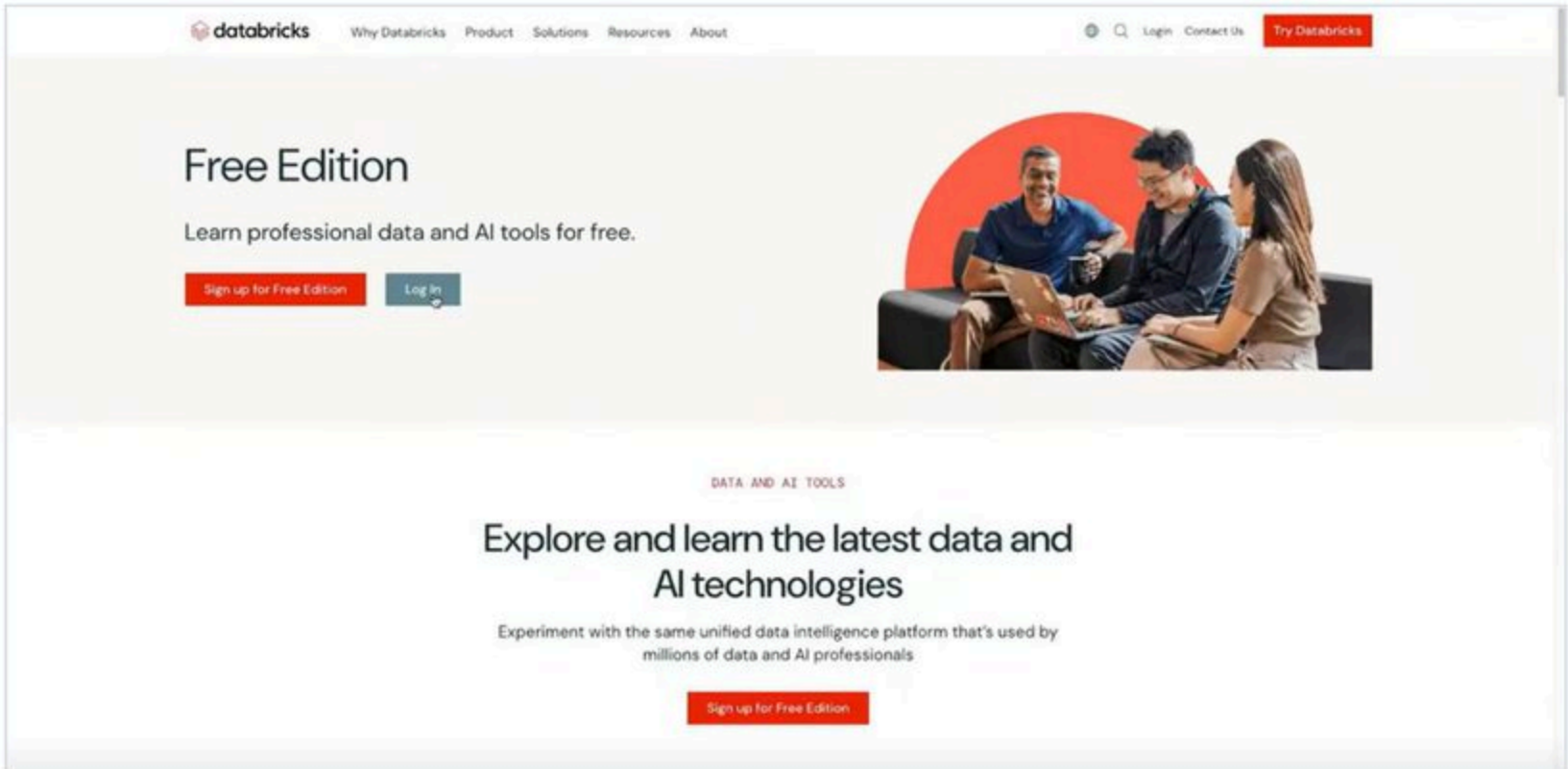
PART 3

Who creates workspaces and clusters?

- Resource creation (workspace, clusters) is usually the ADMIN team's job
- Data engineers get access + permission to use or create compute
- Cluster policies limit size and cost (e.g. max workers)
- In interviews: know the options, even if you never click Create

PART 3

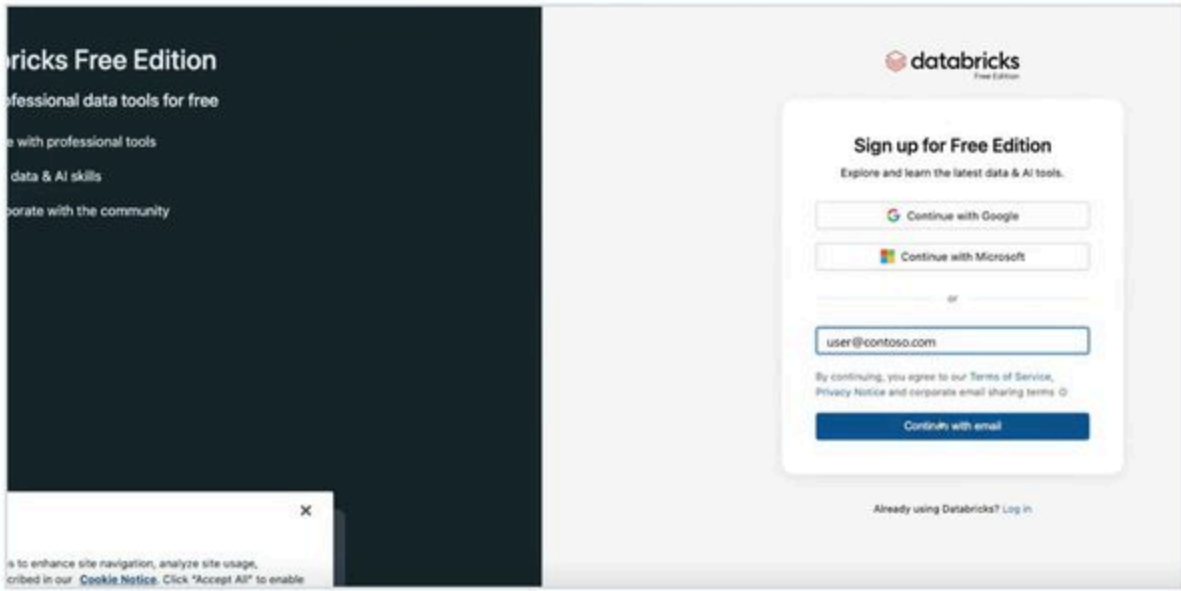
Free Edition – the sign-up page

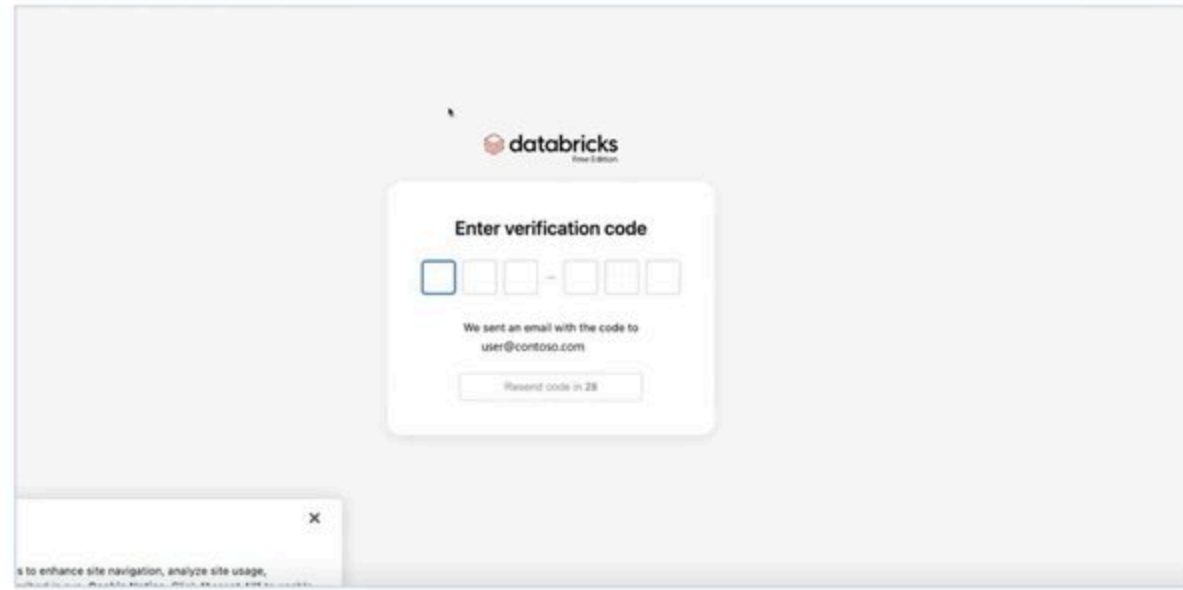


- Search "Databricks Free Edition"
- Click Sign up for Free Edition
- Learn data and AI tools for free

PART 3

Free Edition – sign up with email



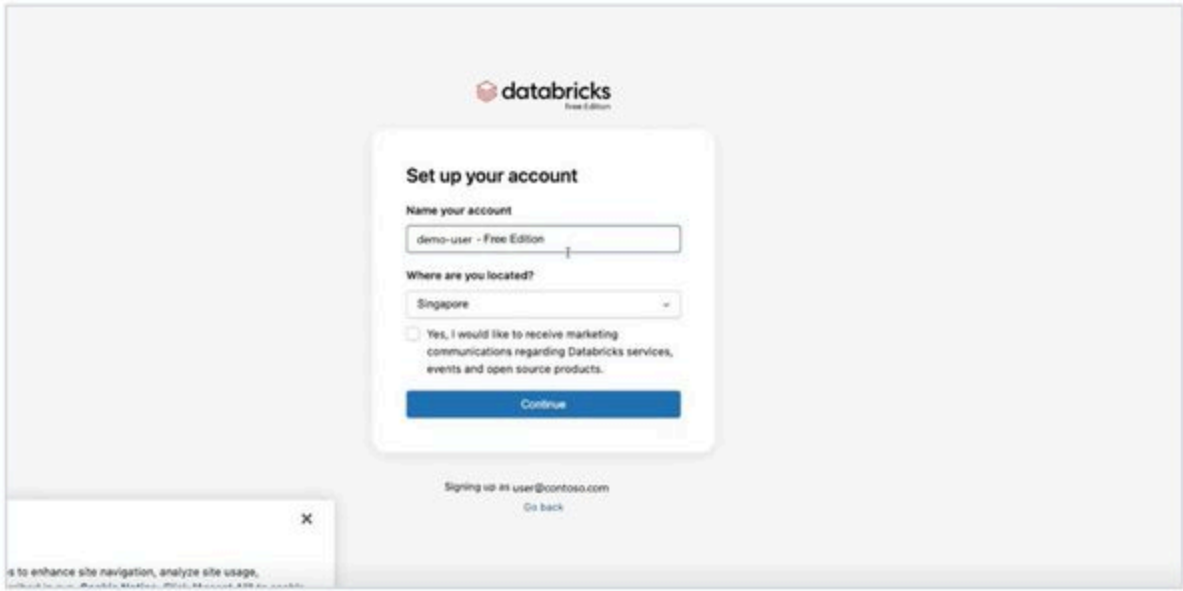


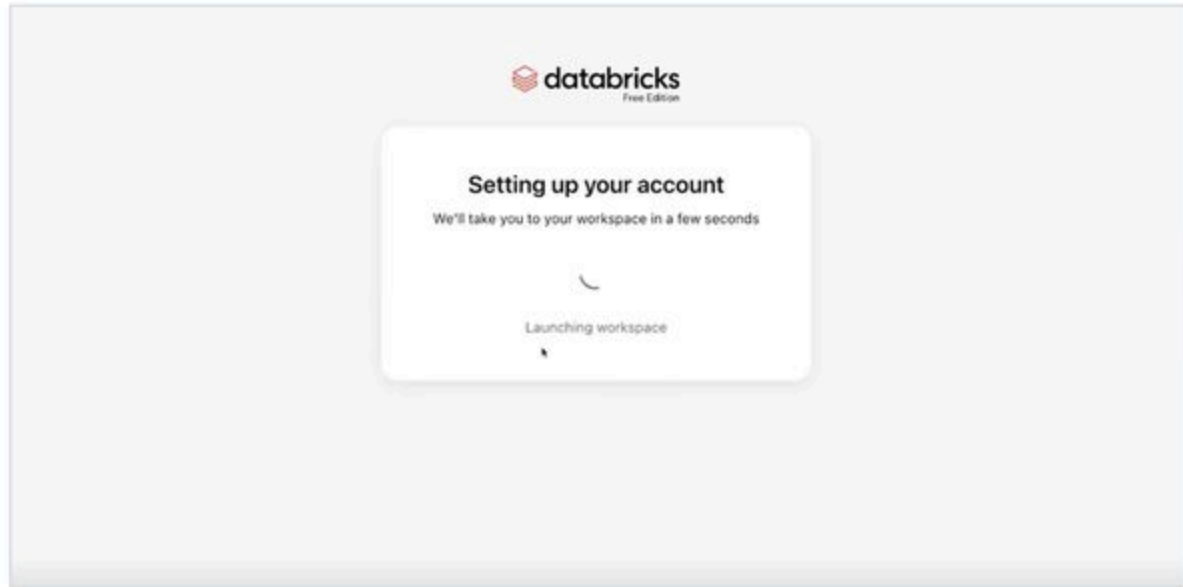
- Continue with Google, Microsoft or an email address
- Enter the 6-character code sent by email
- Use an email you can open now

⚠ Use your own email address. Never share verification codes.

PART 3

Free Edition – set up your account

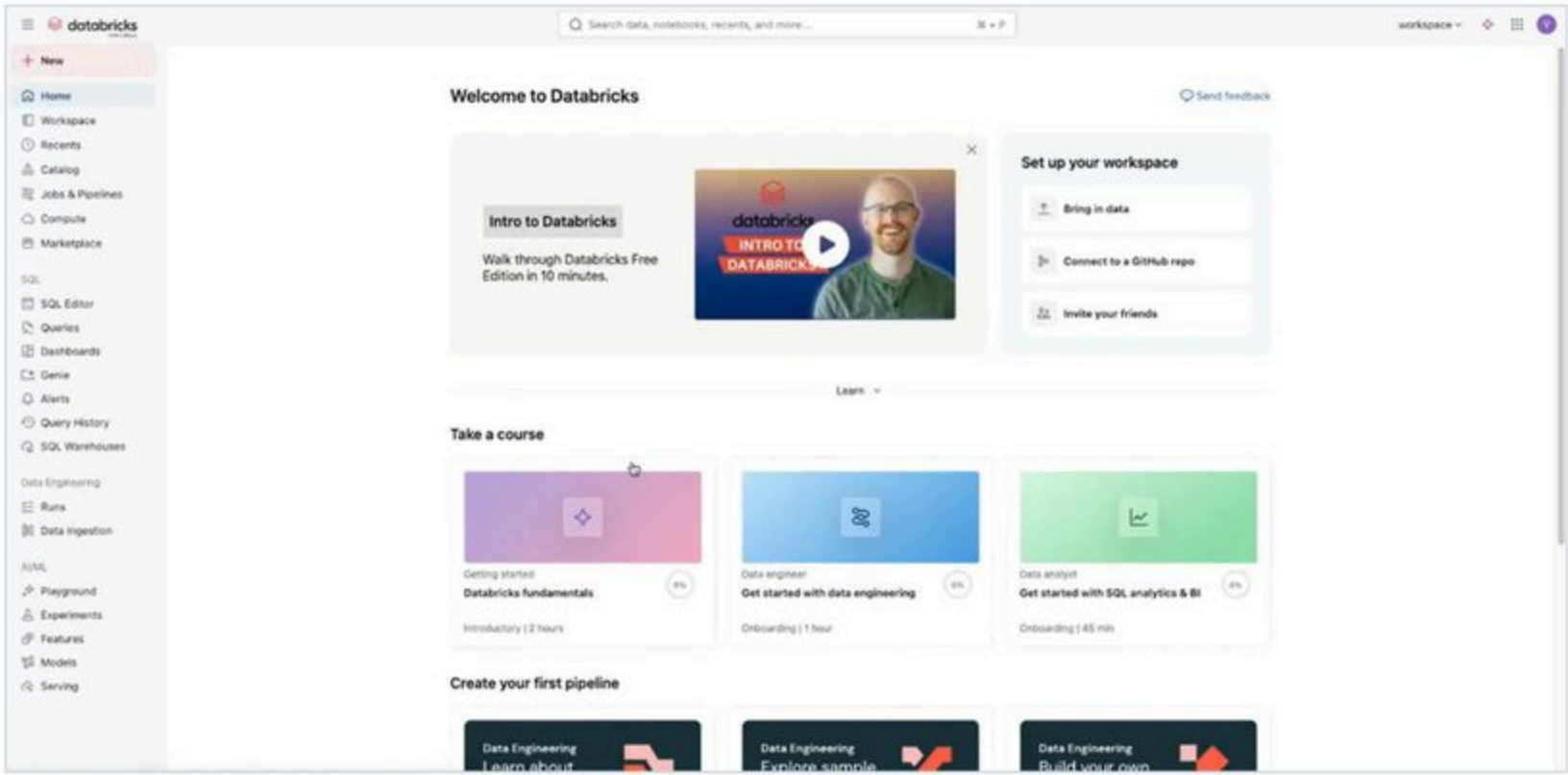




- Name your account
- Choose your country
- Solve the "are you human" puzzle if asked
- Wait for "Launching workspace"

PART 3

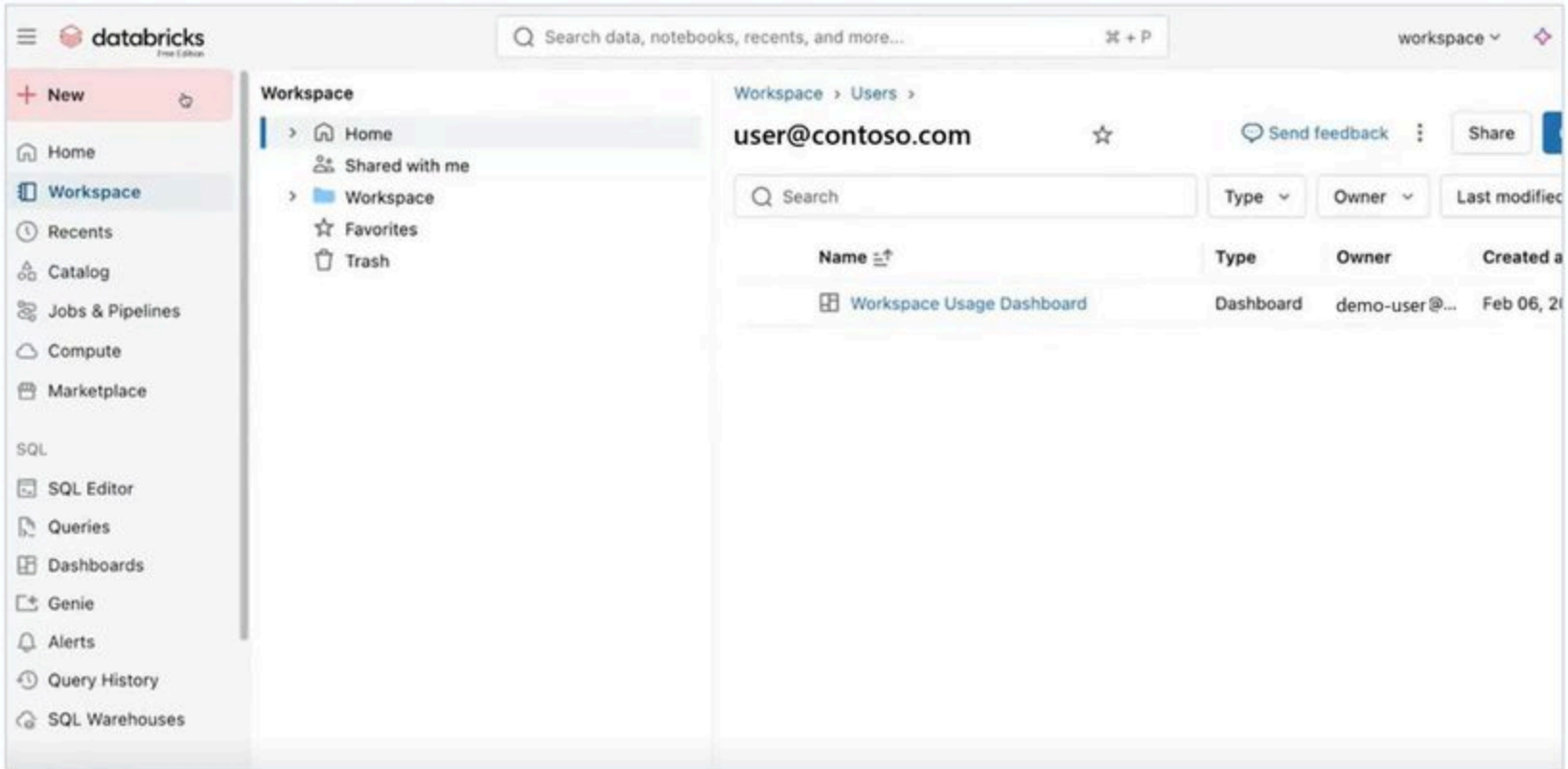
Welcome to your Databricks workspace



- Left menu: Workspace, Catalog, Jobs & Pipelines, Compute
- SQL: SQL Editor, Queries, Dashboards, Genie
- Free courses on the home page

PART 3

Workspace folders – your user folder



- Workspace → Users → your email
- Home = your personal folder
- Shared with me, Favorites, Trash

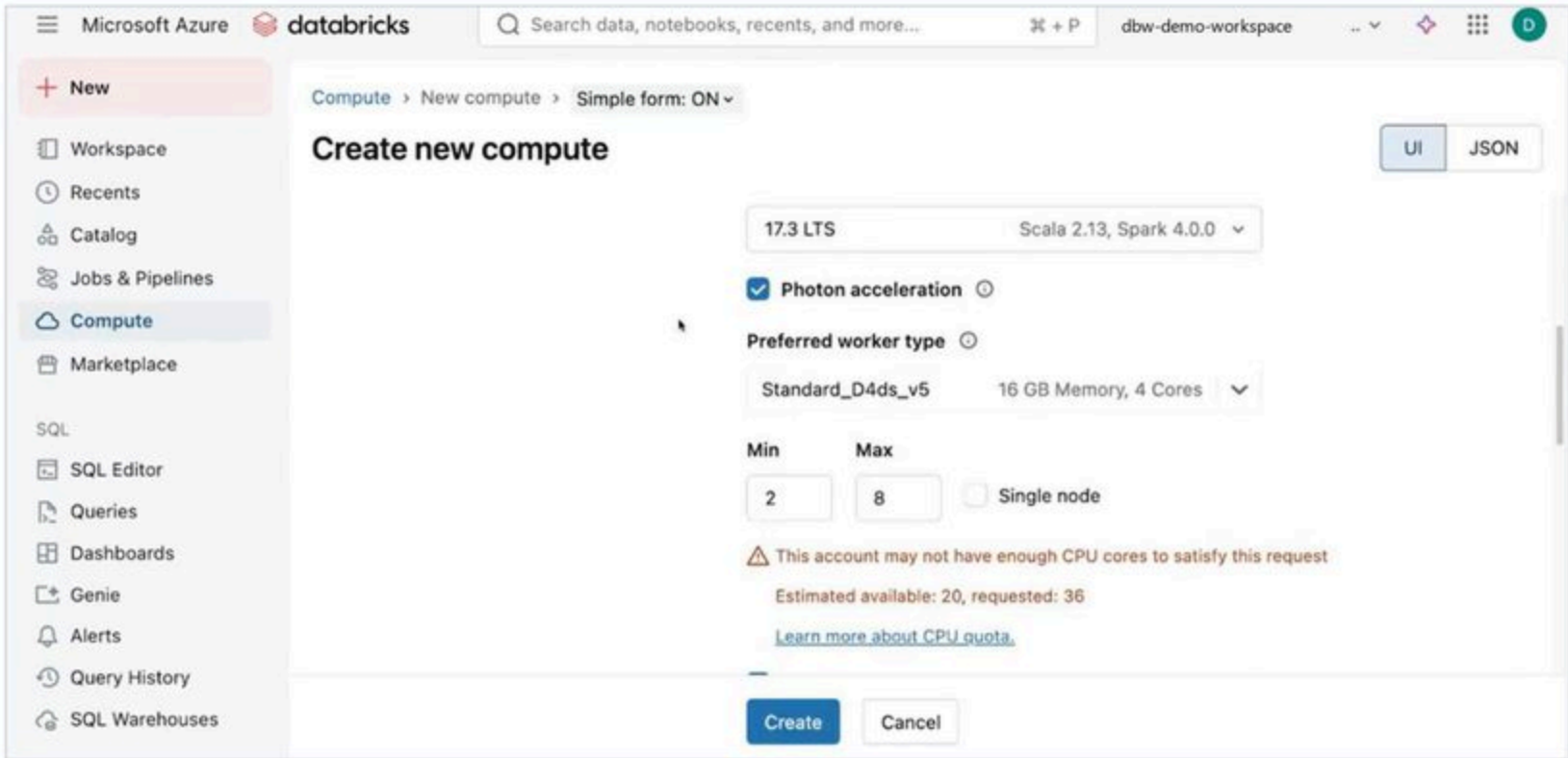
PART 4

Compute: clusters and SQL warehouses

Runtime • Photon • Worker type • Workers and autoscaling • Quotas • SQL warehouses

PART 4

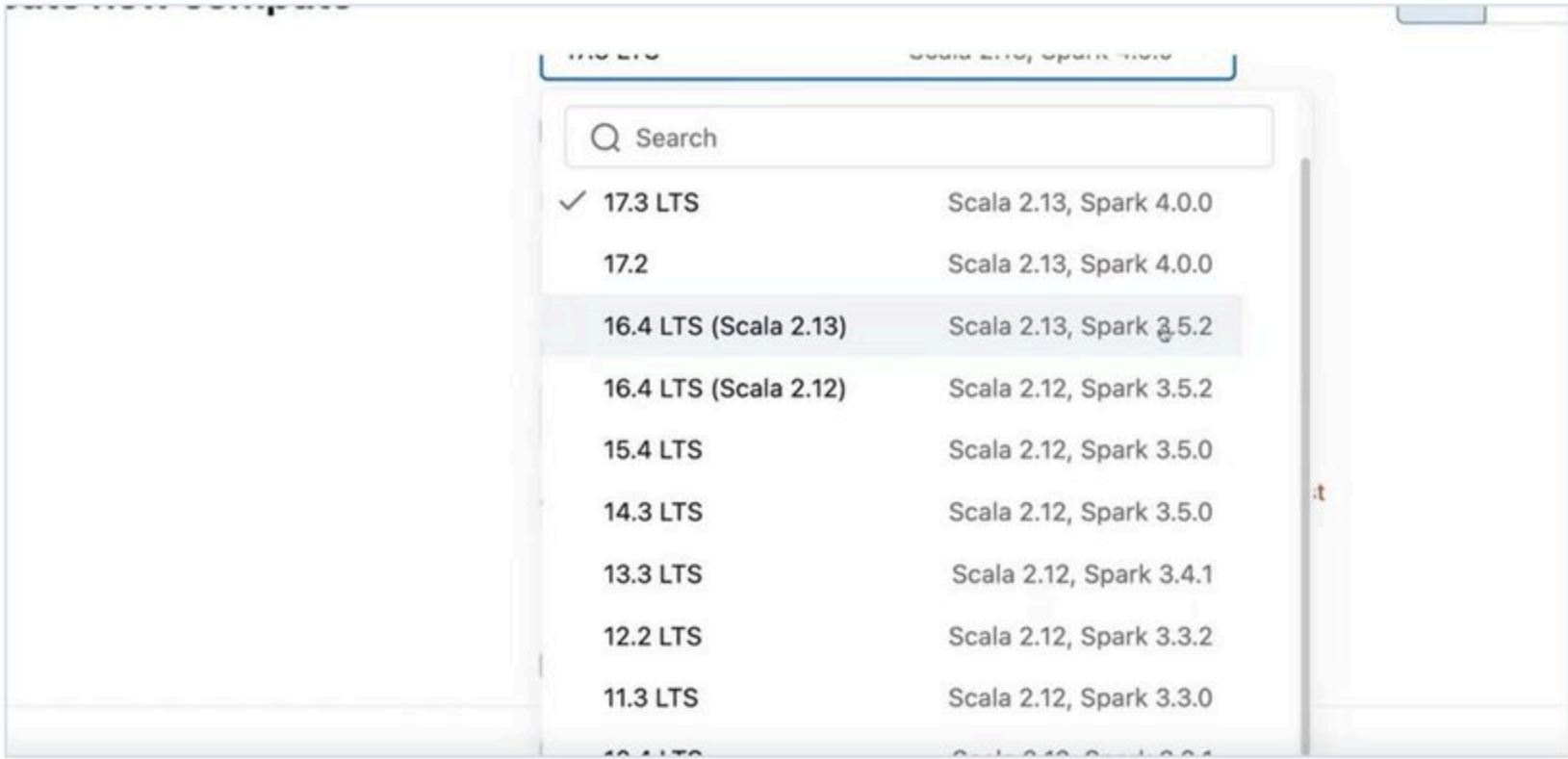
Create new compute (Azure Databricks)



- Compute → Create compute
- Databricks Runtime version
- Photon acceleration
- Worker type + min / max workers

PART 4

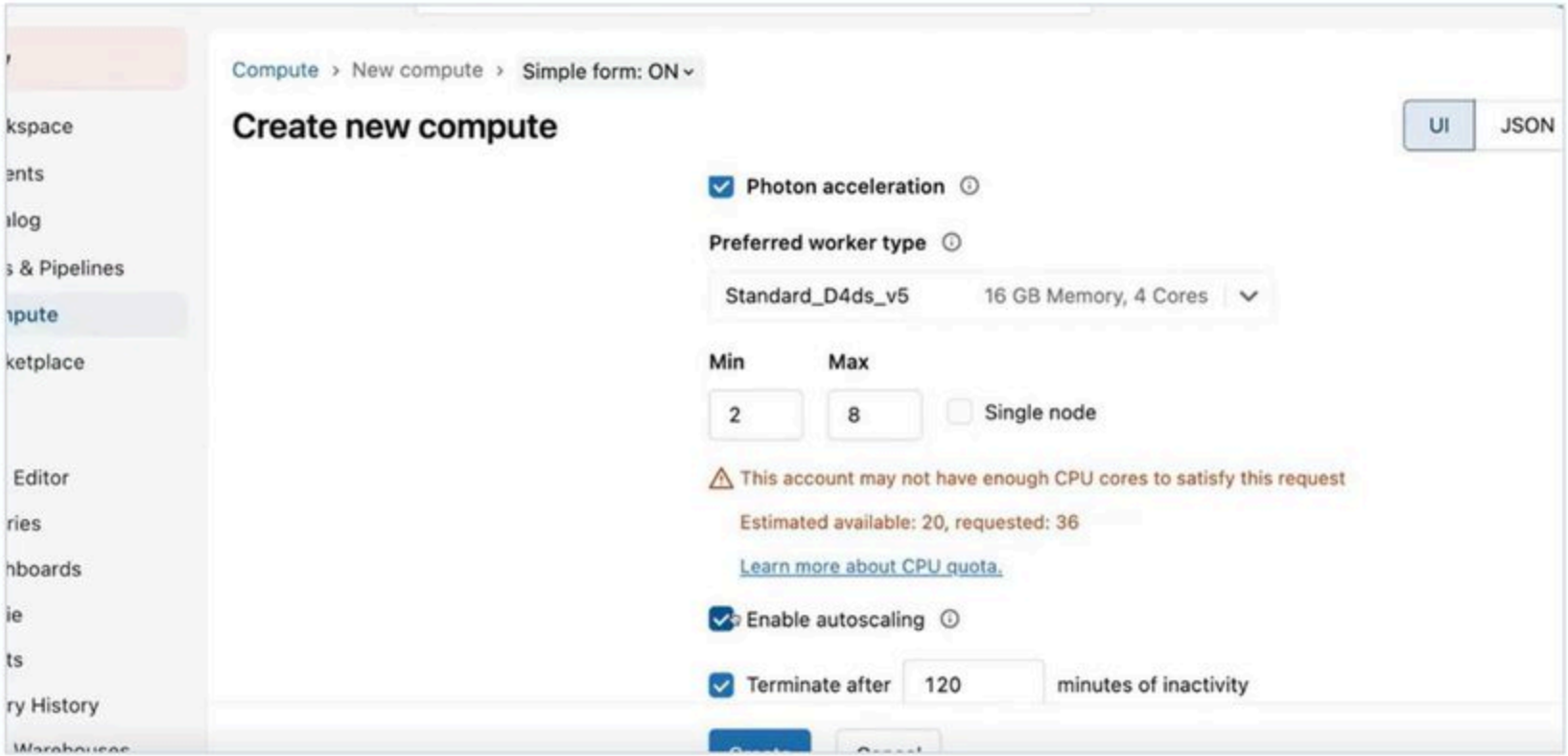
Databricks Runtime versions



- Each runtime = a Spark + Scala version
- 17.3 LTS → Spark 4.0.0
- 16.4 LTS → Spark 3.5.2
- Prefer LTS (Long Term Support)

PART 4

Workers, autoscaling and auto-termination



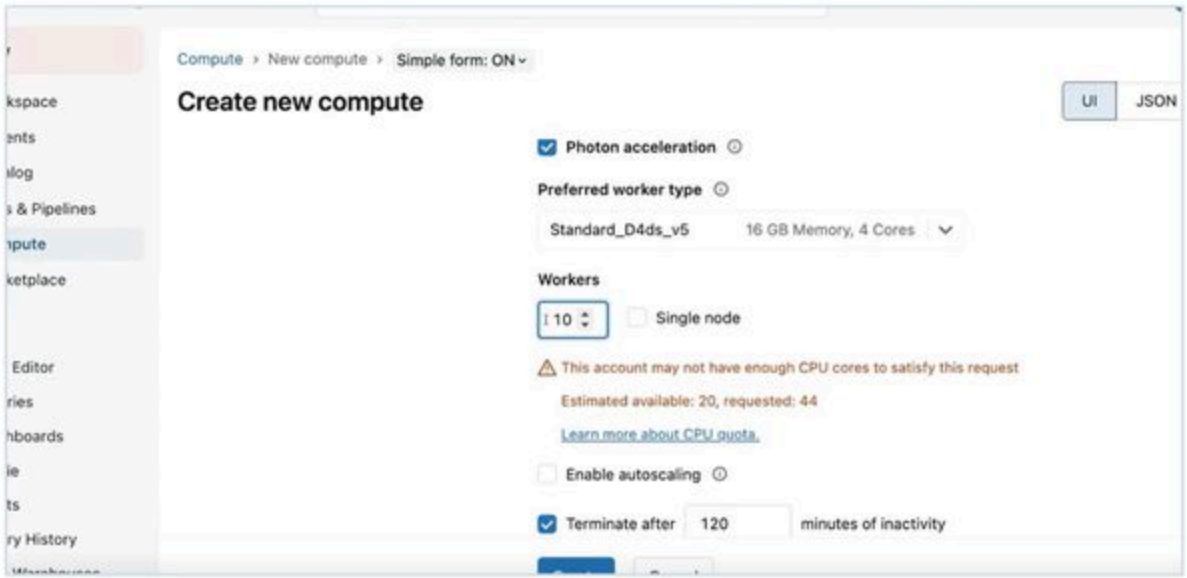
- Min 2 – Max 8 workers
- Enable autoscaling
- Terminate after 120 minutes of inactivity

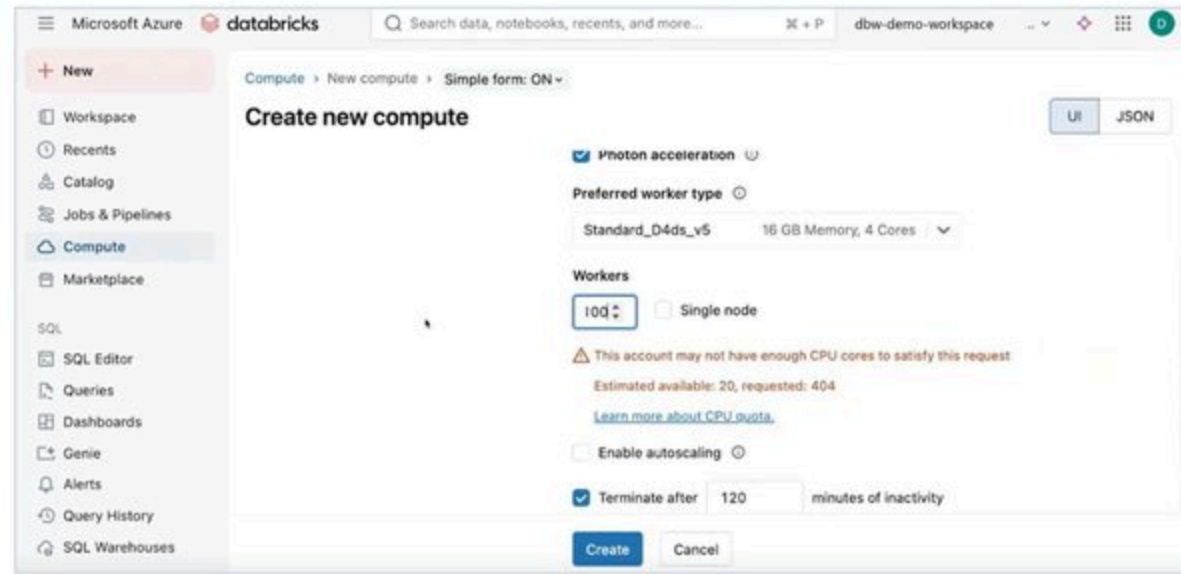
⚠ **Idle clusters still cost money. Always set auto-termination.**

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

PART 4

CPU quota – why big clusters fail



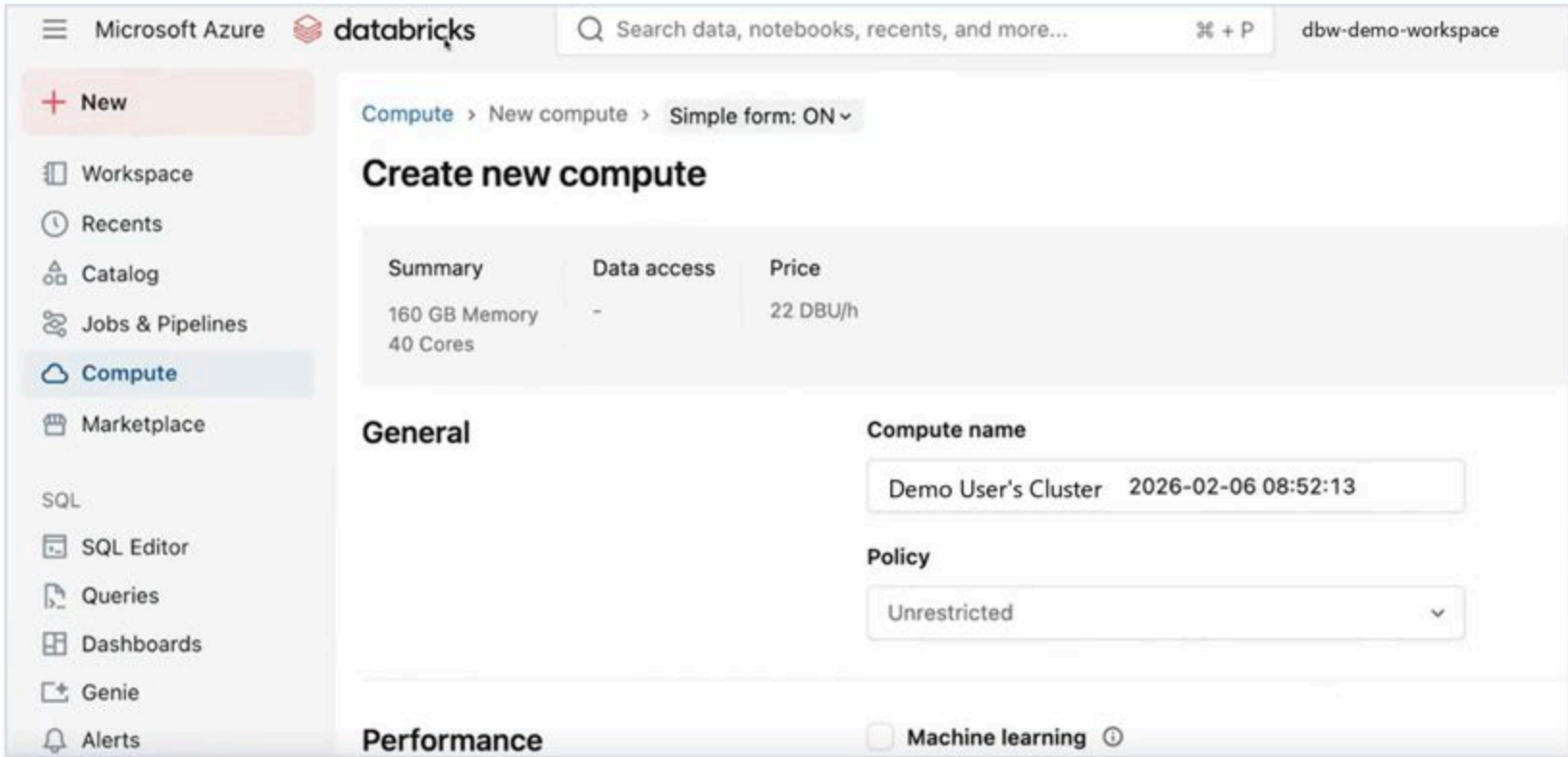


- 10 workers × 4 cores + driver = 44 cores
- 100 workers = 404 cores requested

- Subscription quota here: 20 cores
- Request a quota increase or use fewer workers

PART 4

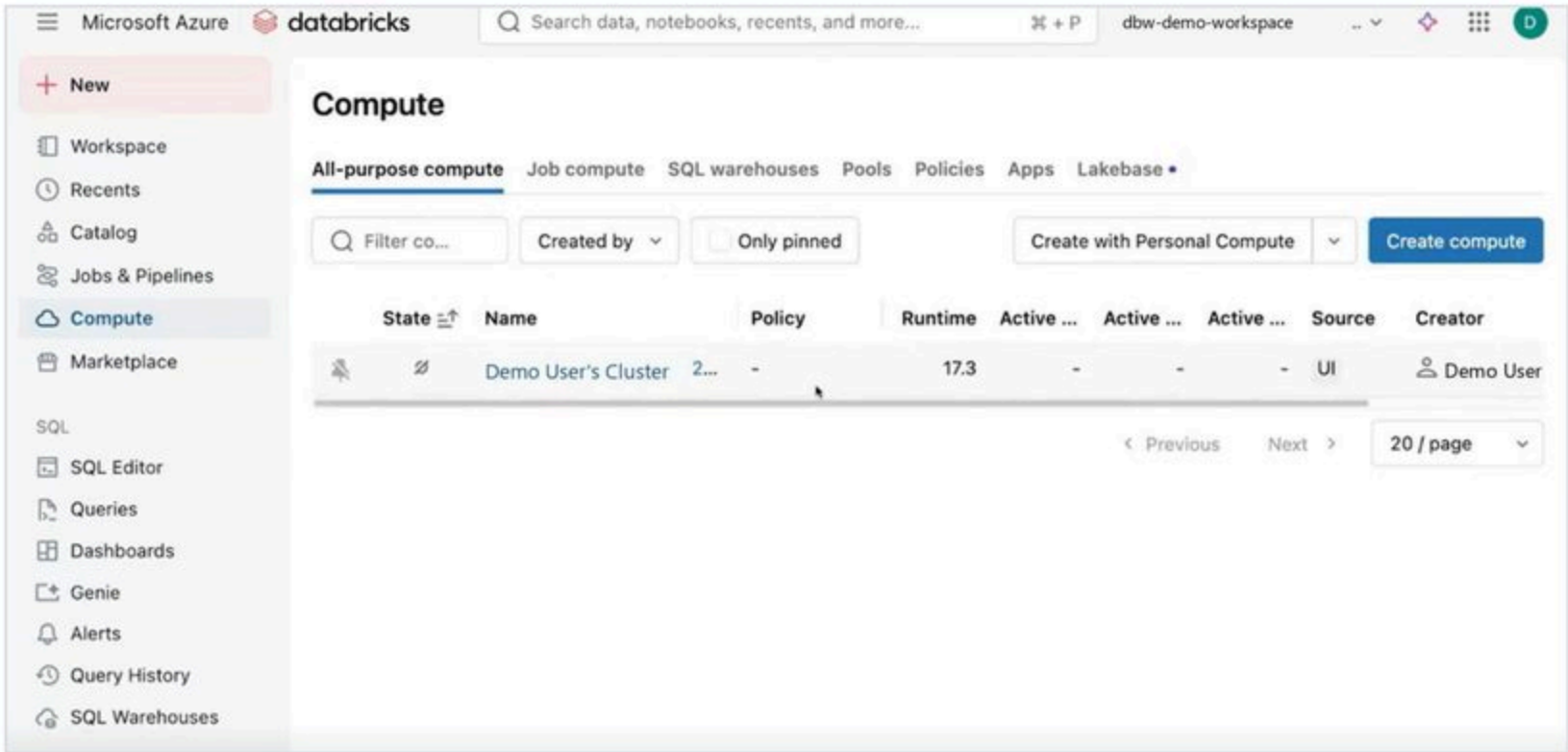
Compute summary and price



- Summary: memory and cores
- Price in DBU per hour
- Compute name + policy
- Unrestricted policy = any size

PART 4

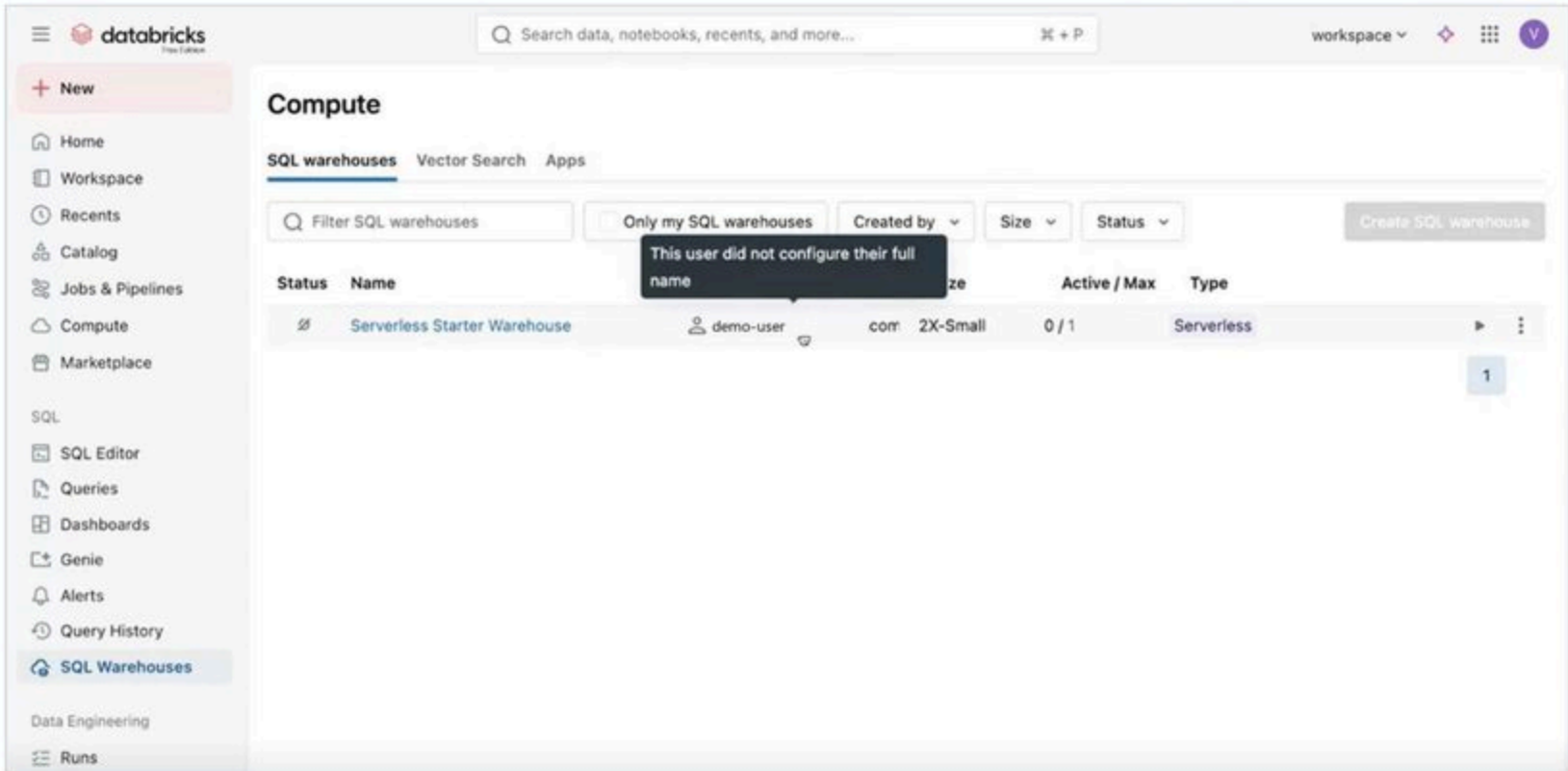
Compute list – all-purpose compute



- All-purpose compute: interactive notebooks
- Job compute: scheduled jobs
- SQL warehouses: SQL editor & dashboards
- State, runtime, creator

PART 4

Free Edition – serverless SQL warehouse



- Free Edition: serverless compute only
- Serverless Starter Warehouse (2X-Small)
- Starts in seconds, no cluster to manage

PART 4

Compute options at a glance

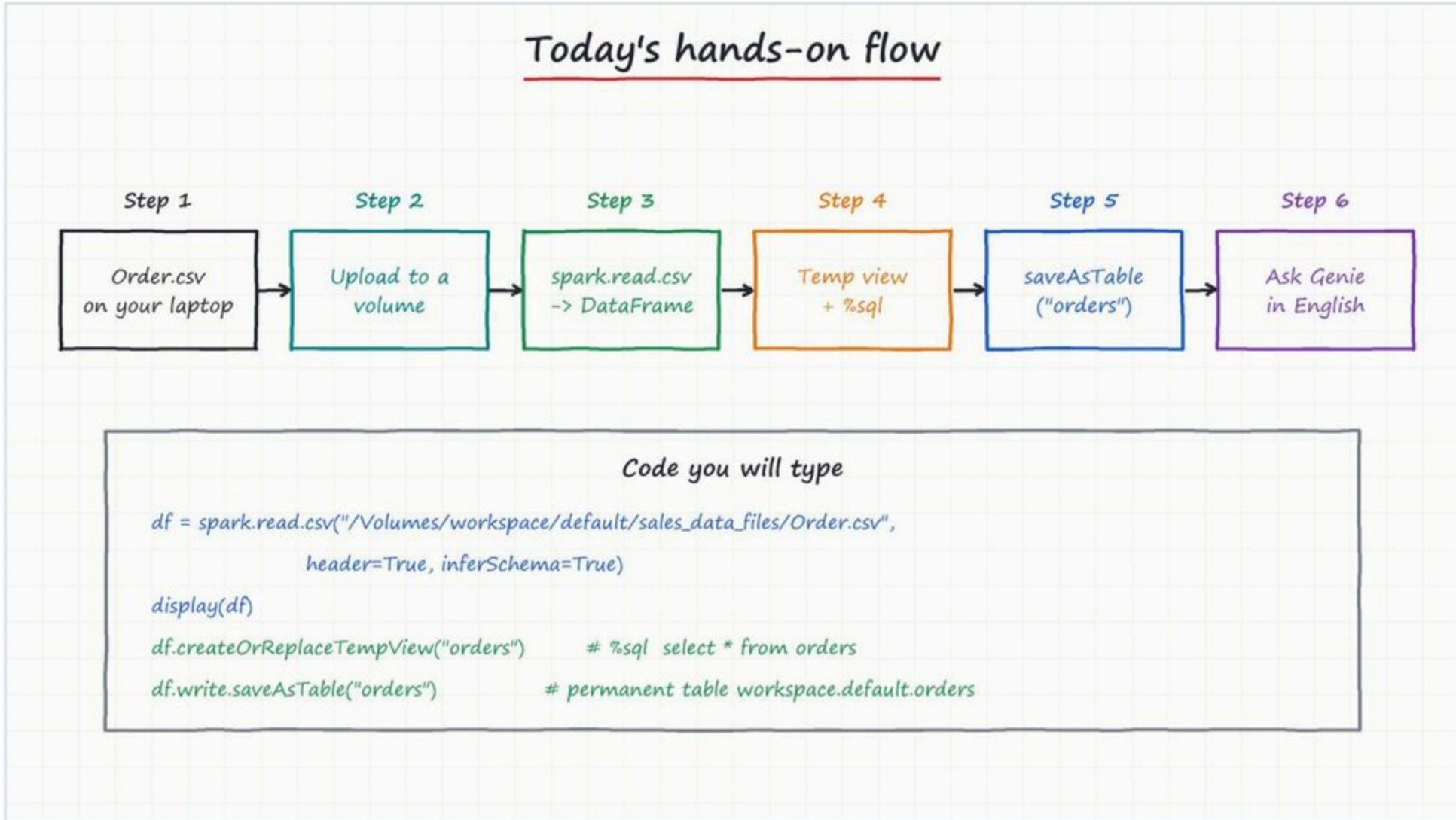
Option	Use it for
All-purpose cluster	Interactive development in notebooks
Job cluster	Production jobs – created per run, then terminated
Serverless compute	Notebooks/jobs without managing clusters
SQL warehouse	SQL editor, dashboards, BI tools, Genie
Photon	Faster SQL / DataFrame execution
Autoscaling + auto-termination	Control performance and cost

LAB 1

Upload Order.csv to a volume

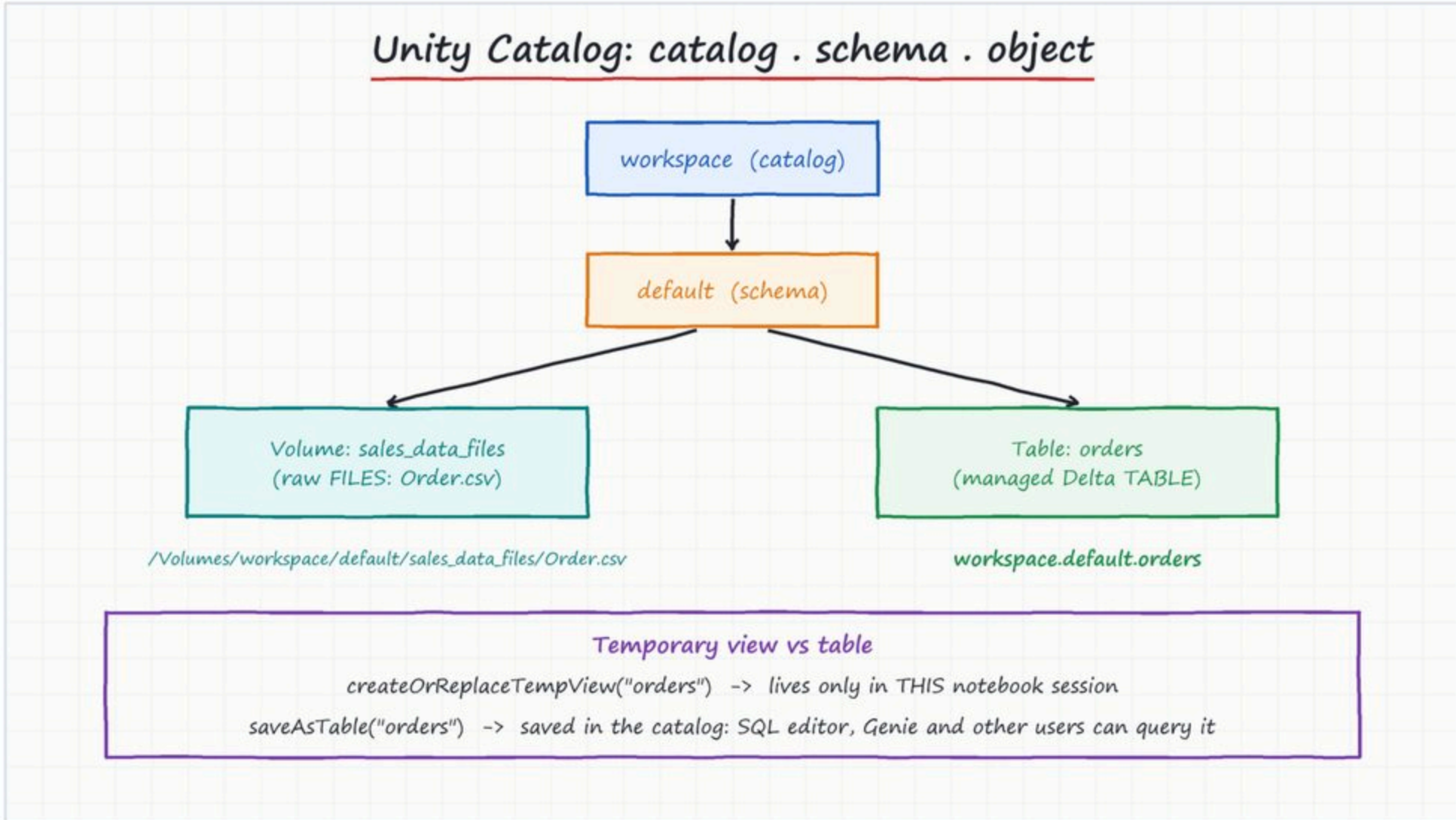
Unity Catalog • catalog.schema.volume • Upload a file

Today's hands-on flow



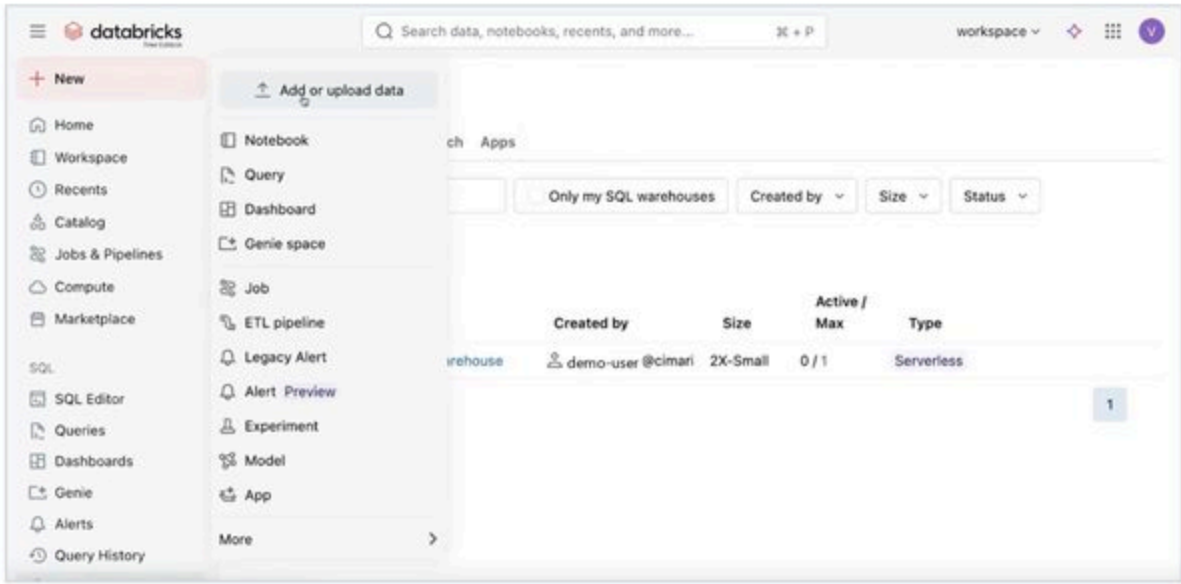
Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

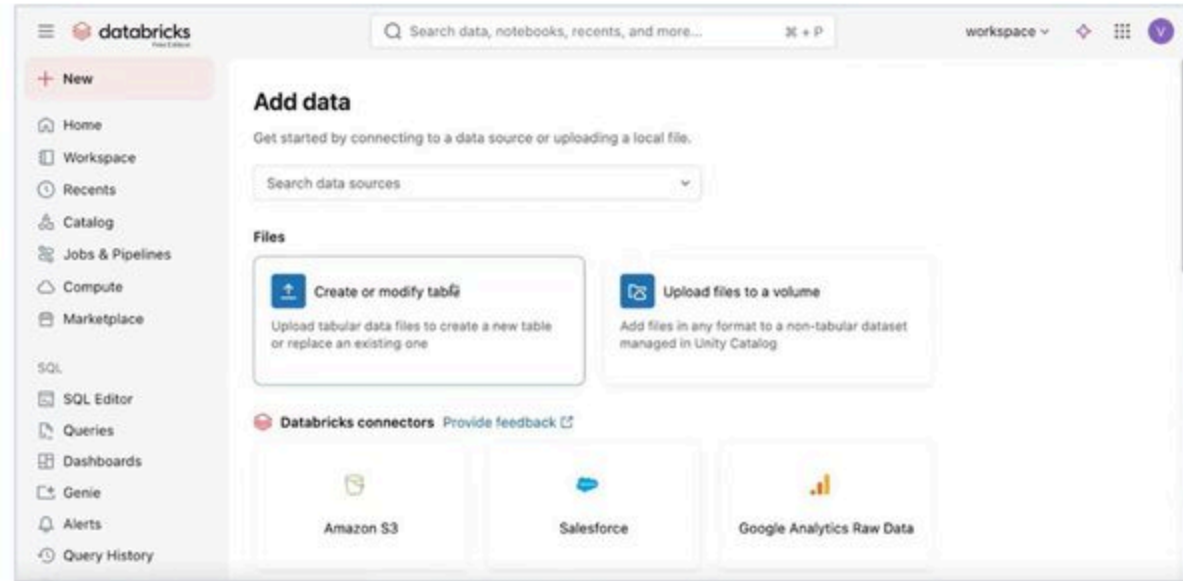
Unity Catalog: catalog . schema . object



LAB 1

New → Add or upload data

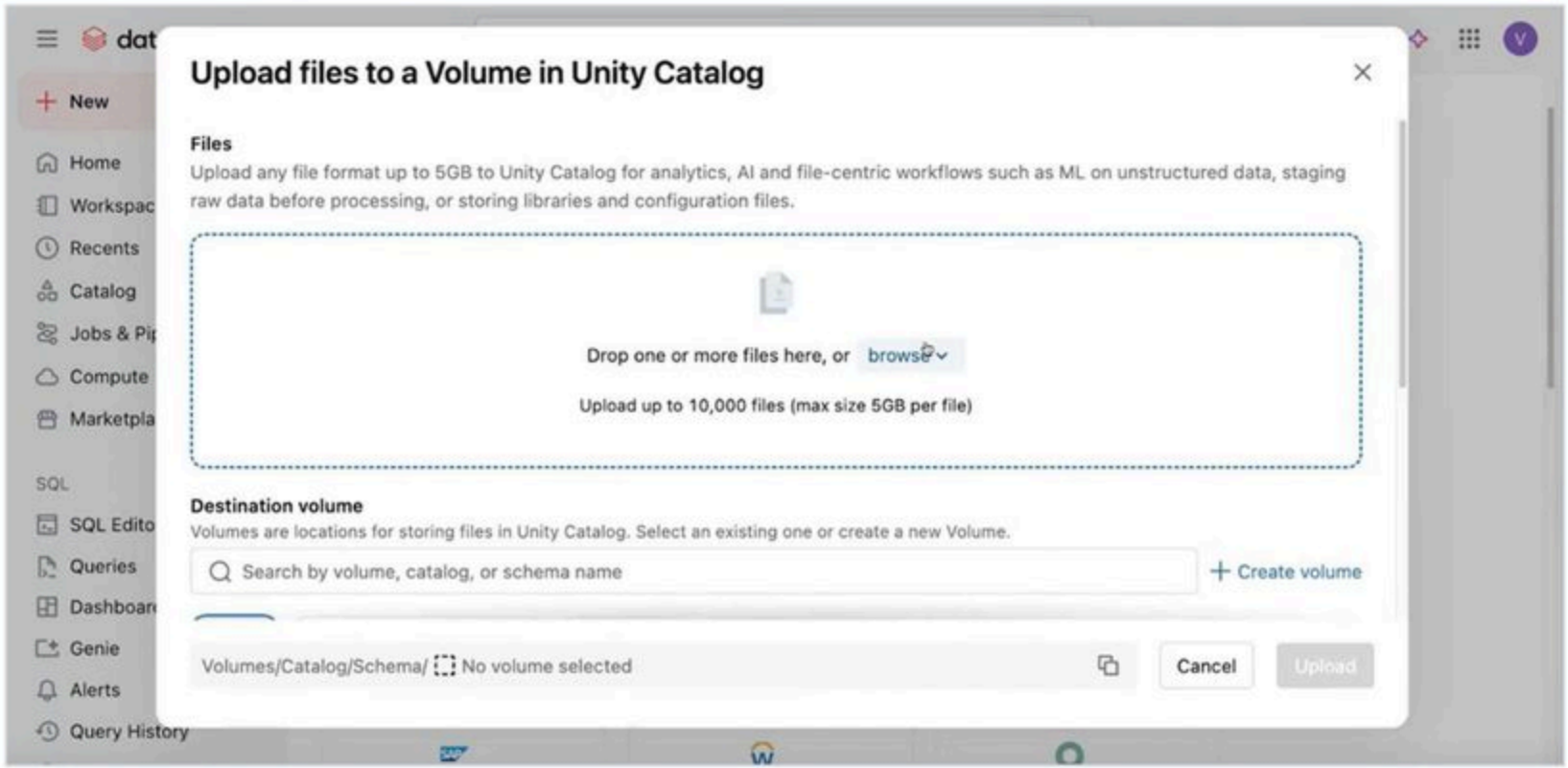




- + New → Add or upload data
- Create or modify table: tabular files → table
- Upload files to a volume: any file type
- Choose Upload files to a volume

LAB 1

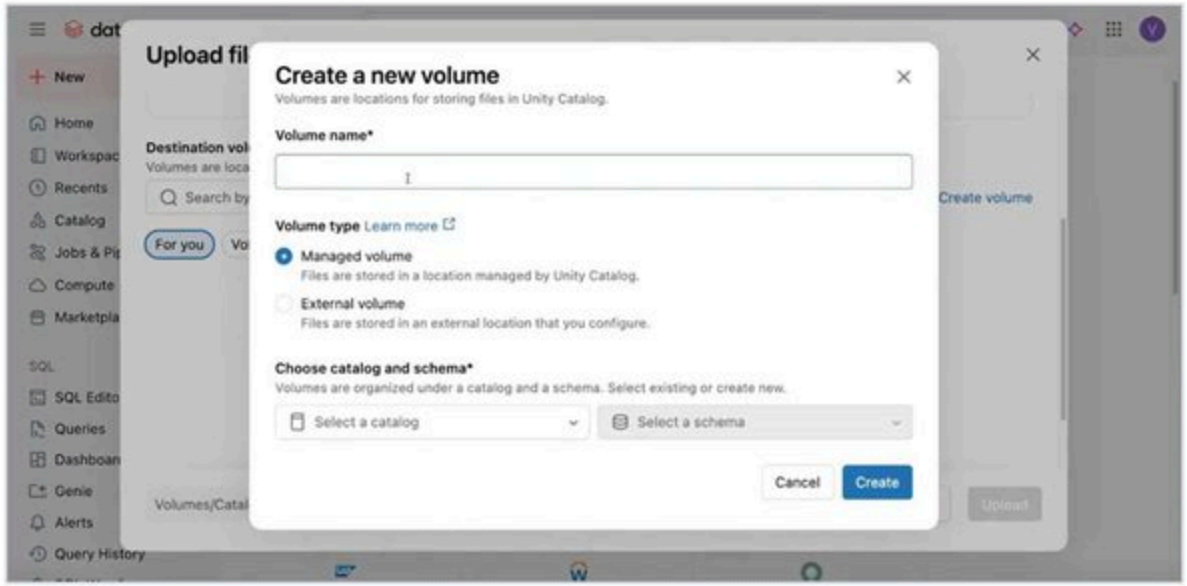
Upload files to a volume

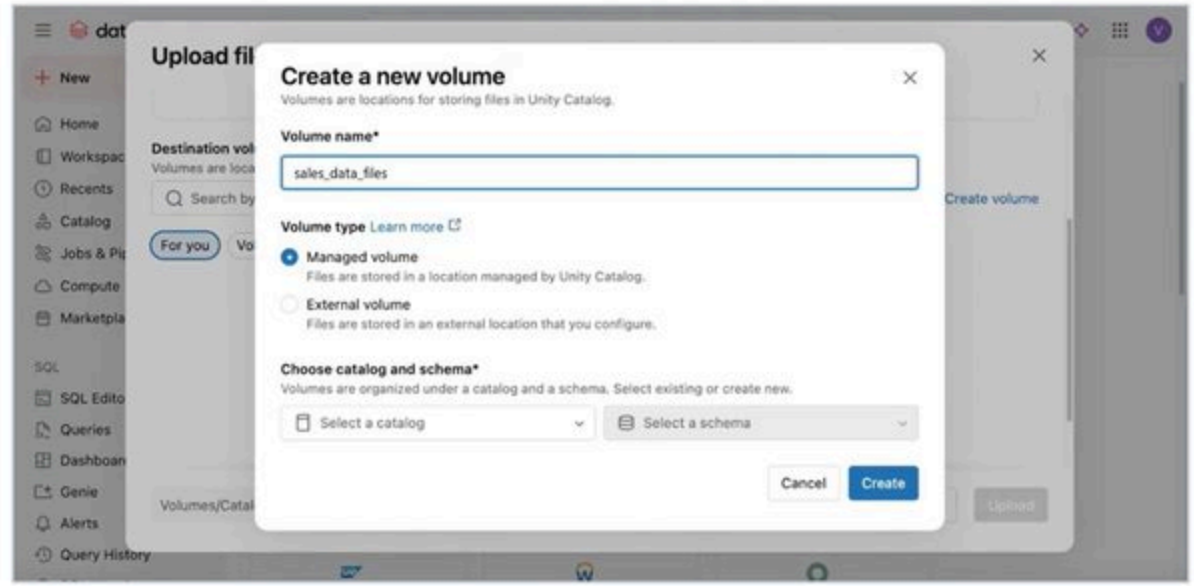


- Drop files or browse
- Up to 10,000 files, 5 GB each
- Destination volume: none yet
- Click + Create volume

LAB 1

Create a new volume





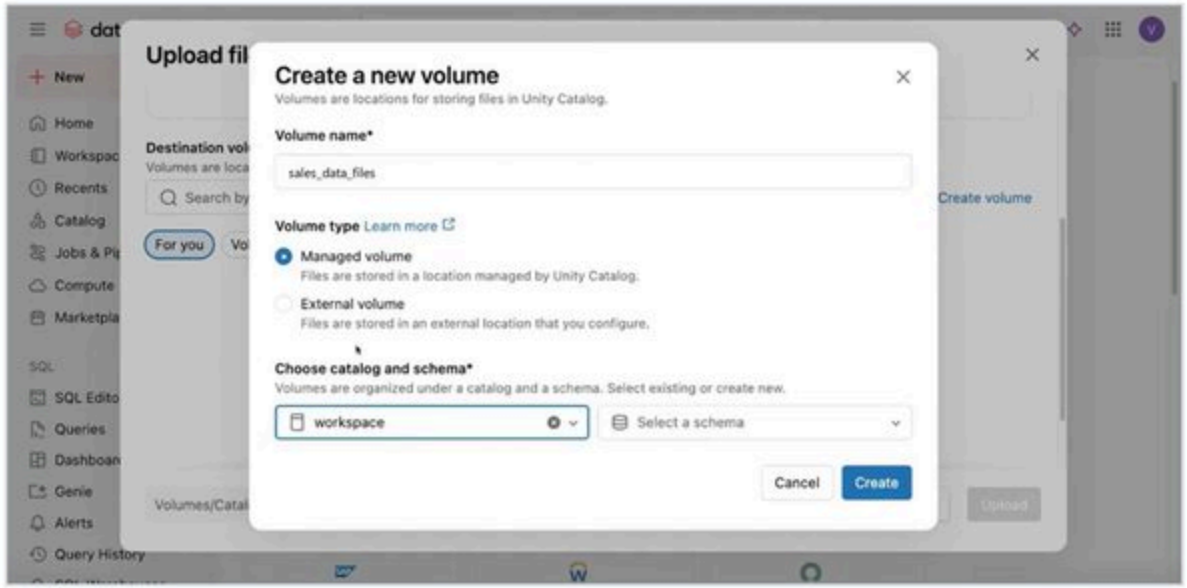
- Volume name: sales_data_files
- Managed volume (default)
- Catalog: workspace
- Schema: default

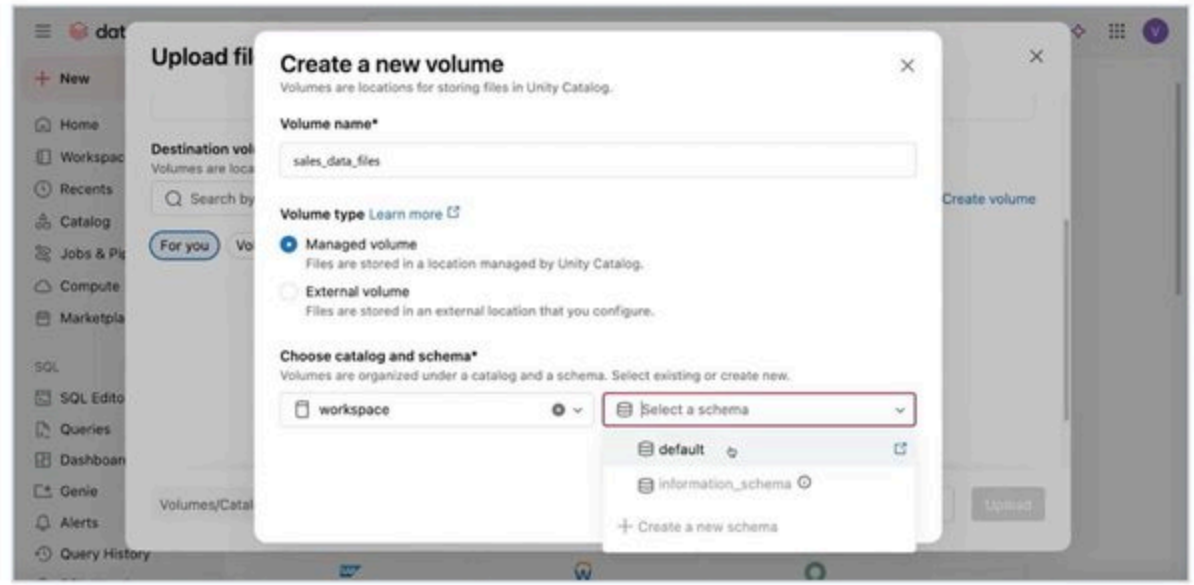
⚠ Use lower case and underscores in Unity Catalog names – they are easier to type in code.

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

LAB 1

Choose catalog and schema

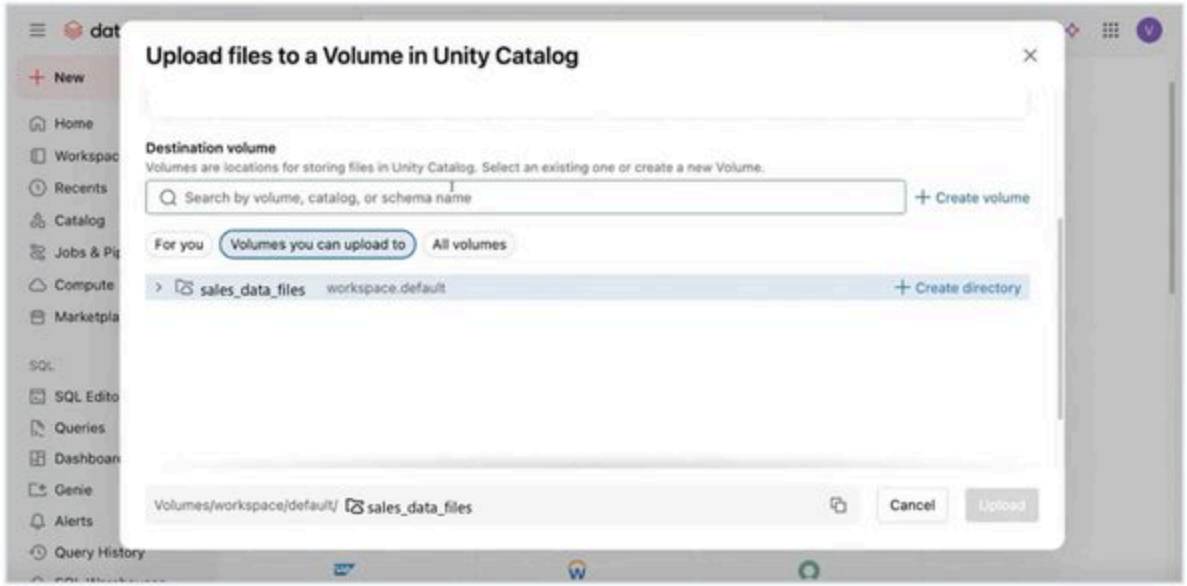


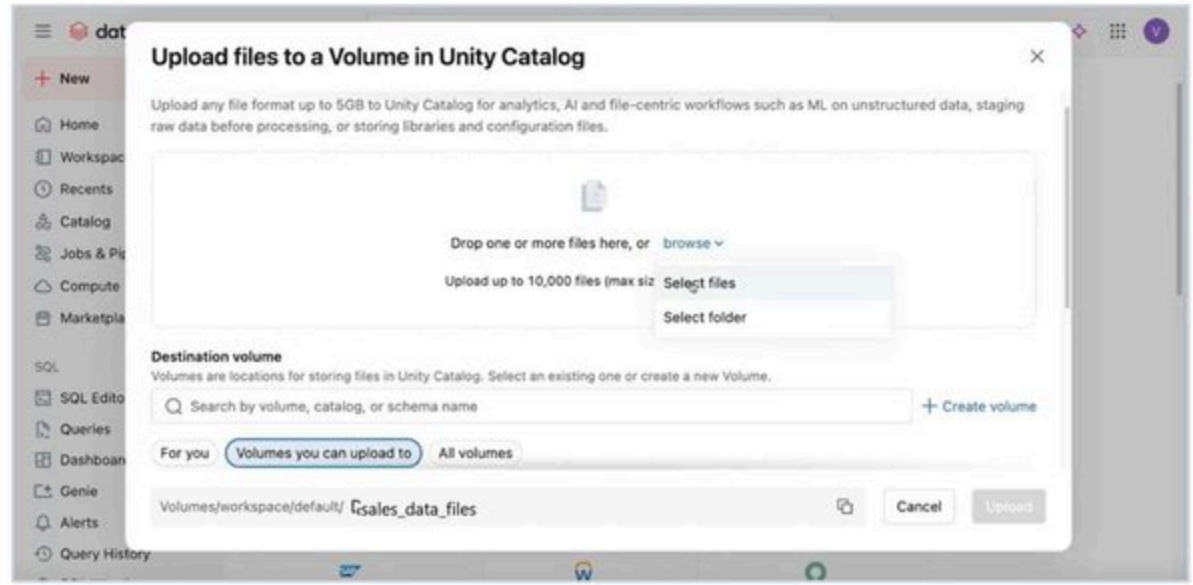


- Select catalog → workspace
- Select schema → default
- Error "A schema is required" until both are set
- Click Create

LAB 1

Volume ready – select the file

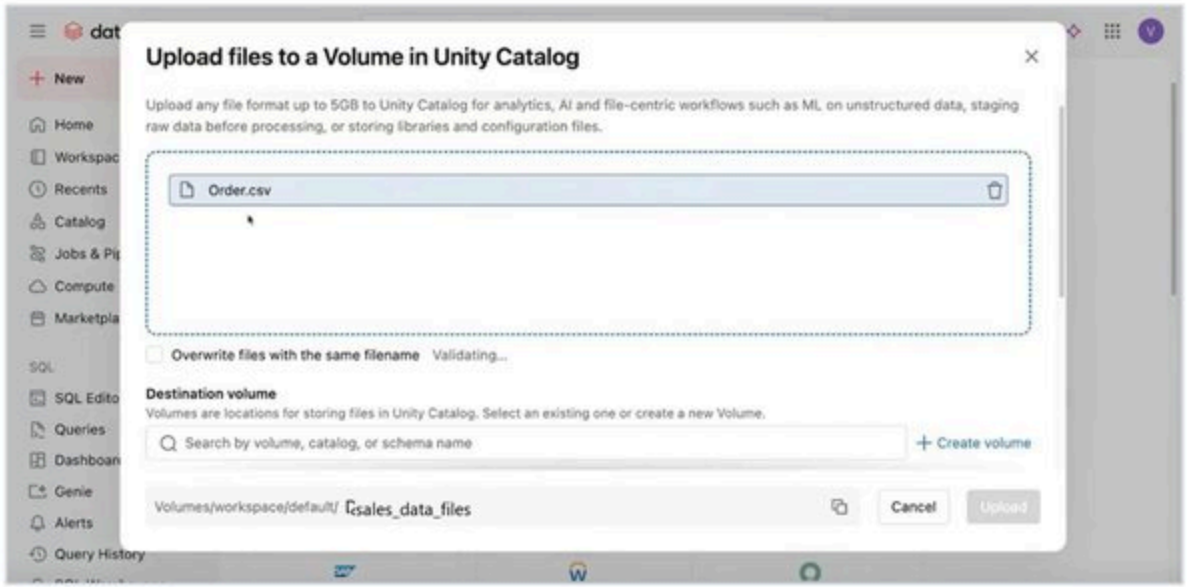


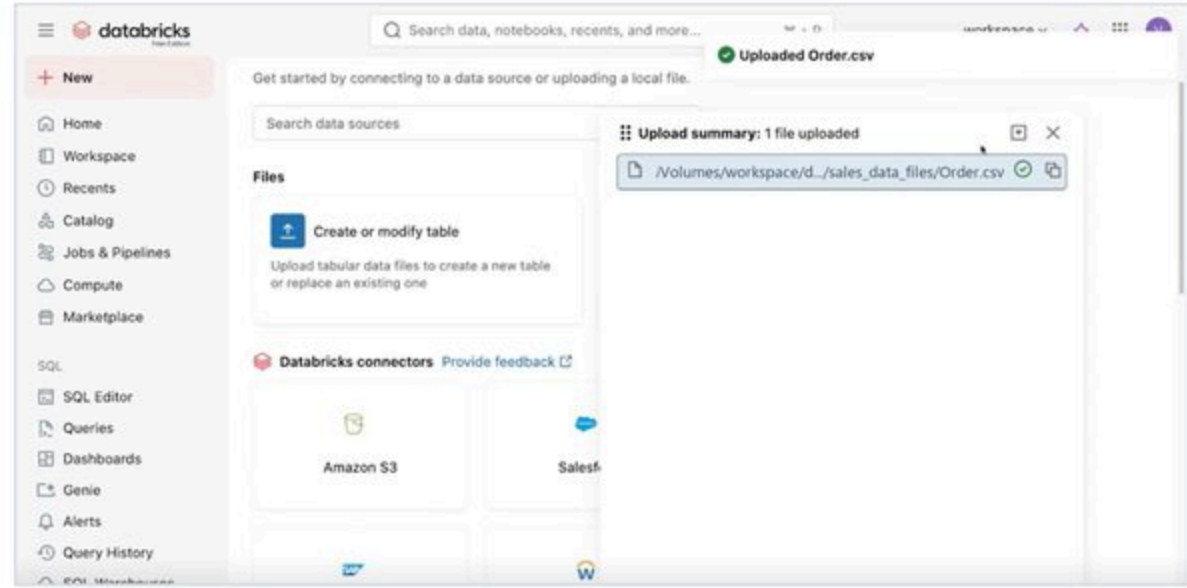


- Destination: /Volumes/workspace/default/sales_data_files
- browse → Select files
- Pick Order.csv from your laptop

LAB 1

Upload Order.csv

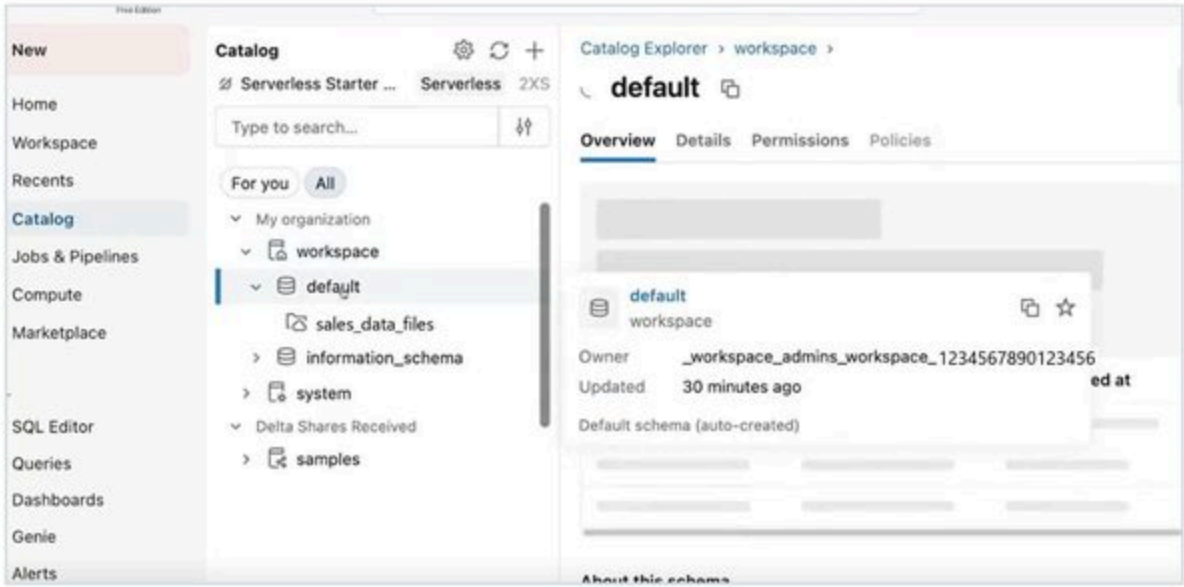


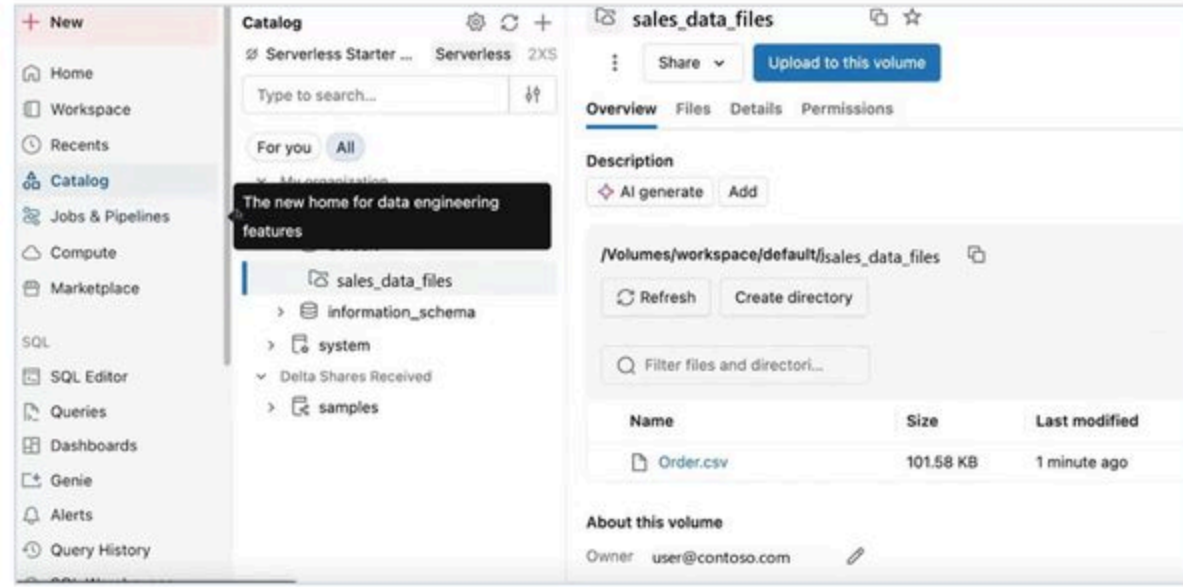


- Order.csv listed → Upload
- Upload summary: 1 file uploaded
- Copy the file path for later

LAB 1

Find the volume in Catalog Explorer

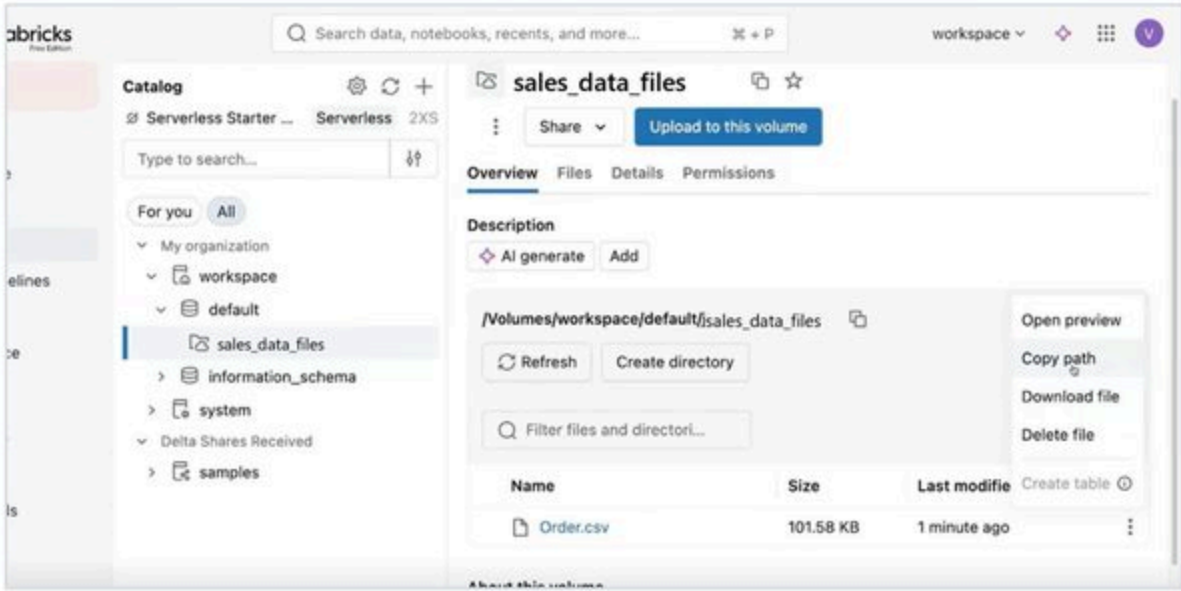


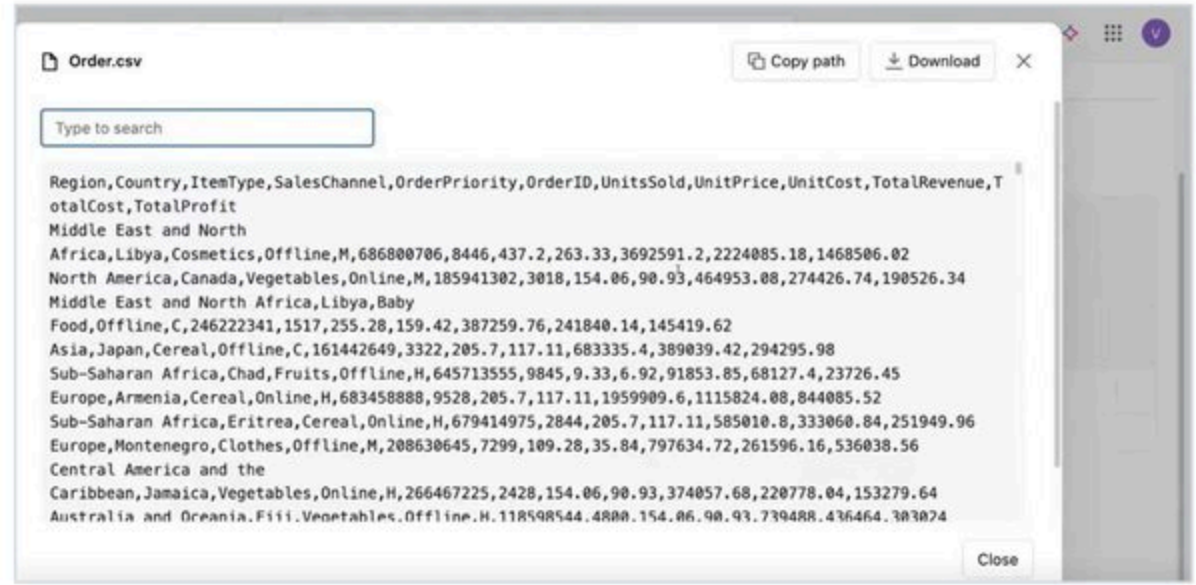


- Catalog → workspace → default → sales_data_files
- Overview shows the volume path
- Files: Order.csv, 101.58 KB
- Owner = you

LAB 1

Preview the file and copy its path





- File menu: Open preview, Copy path, Download, Delete
- Preview shows the raw CSV text
- First line = header (column names)
- 12 columns, comma-separated

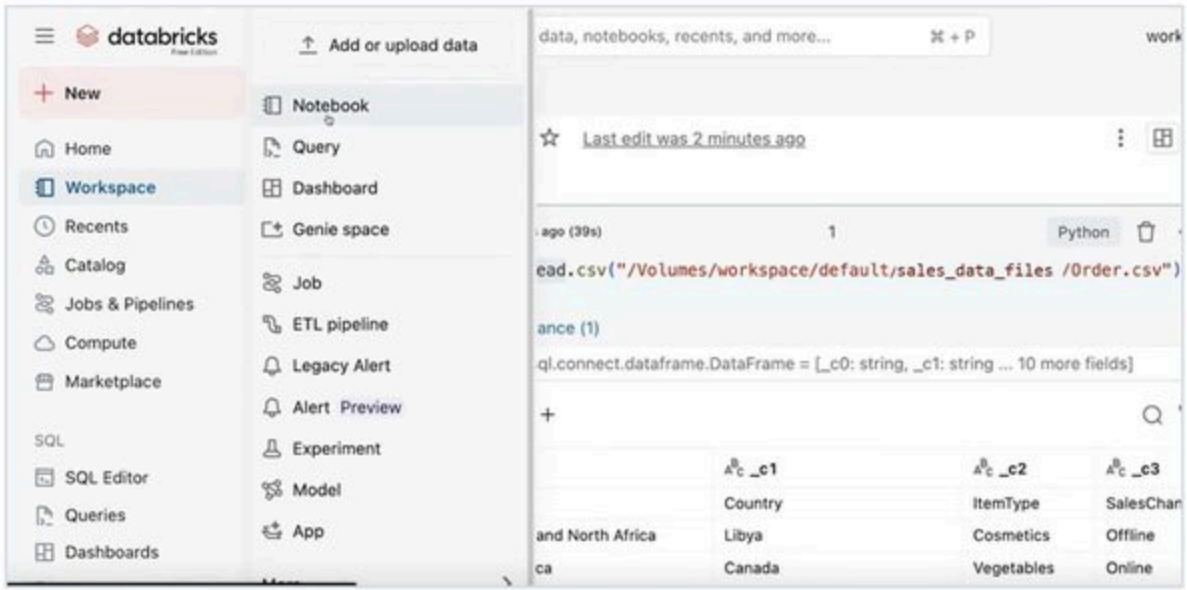
LAB 2

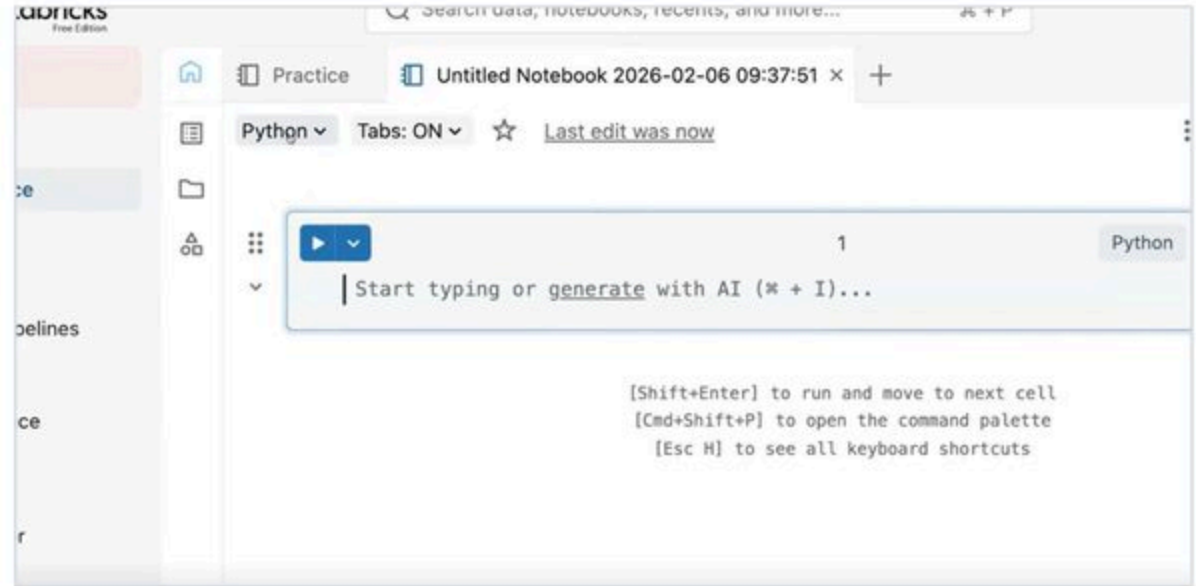
Read the CSV with PySpark

Notebook • spark.read.csv • header • inferSchema • display

LAB 2

Create a notebook

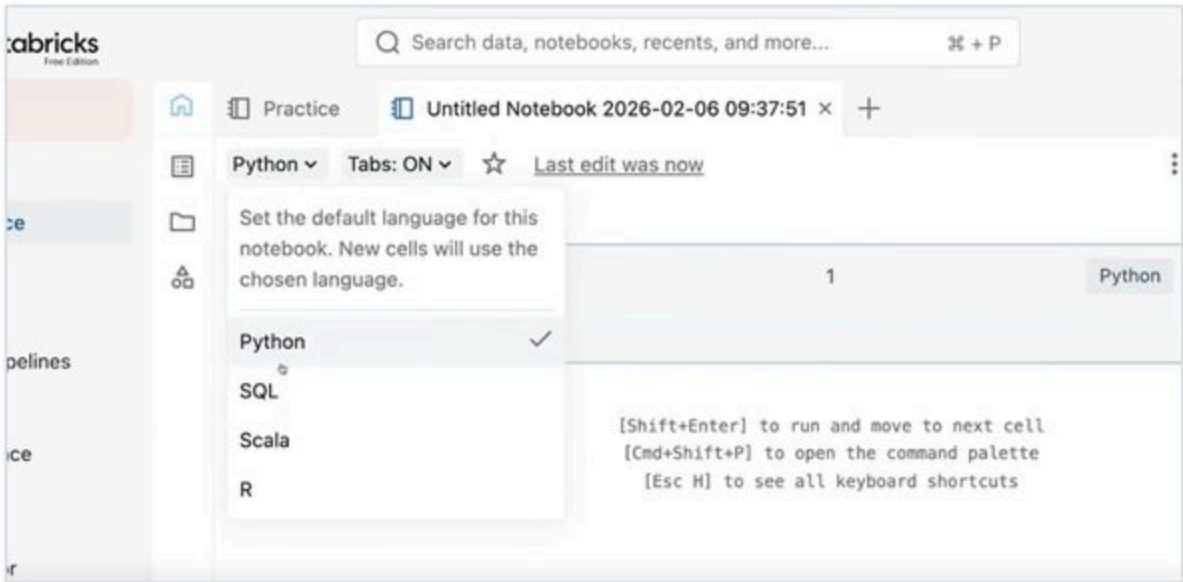


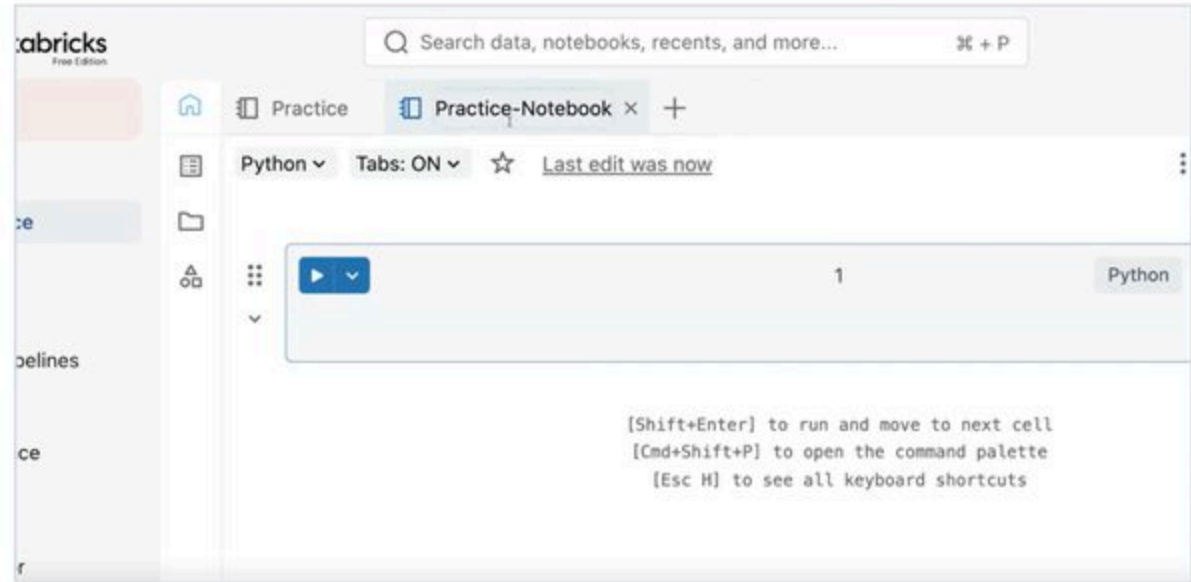


- + New → Notebook
- Opens "Untitled Notebook <date>"
- Cells run with Shift+Enter
- Serverless compute attaches automatically

LAB 2

Default language and notebook name



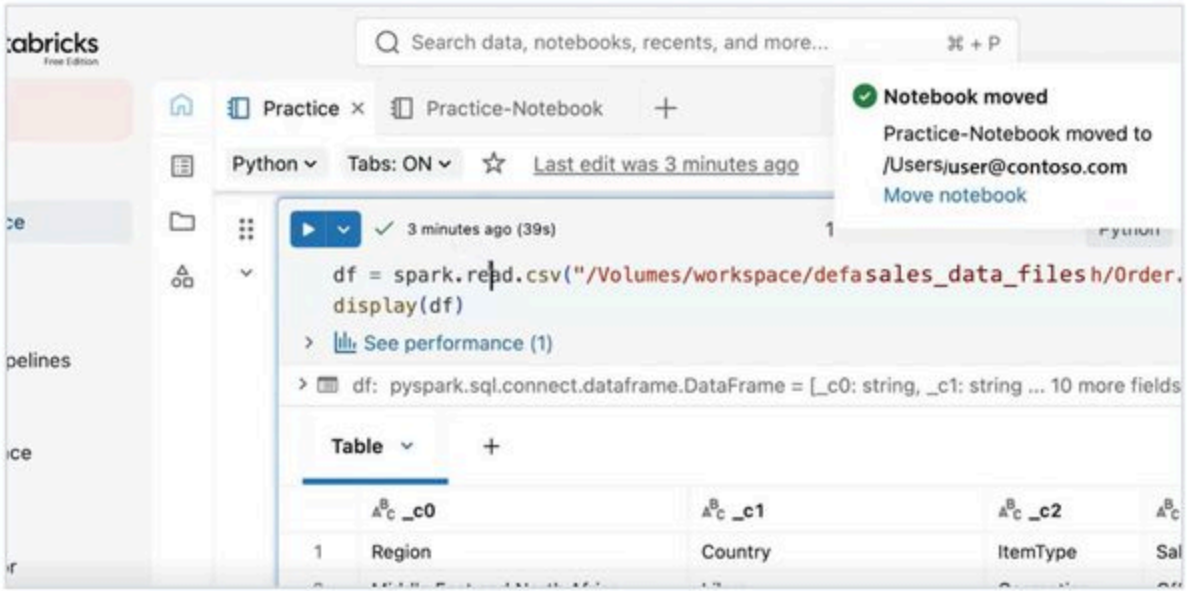


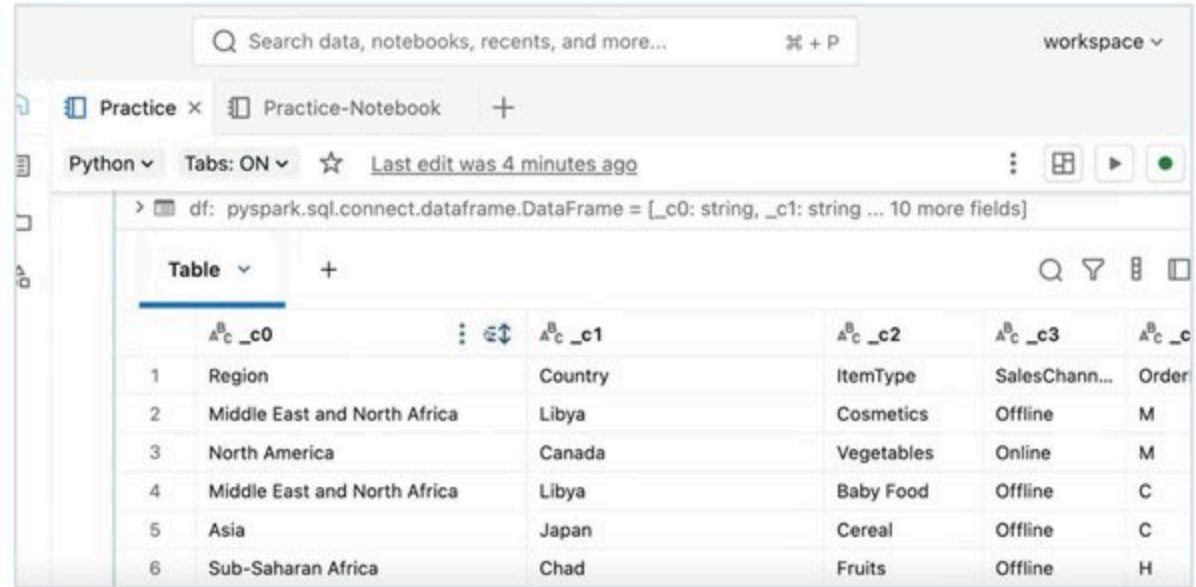
- Default language: Python, SQL, Scala or R
- Keep Python

- Rename: click the title → Practice-Notebook

LAB 2

First read: spark.read.csv(path)

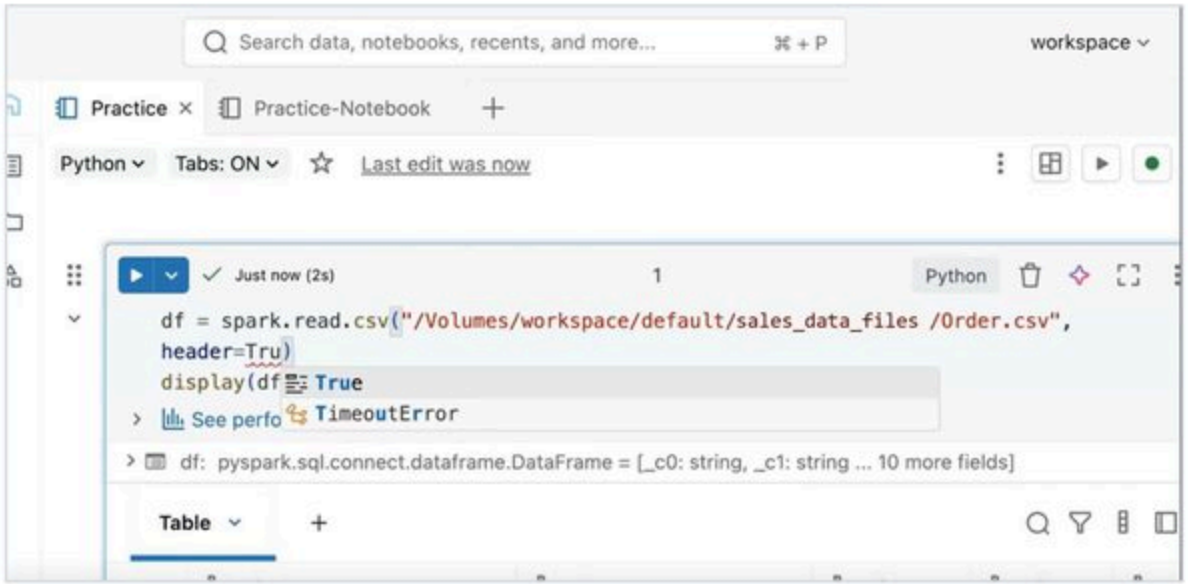


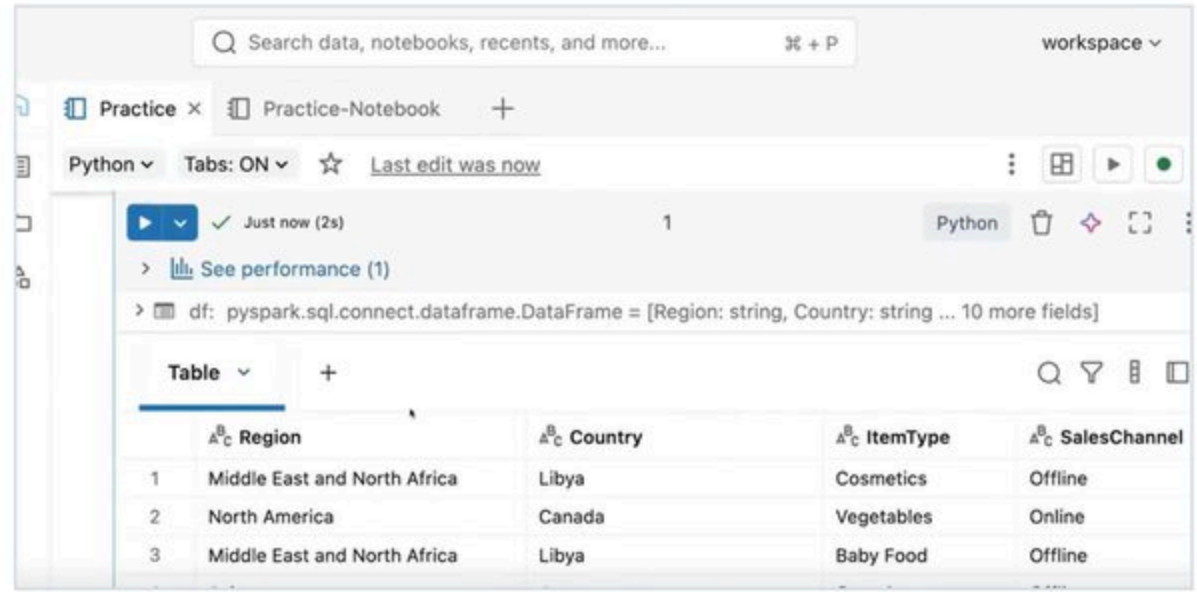


- df = spark.read.csv("/Volumes/.../Order.csv")
- display(df) shows a table
- Columns are _c0, _c1, _c2 ...
- Header line appears as row 1!

LAB 2

Fix the header: header=True



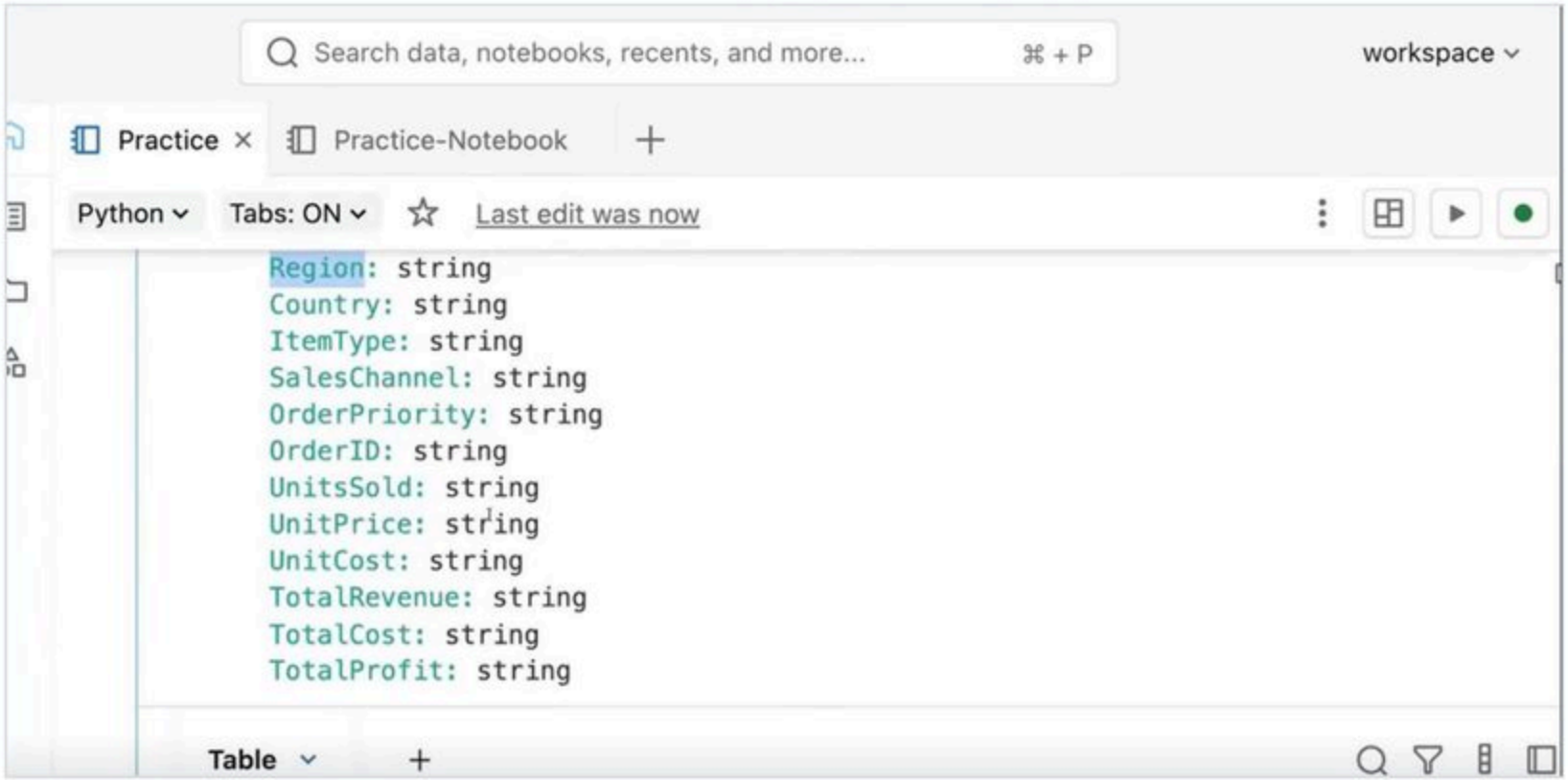


- Add header=True
- Column names come from the first line
- Row 1 is now real data

⚠ Python is case-sensitive: header=True works, header=true fails.

LAB 2

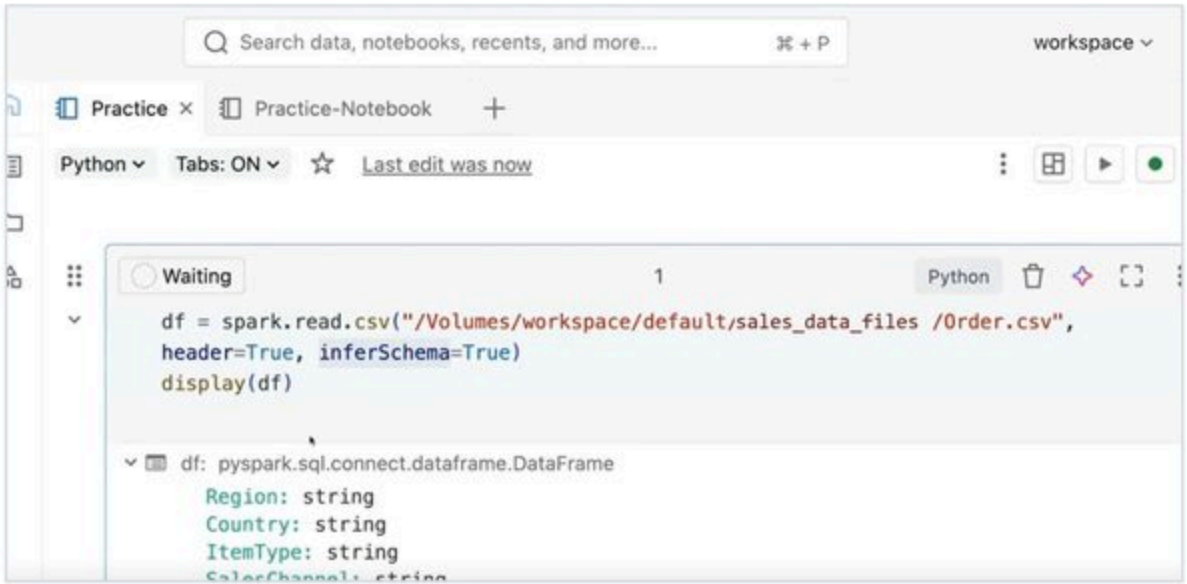
Check the schema – everything is a string

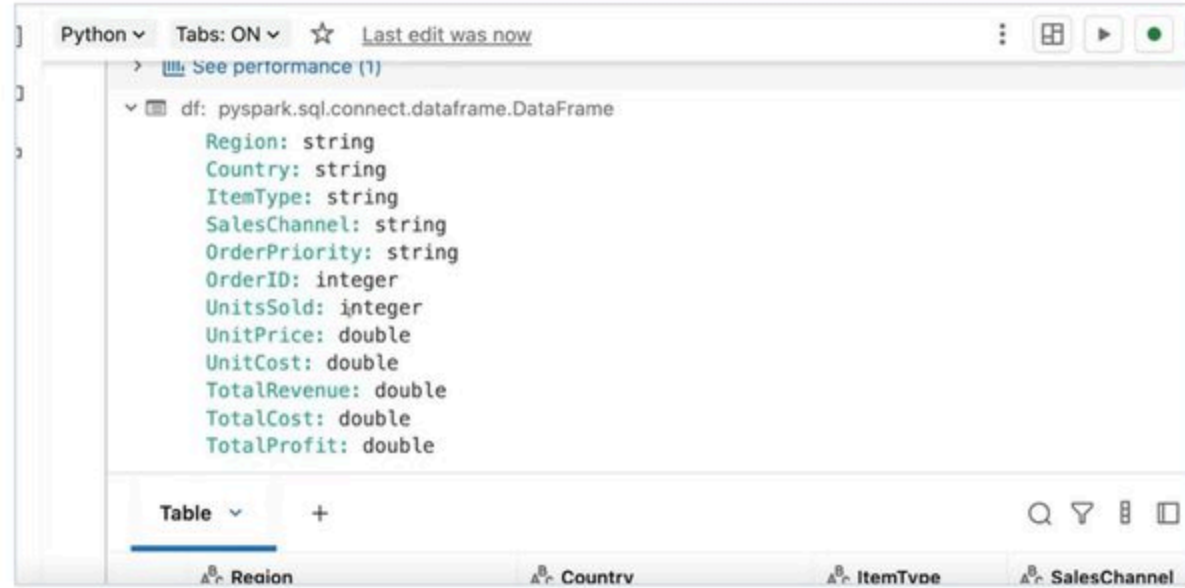


- Expand df: pyspark.sql.DataFrame
- Every column is string
- UnitsSold, UnitPrice … should be numbers
- Strings cannot be summed correctly

LAB 2

Detect data types: inferSchema=True



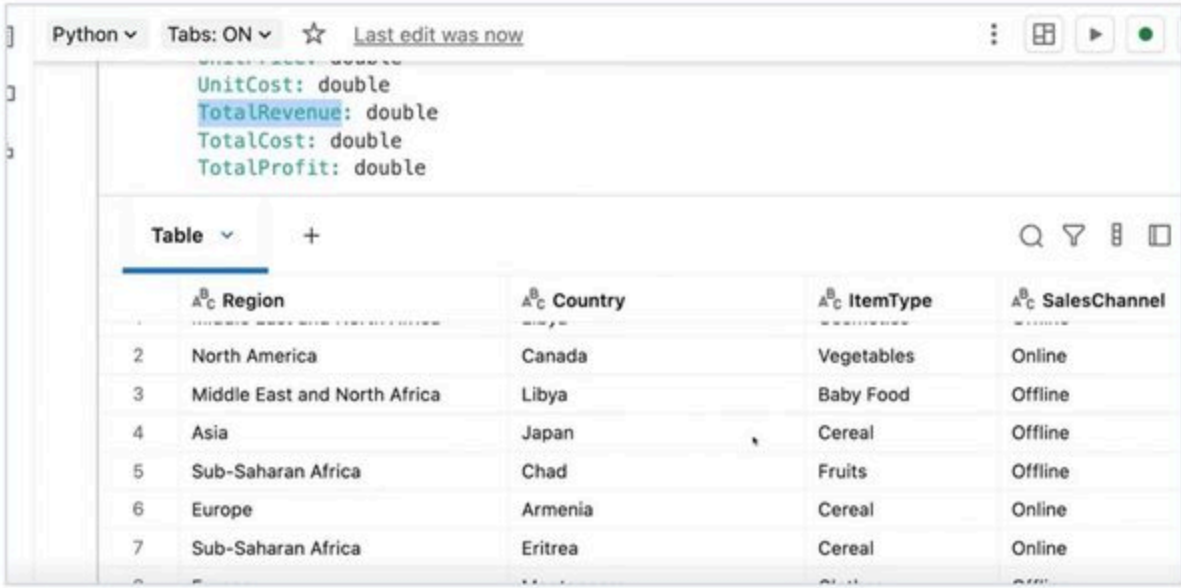


- Add inferSchema=True
- OrderID, UnitsSold → integer
- UnitPrice, TotalRevenue … → double
- Spark reads the data once more to guess types

⚠ Production tip: define the schema yourself (StructType or a DDL string) – faster and safer than inferSchema.

LAB 2

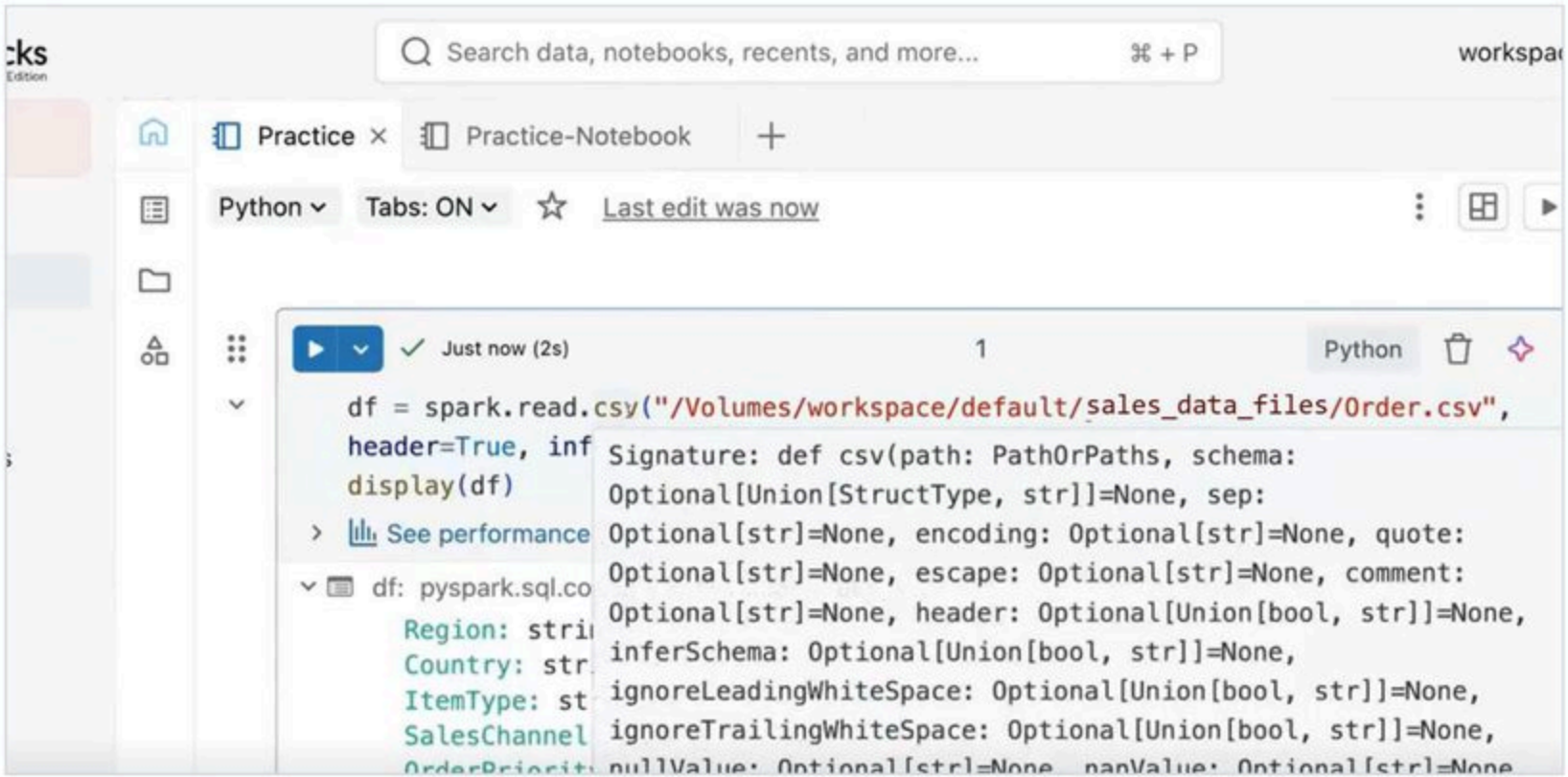
Result table and type icons





- display() = interactive table
- Icons show the type: ABC string, 123 int, 1.2 double
- Hover a column icon to see the type
- Sort, filter, add visualisations with +

Code help: function signature



- Hover a function or press Tab
- Shows all parameters of csv()
- sep, encoding, quote, escape, header, inferSchema ...

spark.read.csv – options you will use most

Option	What it does
header=True	First line contains the column names
inferSchema=True	Detect integer / double / date types
sep=";"	Use another separator (default is a comma)
schema="OrderID INT, Region STRING ..."	Give the types yourself (fast, reliable)
nullValue="NA"	Treat a text value as NULL
spark.read.format("csv").option(...).load(path)	Same thing, longer syntax

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

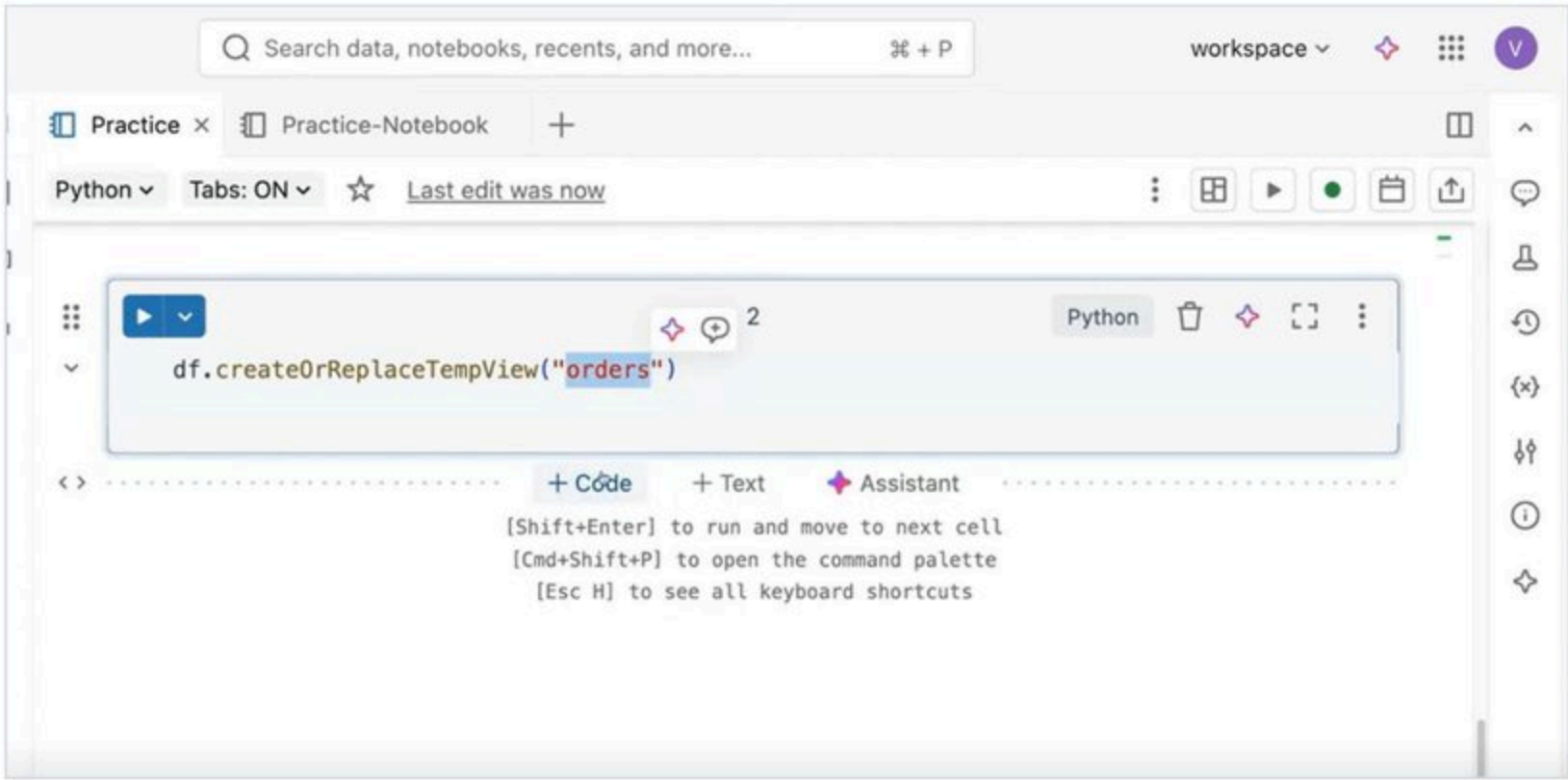
LAB 3

Spark SQL and the AI assistant

Temp view • %sql • _sqldf • Generate with AI • Diagnose error

LAB 3

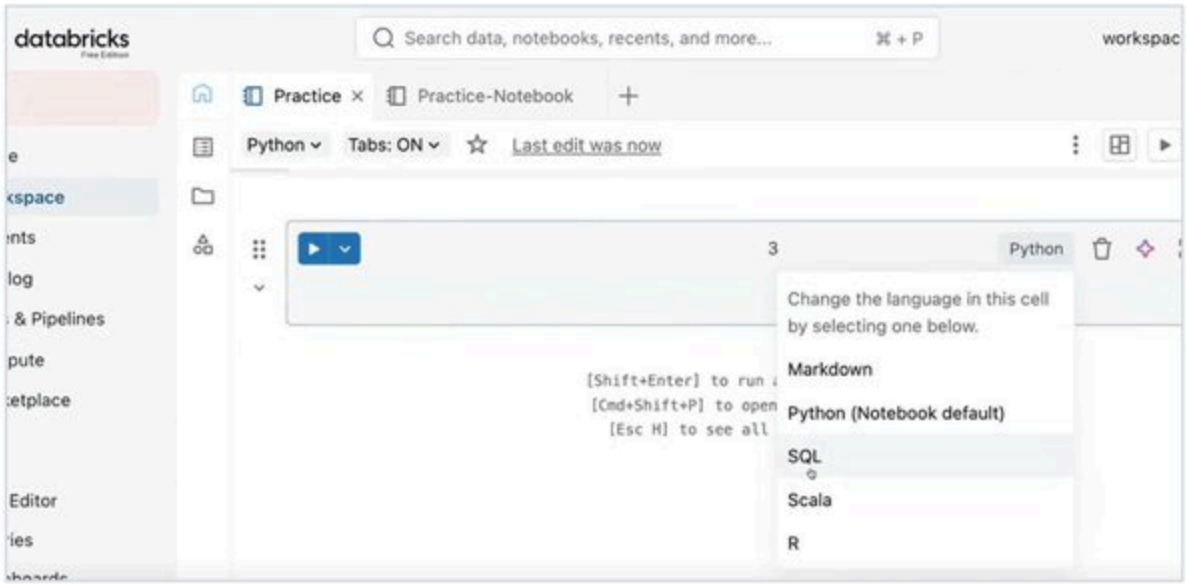
Register a temporary view

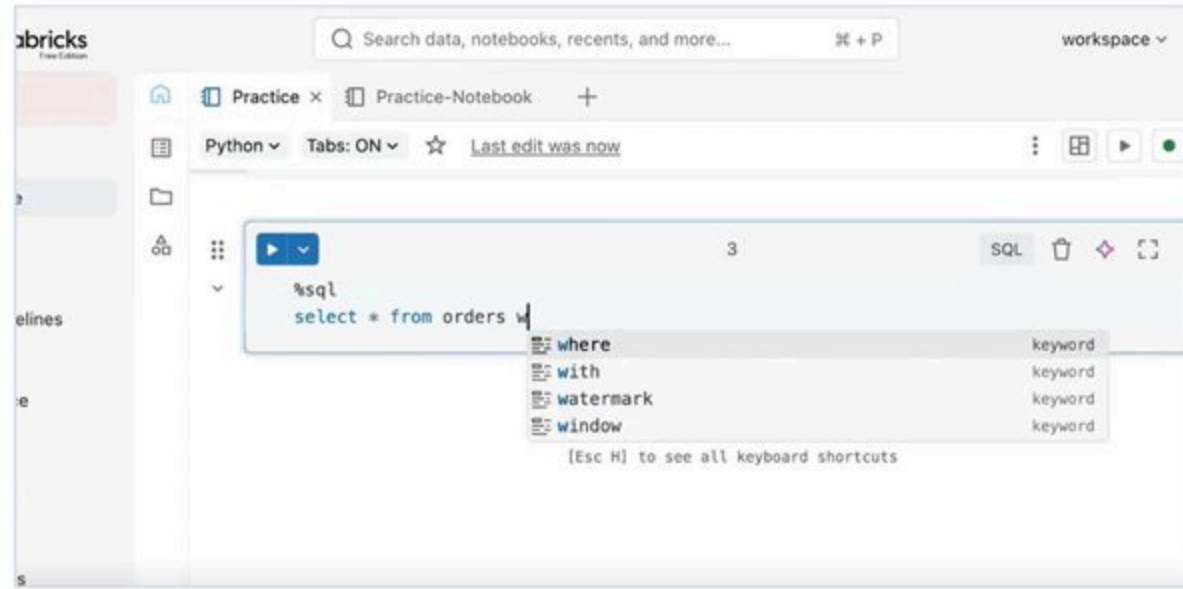


- df.createOrReplaceTempView("orders")
- Gives the DataFrame a SQL name
- Lives only in this notebook session

LAB 3

Change a cell to SQL with %sql

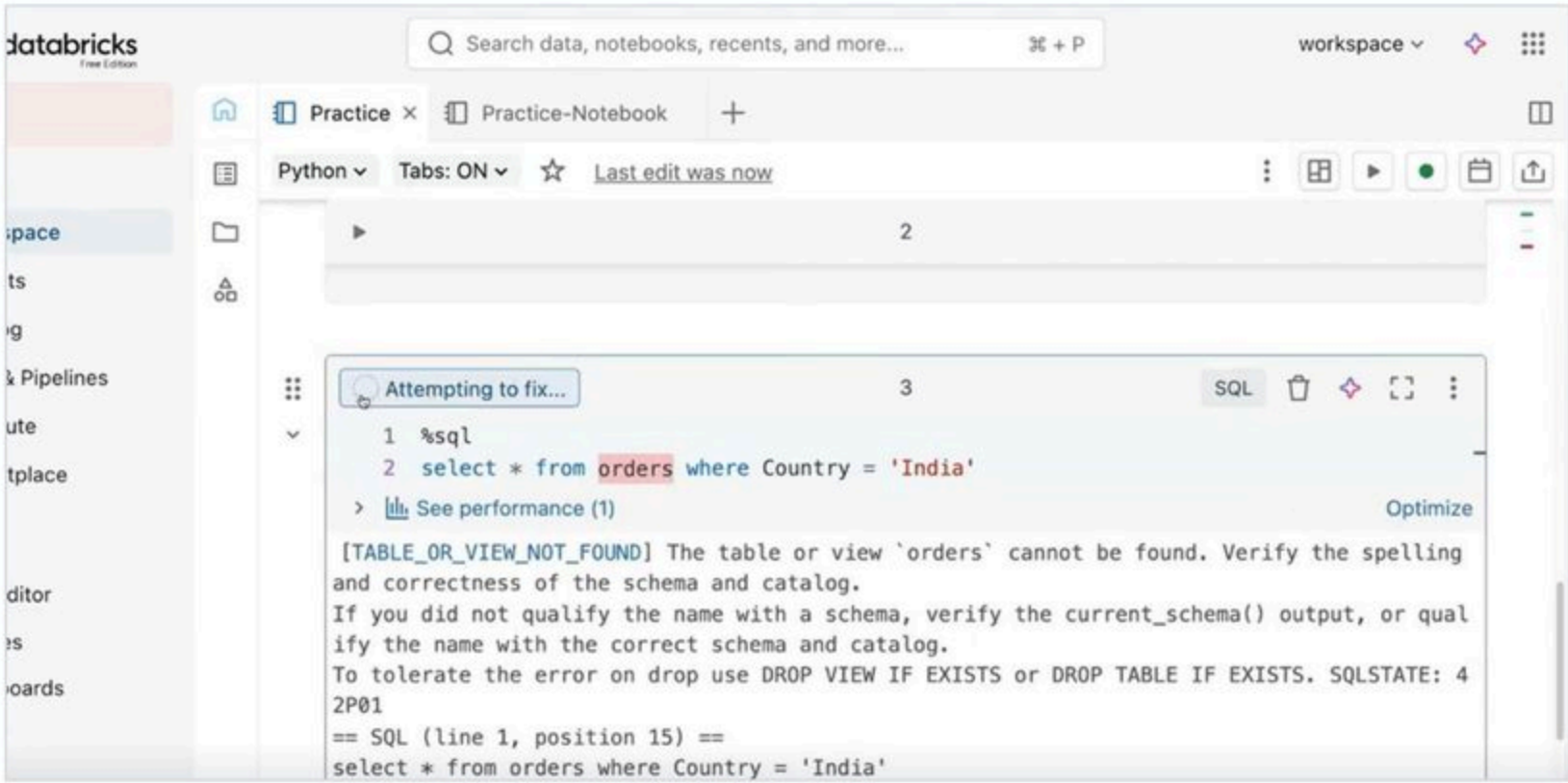




- Cell language menu: Markdown, Python, SQL, Scala, R
- Or type %sql on the first line
- Auto-complete helps with keywords

LAB 3

Error: TABLE_OR_VIEW_NOT_FOUND

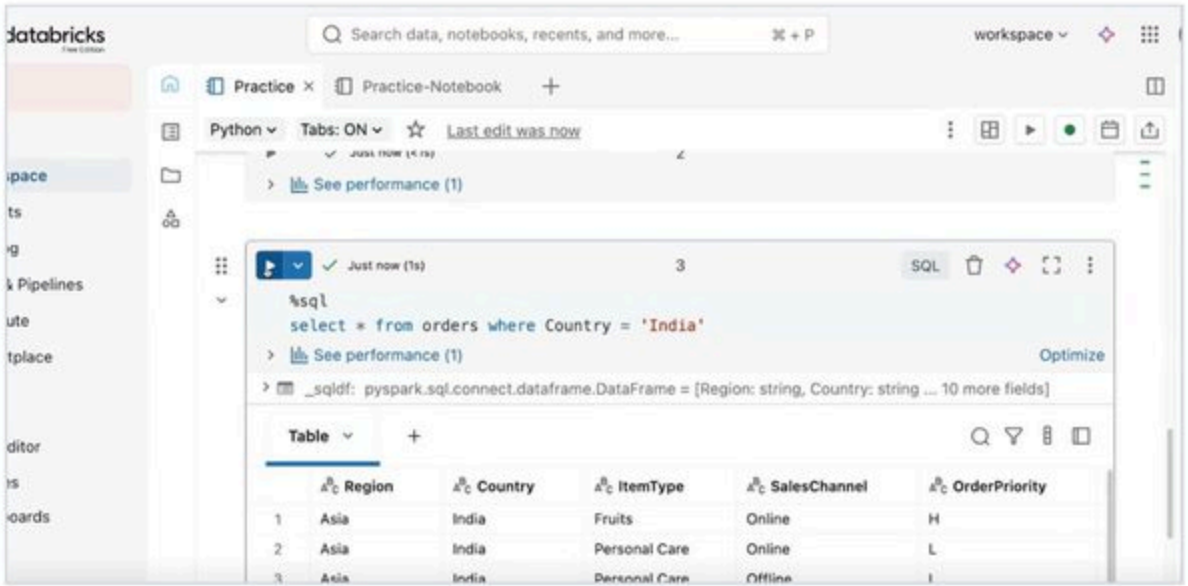


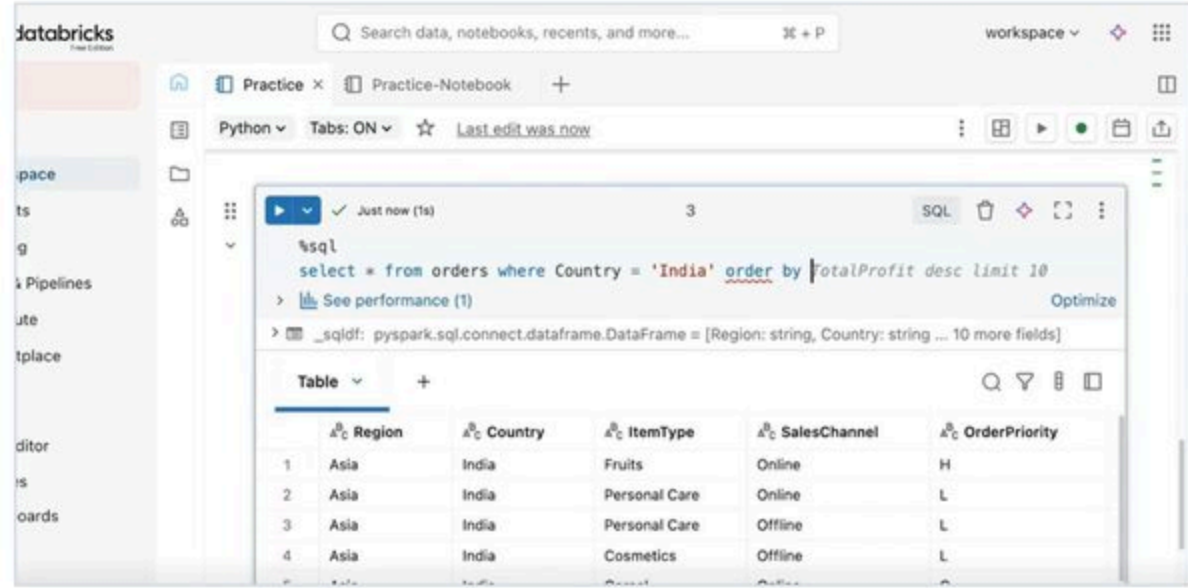
- The view "orders" cannot be found
- Cause: the createOrReplaceTempView cell was not run
- Fix: run cells in order (cell 2, then cell 3)

⚠ Before you debug the code, check: did I run the cells above?

LAB 3

SQL result – orders from India



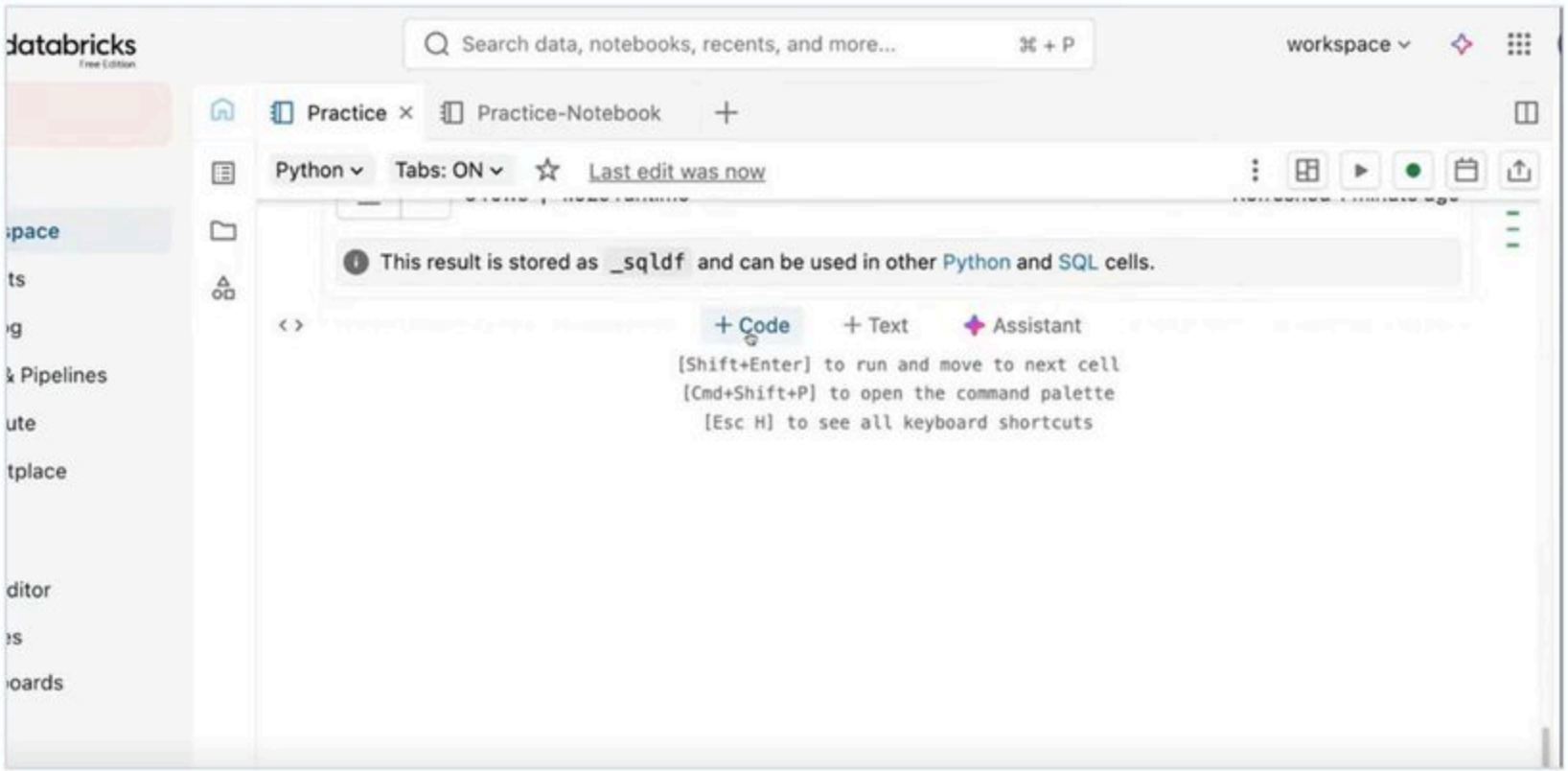


- where Country = 'India'
- order by TotalProfit desc limit 10

- Text values in single quotes
- Result is also a DataFrame: _sqldf

LAB 3

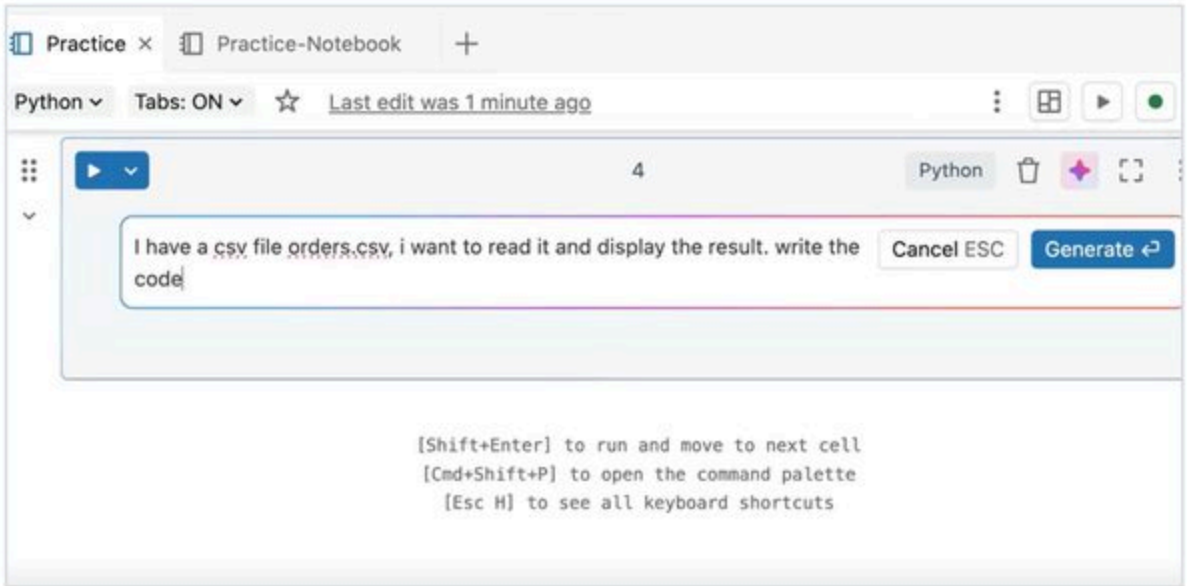
Python and SQL work together: _sqldf

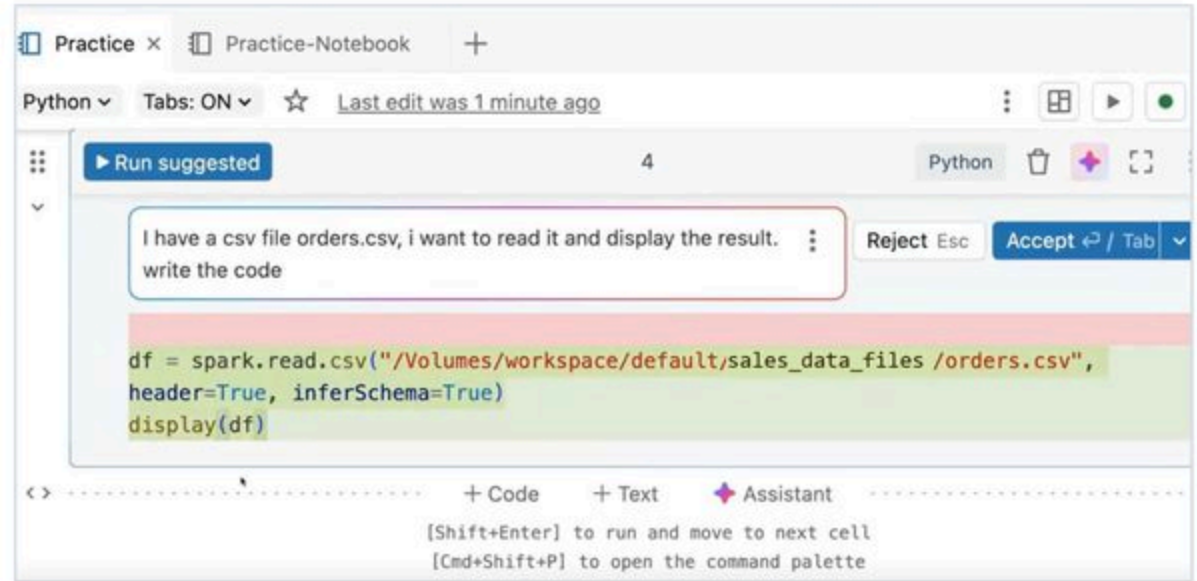


- SQL result stored as _sqldf
- Use it in the next Python cell
- Mix languages in one notebook

LAB 3

Generate code with the AI assistant

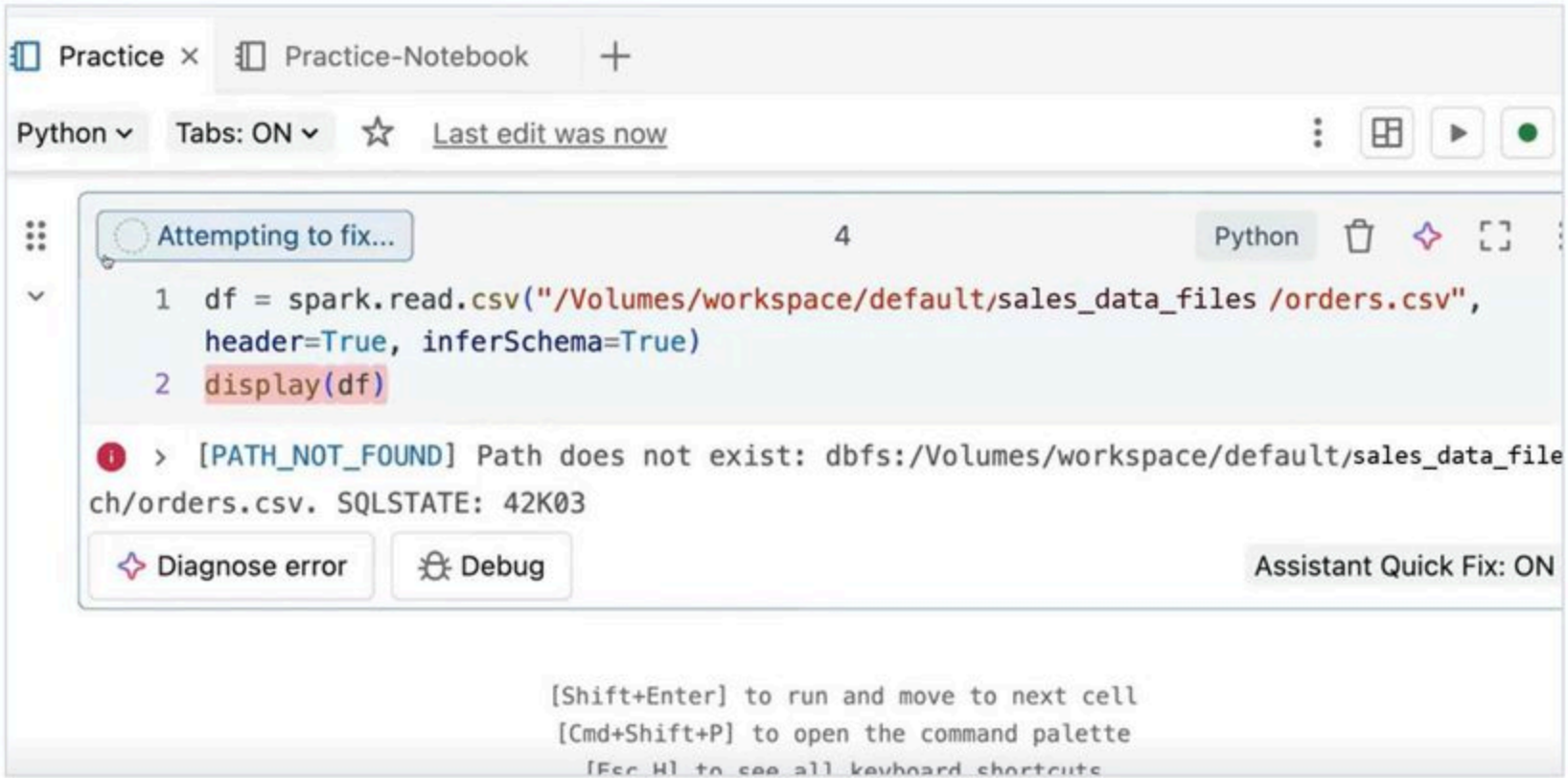




- In an empty cell: Cmd/Ctrl + I
- Prompt: "I have a csv file orders.csv, read it and display it"
- Review the code → Accept or Reject
- Run suggested

LAB 3

AI code fails: PATH_NOT_FOUND

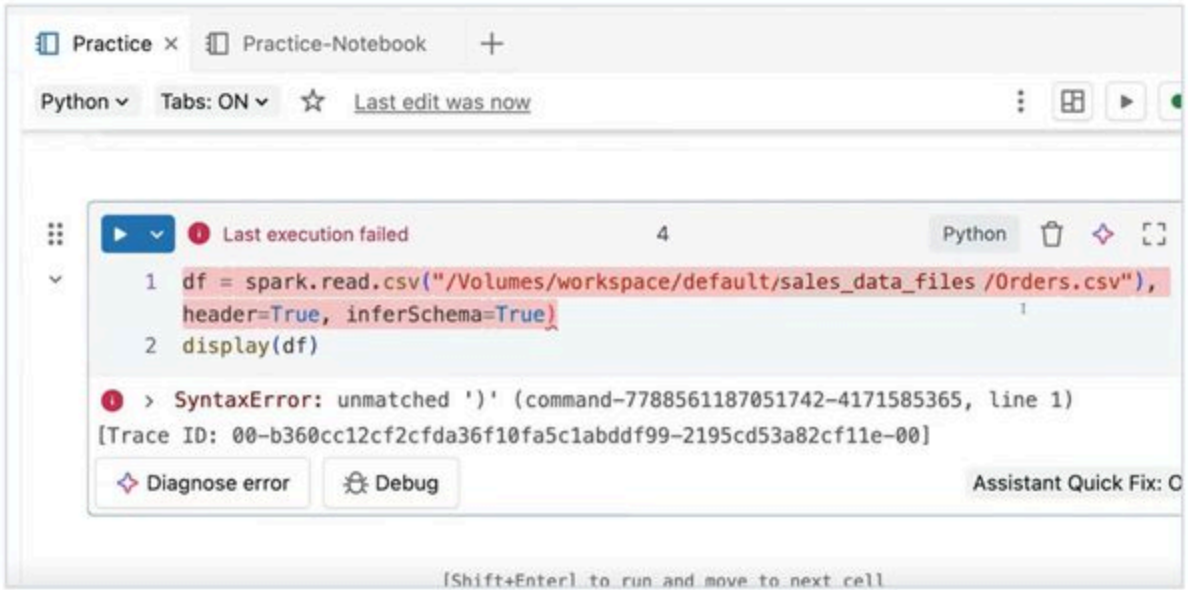


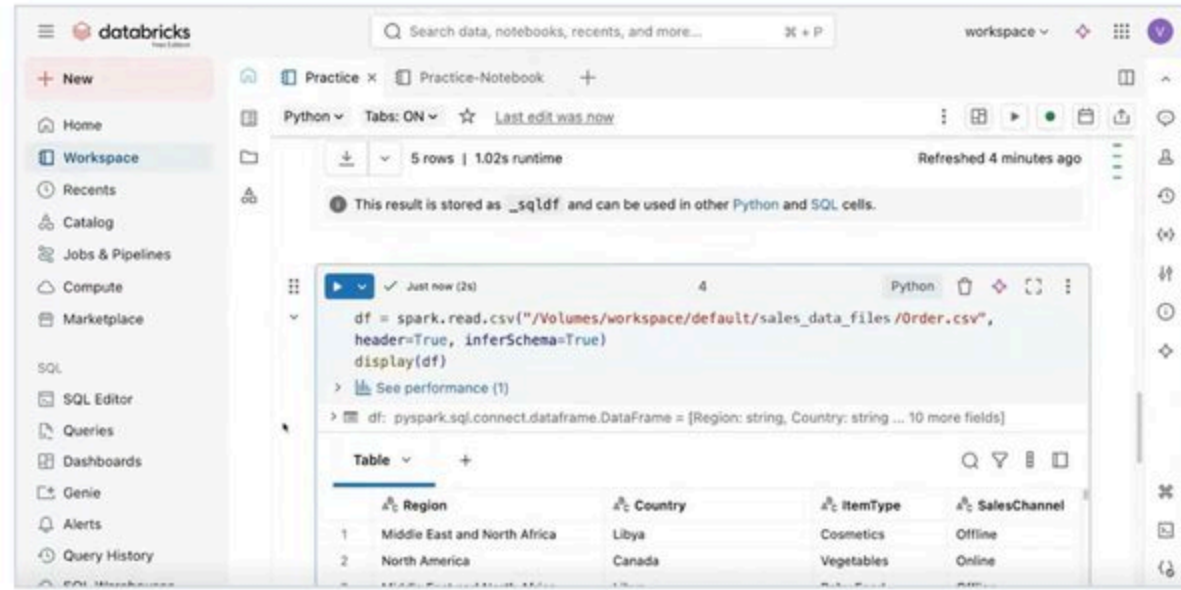
- The prompt said orders.csv
- The real file is Order.csv
- Paths are case-sensitive and exact
- Fix the file name, run again

⚠ PATH_NOT_FOUND = check spelling, upper/lower case and the volume name in the path.

LAB 3

Typical errors while editing





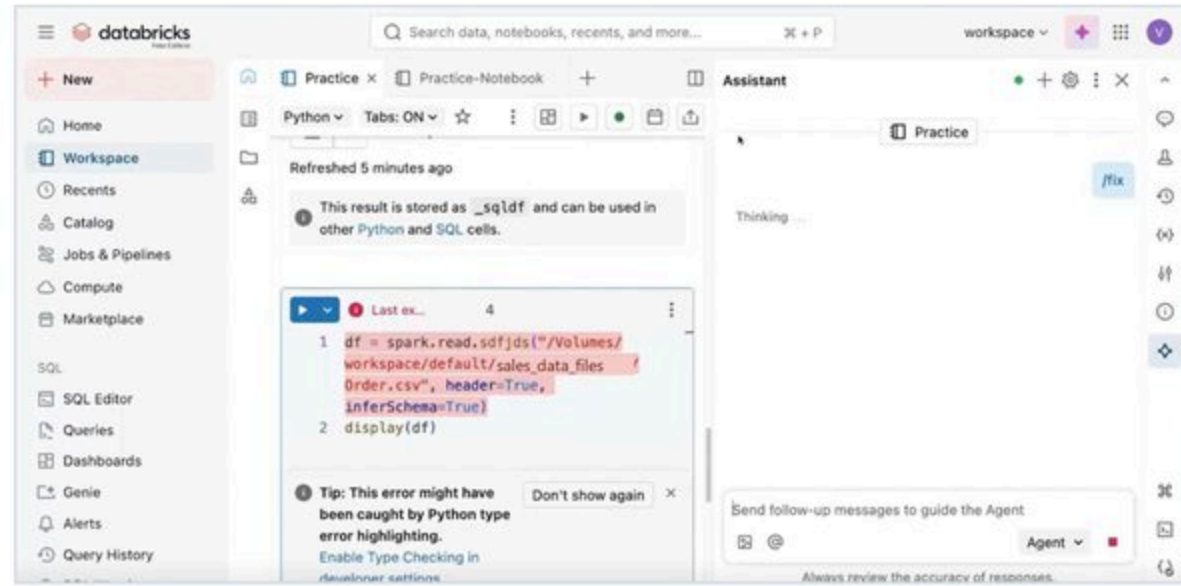
- SyntaxError: unmatched ')' – an extra bracket
- Fix brackets and quotes, run again

- Working version reads 1,000 rows

LAB 3

Diagnose error with the Assistant



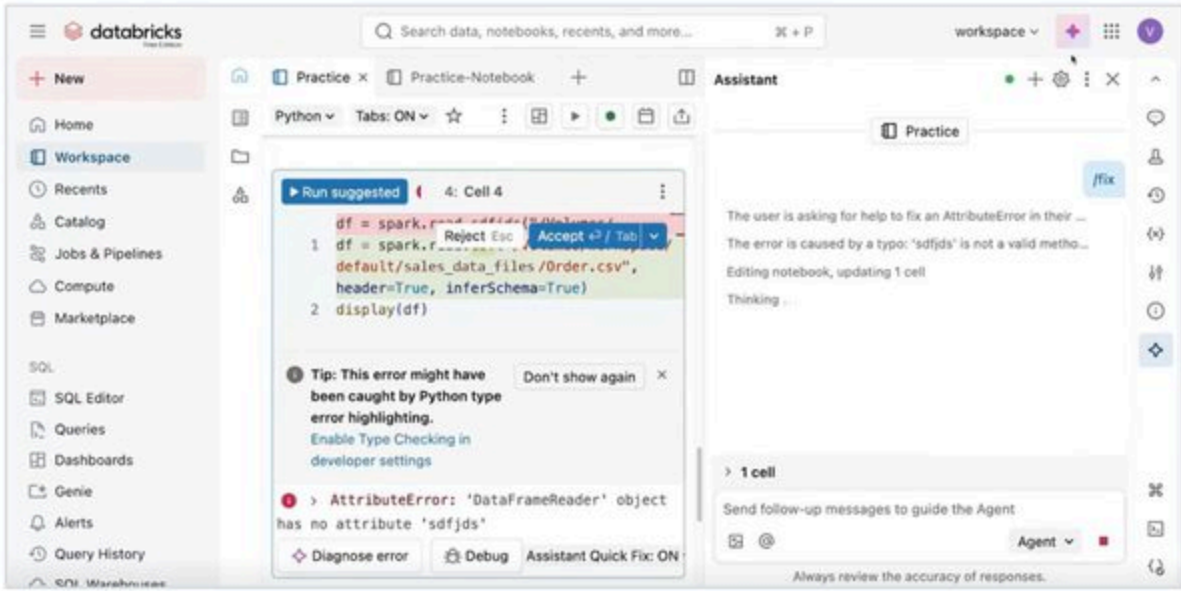


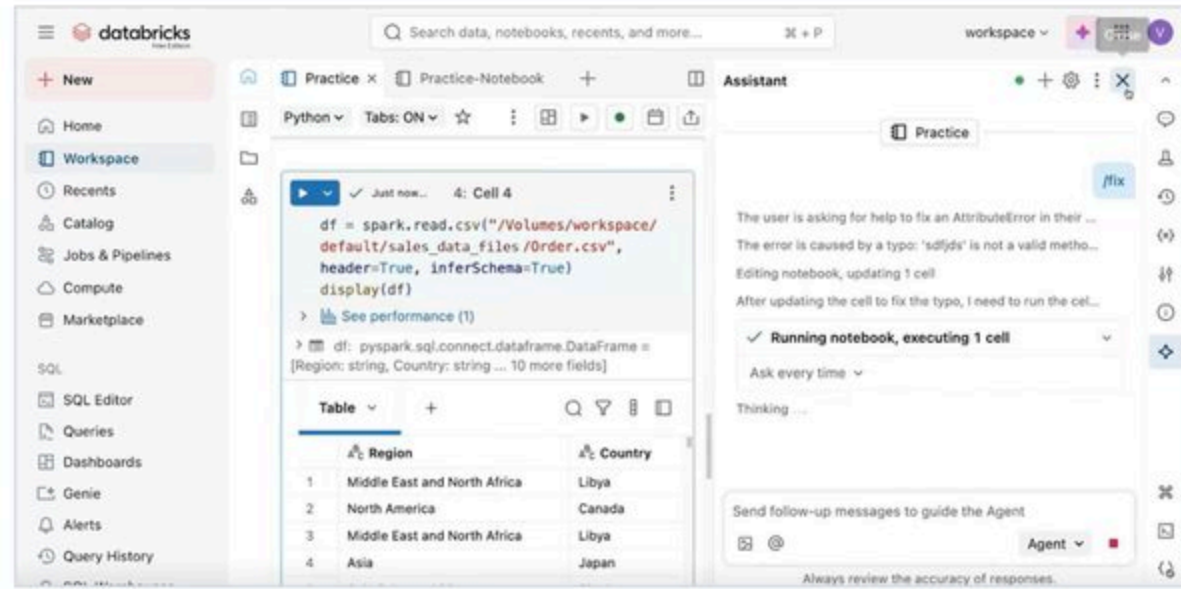
- spark.read.sdfjds(...) → AttributeError
- 'DataFrameReader' has no attribute 'sdfjds'

- Click Diagnose error
- Assistant explains and proposes a fix (/fix)

LAB 3

Accept the fix and run





- Suggested change shown in the cell
- Accept → the Assistant runs the cell
- Result appears again
- Always review AI changes

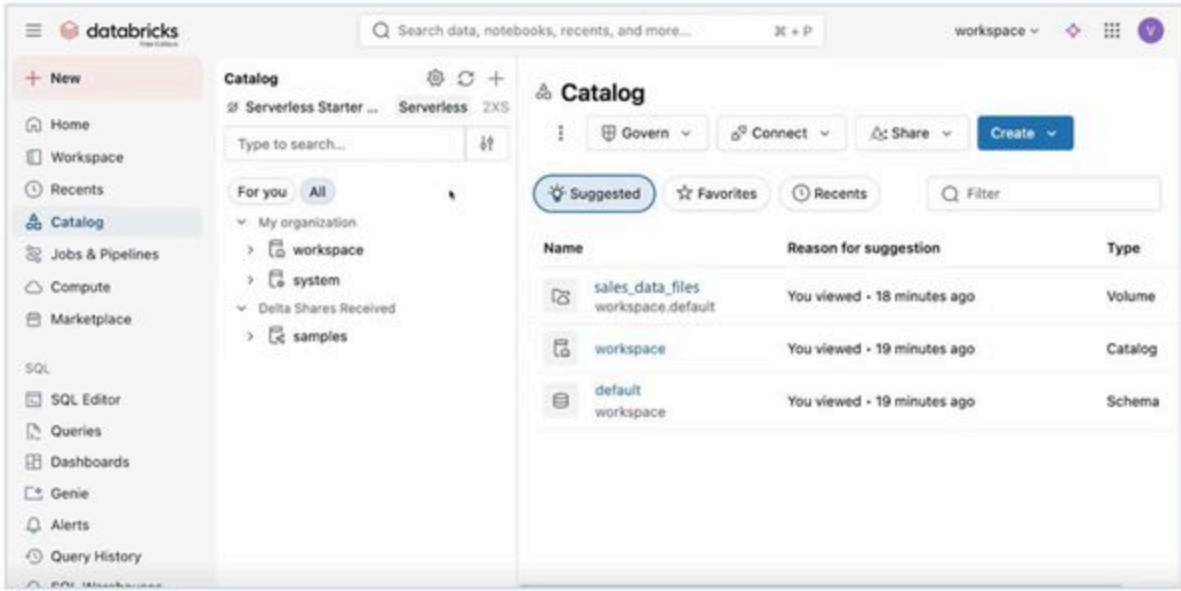
LAB 4

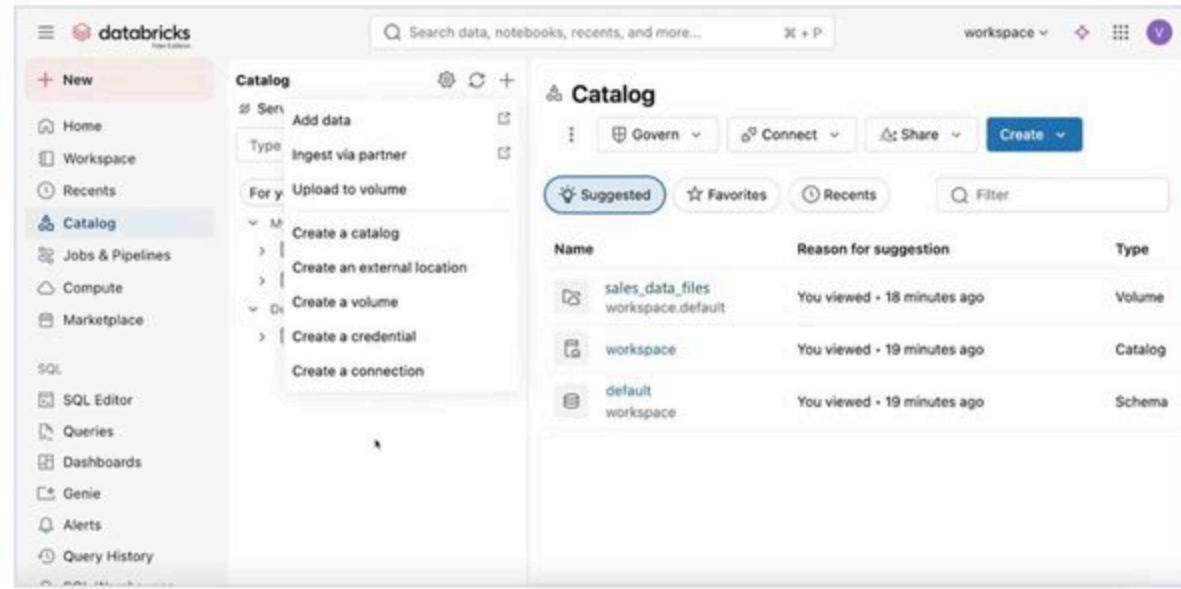
Create a table in Databricks

Permissions • SQL editor • saveAsTable • Catalog Explorer

LAB 4

Catalog Explorer and the Create menu



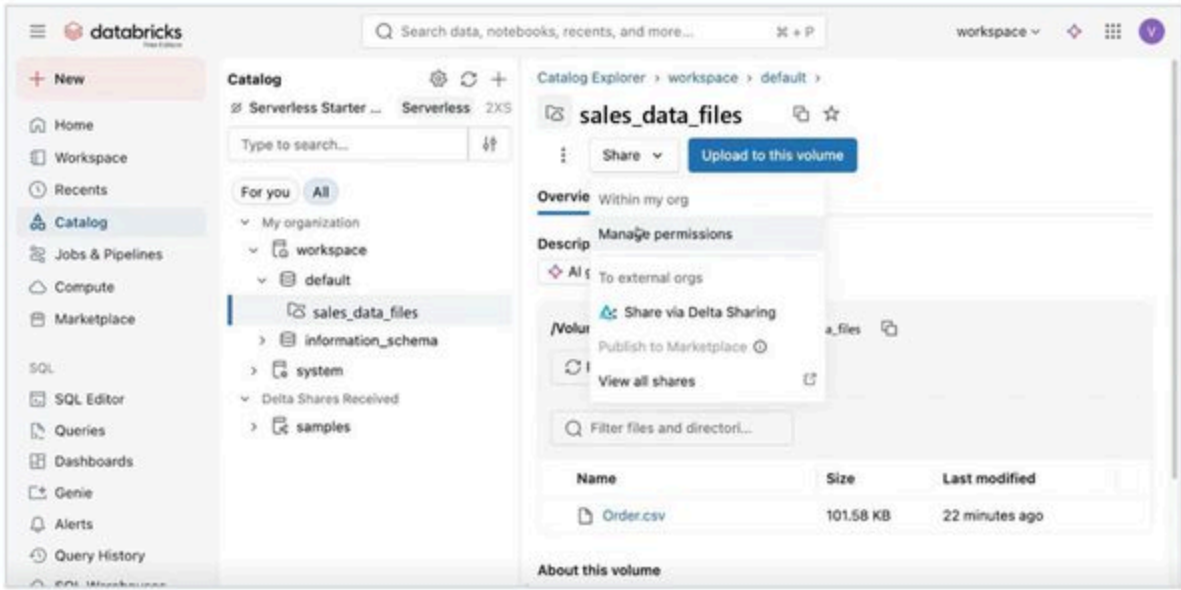


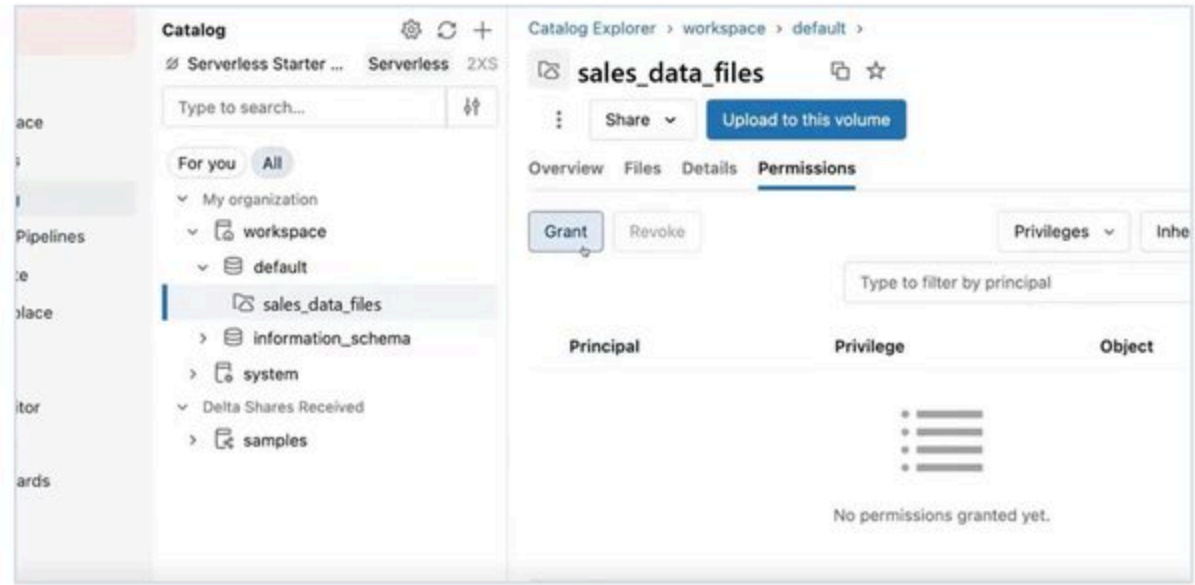
- Suggested: recently used objects
- Create: catalog, volume, external location, credential, connection

- Our volume sales_data_files is listed

LAB 4

Manage permissions on a volume

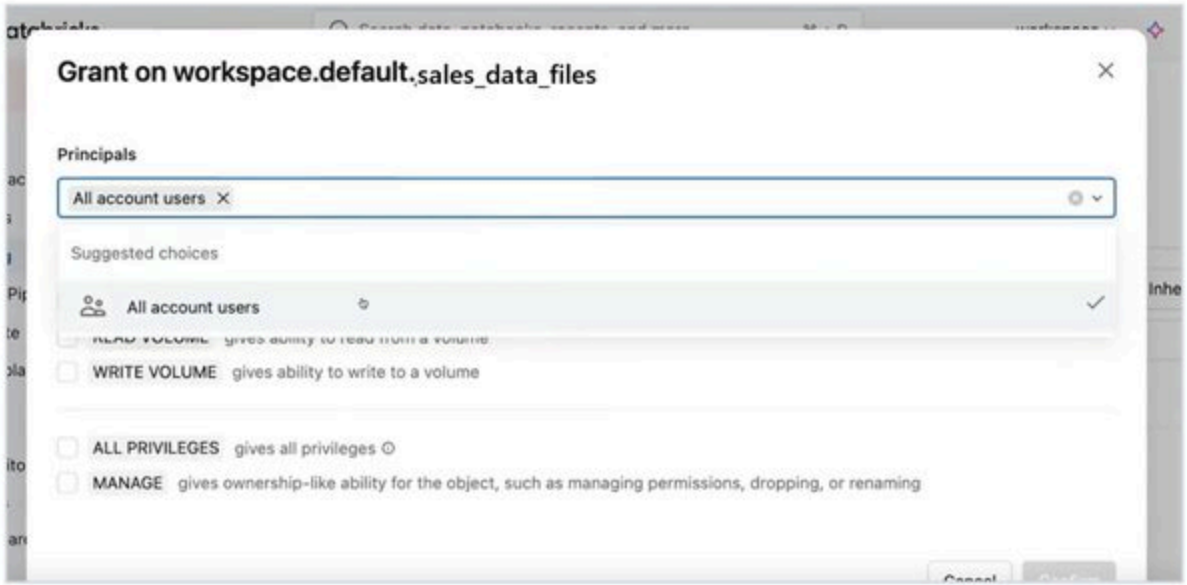


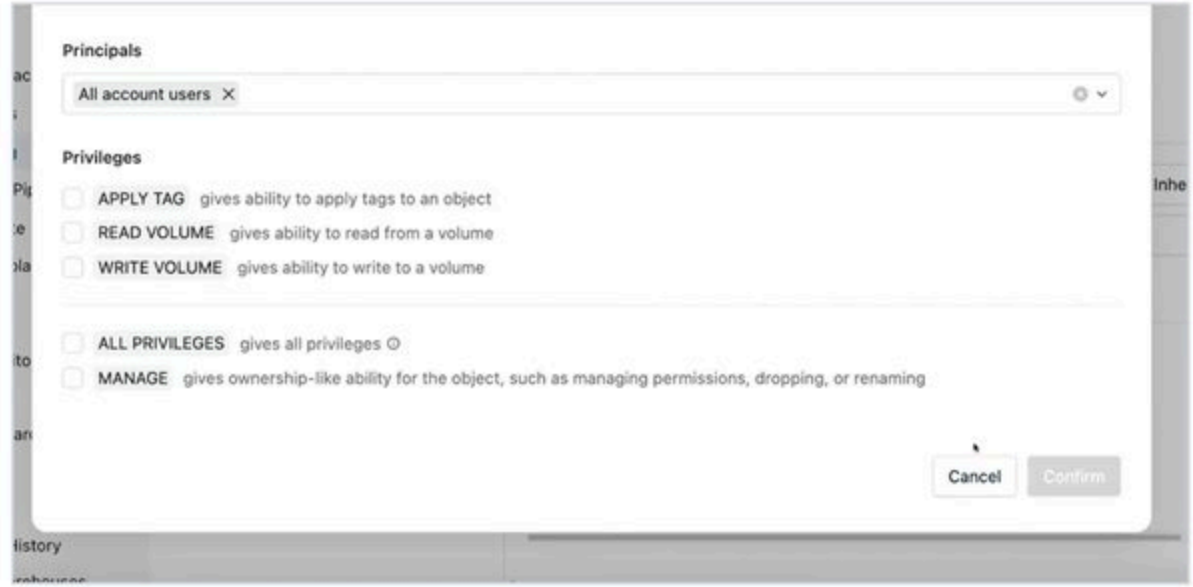


- Share → Manage permissions
- Permissions tab → Grant
- No permissions granted yet = only the owner

LAB 4

Grant privileges

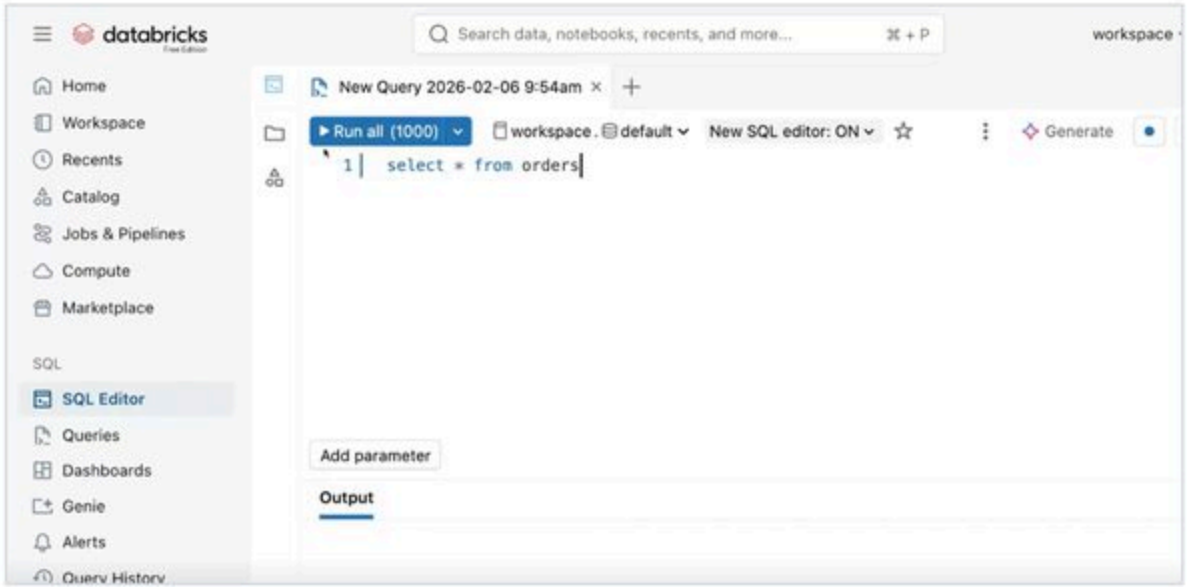


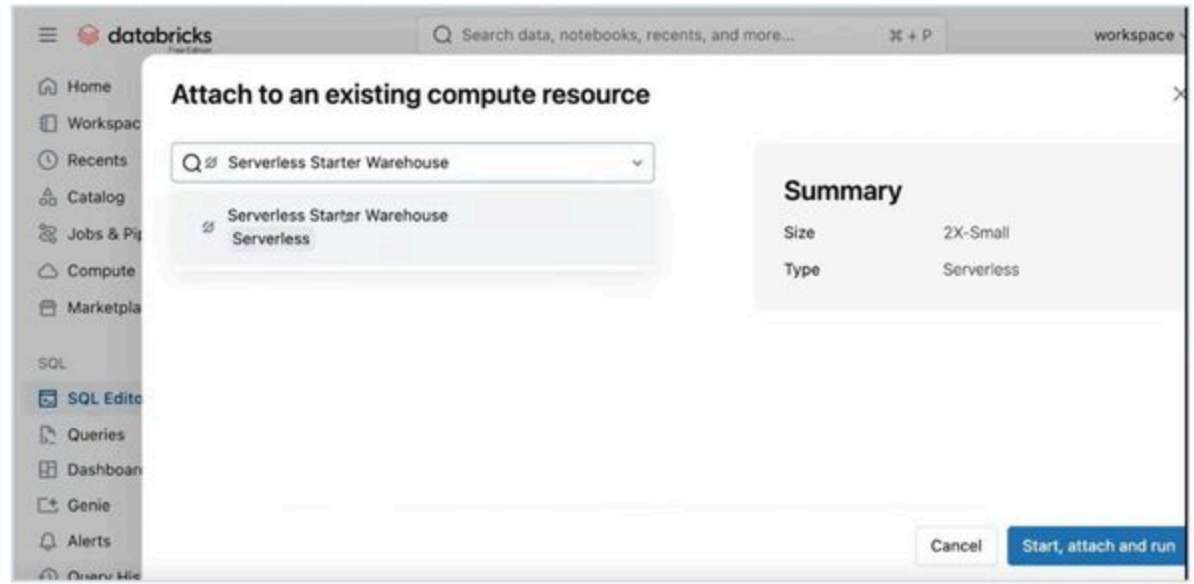


- Principal: a user, group or All account users
- READ VOLUME / WRITE VOLUME
- ALL PRIVILEGES, MANAGE
- Grant the minimum needed

LAB 4

SQL editor: query the temp view?



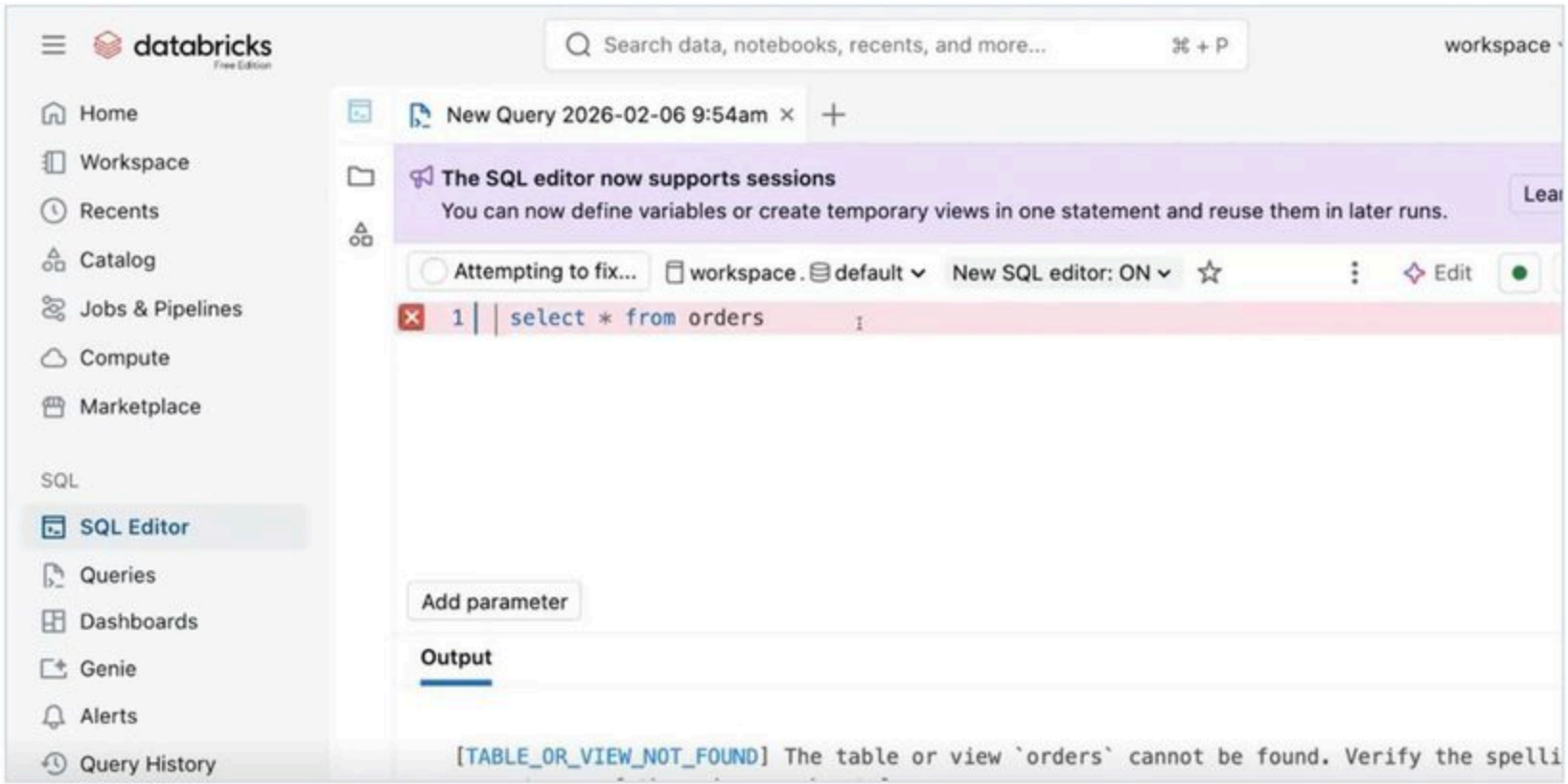


- SQL Editor → New query
- select * from orders

- Attach the Serverless Starter Warehouse
- Start, attach and run

LAB 4

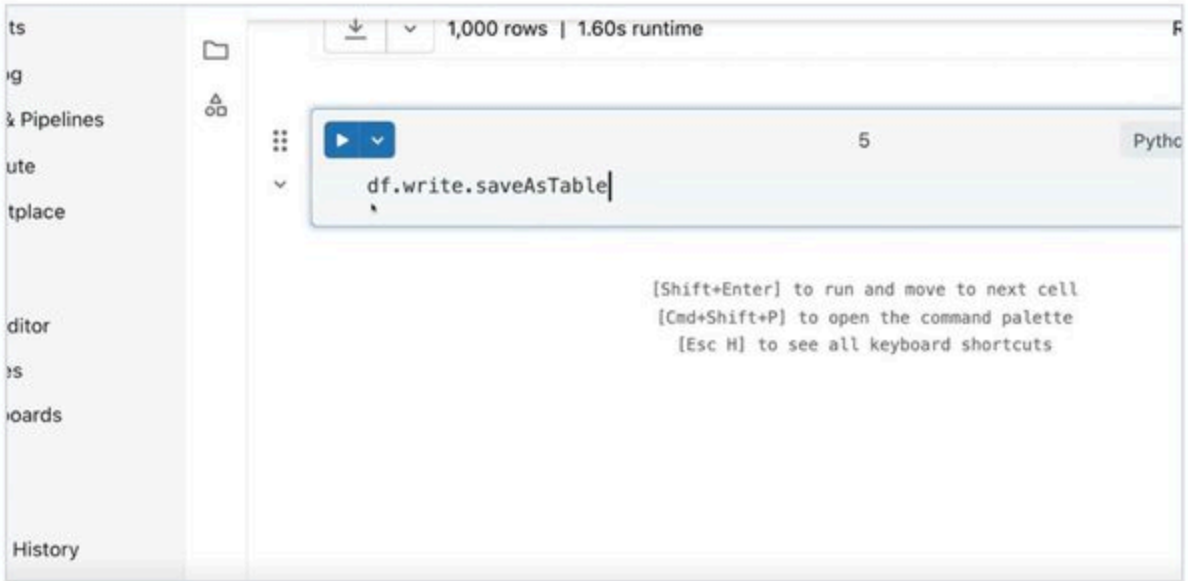
TABLE_OR_VIEW_NOT_FOUND in the SQL editor

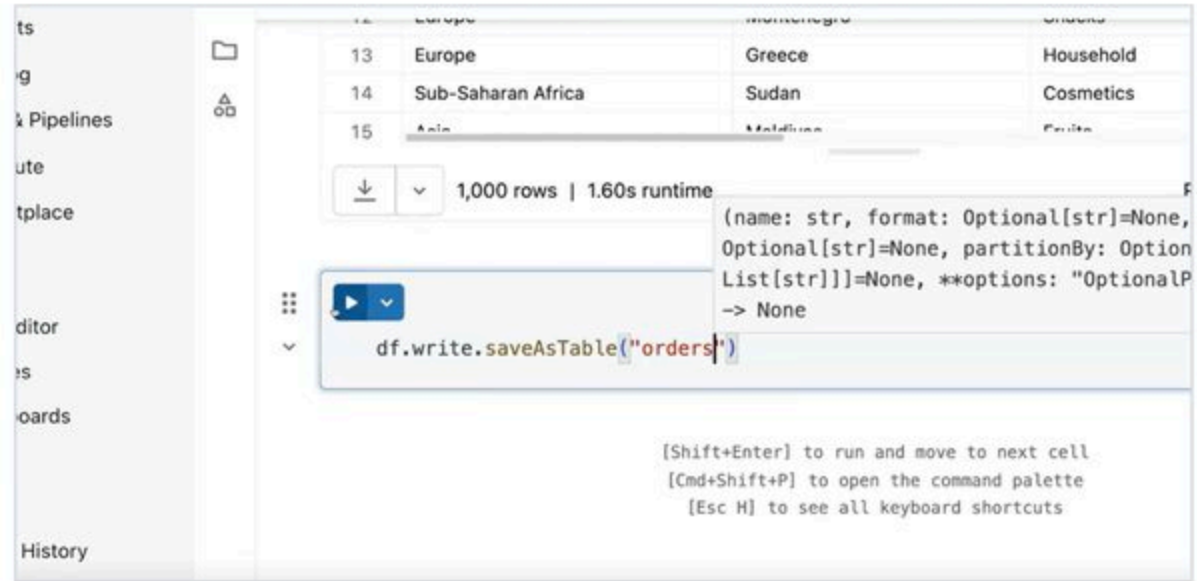


- orders works in the notebook, not here
- A temp view lives only in its notebook session
- Other tools need a real table

LAB 4

Save the DataFrame as a table



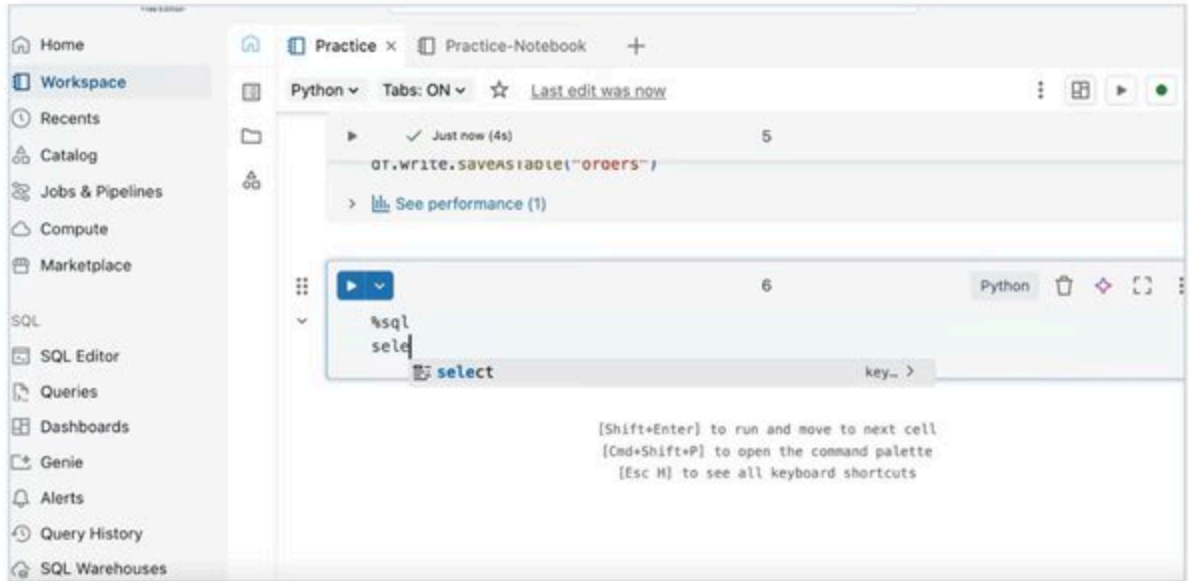


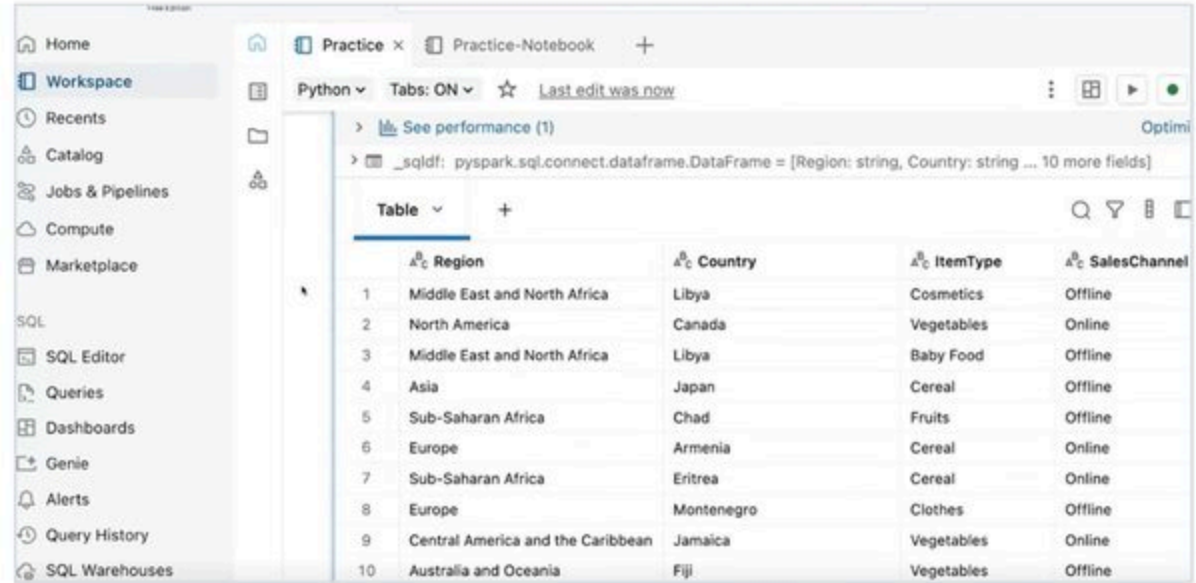
- df.write.saveAsTable("orders")
- Creates workspace.default.orders
- Managed Delta table
- Optional: format, mode, partitionBy

⚠ Run saveAsTable twice and it fails ("table already exists") – use df.write.mode("overwrite").saveAsTable("orders").

LAB 4

Query the new table with %sql



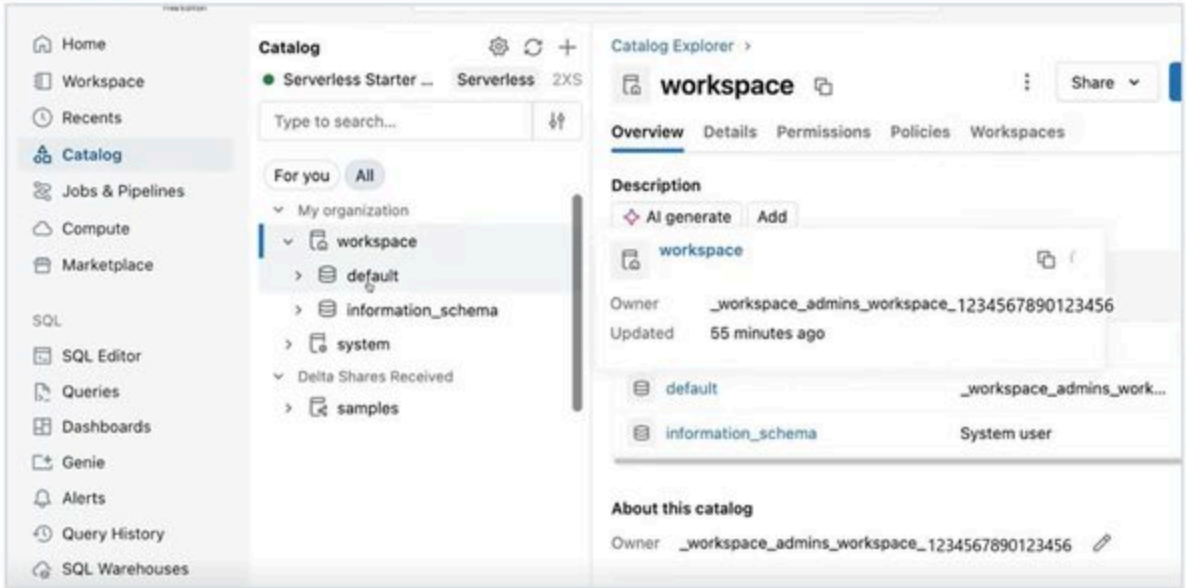


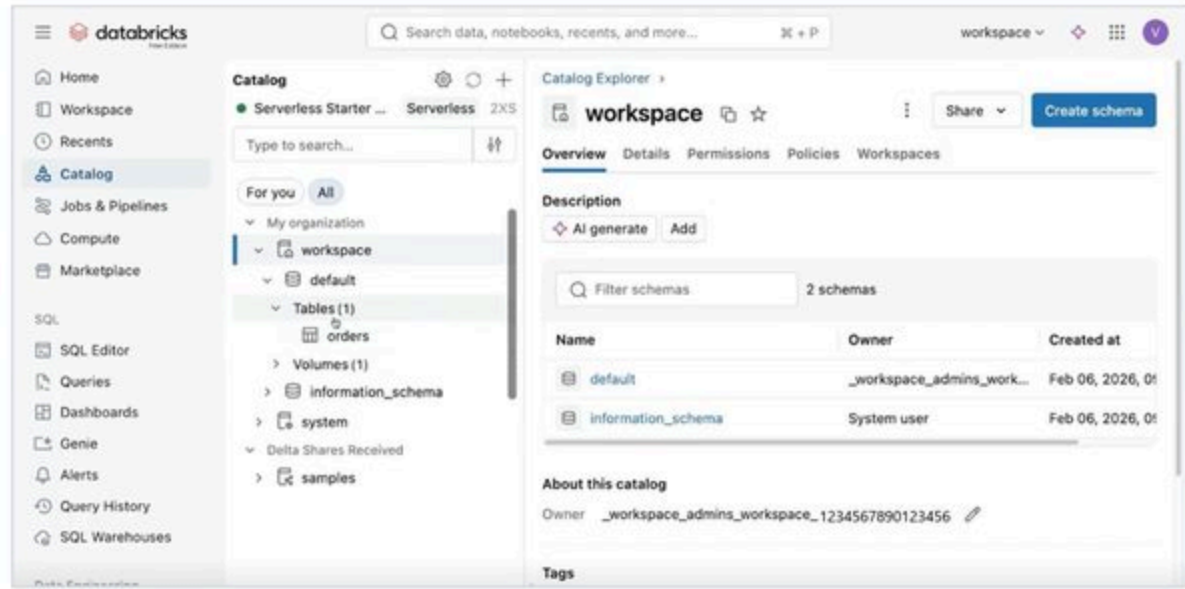
	Region	Country	ItemType	SalesChannel
1	Middle East and North Africa	Libya	Cosmetics	Offline
2	North America	Canada	Vegetables	Online
3	Middle East and North Africa	Libya	Baby Food	Offline
4	Asia	Japan	Cereal	Offline
5	Sub-Saharan Africa	Chad	Fruits	Offline
6	Europe	Armenia	Cereal	Online
7	Sub-Saharan Africa	Eritrea	Cereal	Online
8	Europe	Montenegro	Clothes	Offline
9	Central America and the Caribbean	Jamaica	Vegetables	Online
10	Australia and Oceania	Fiji	Vegetables	Offline

- %sql select * from orders
- Now reads the TABLE (not the view)
- 1,000 rows

LAB 4

The table in Catalog Explorer

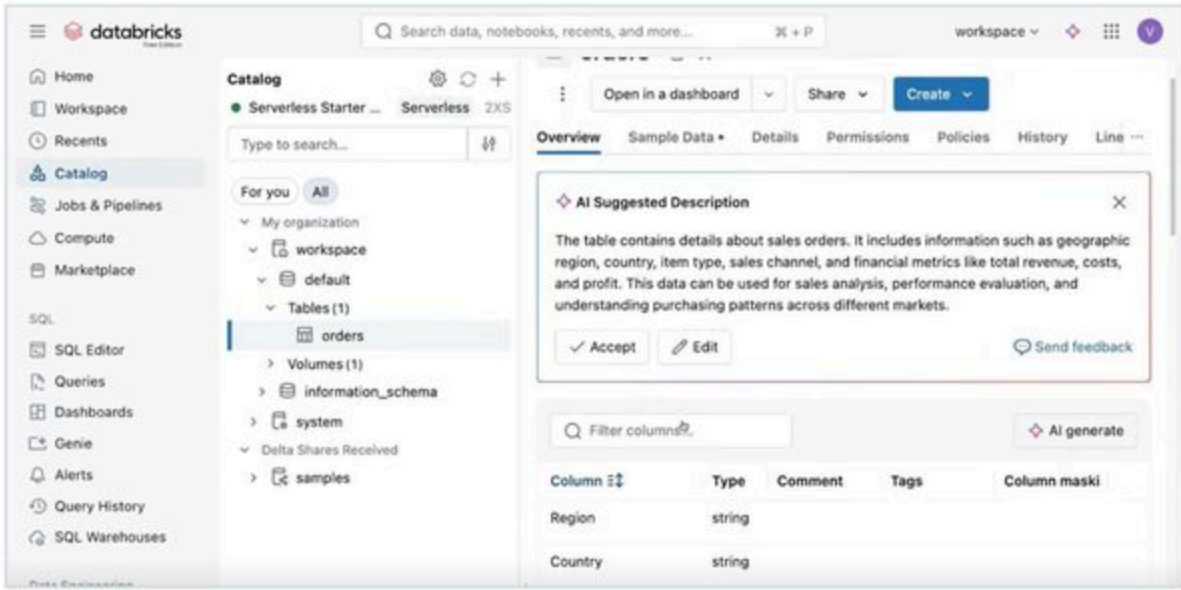


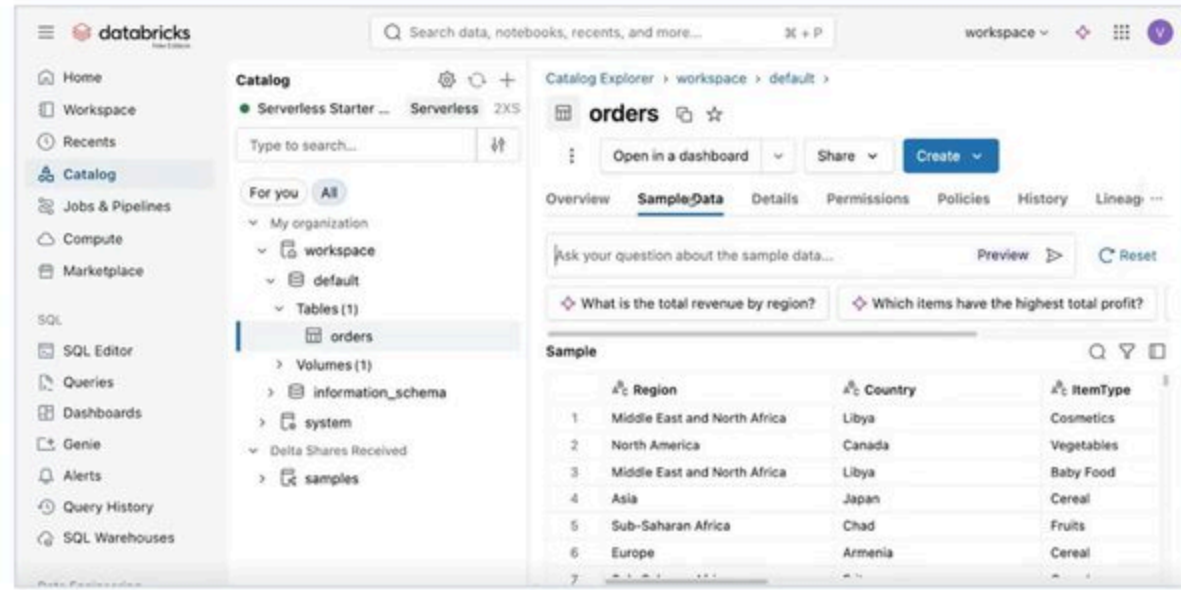


- workspace → default → Tables (1) → orders
- Owner, created date
- Volumes (1) next to Tables (1)

LAB 4

Table overview, AI description and sample data

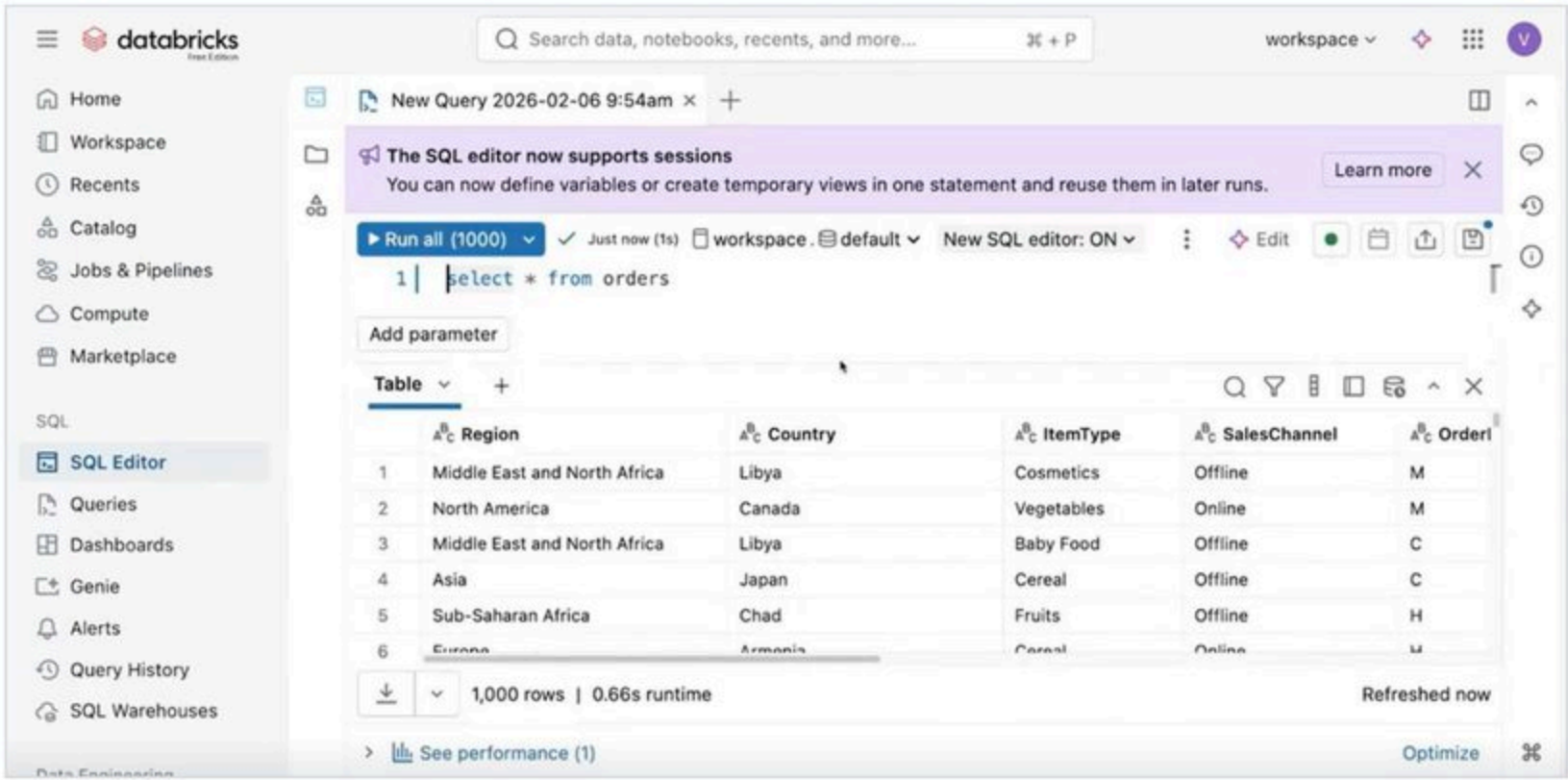




- AI Suggested Description → Accept or Edit
- Columns and types
- Sample Data tab
- Ask questions about the sample data

LAB 4

SQL editor – now it works



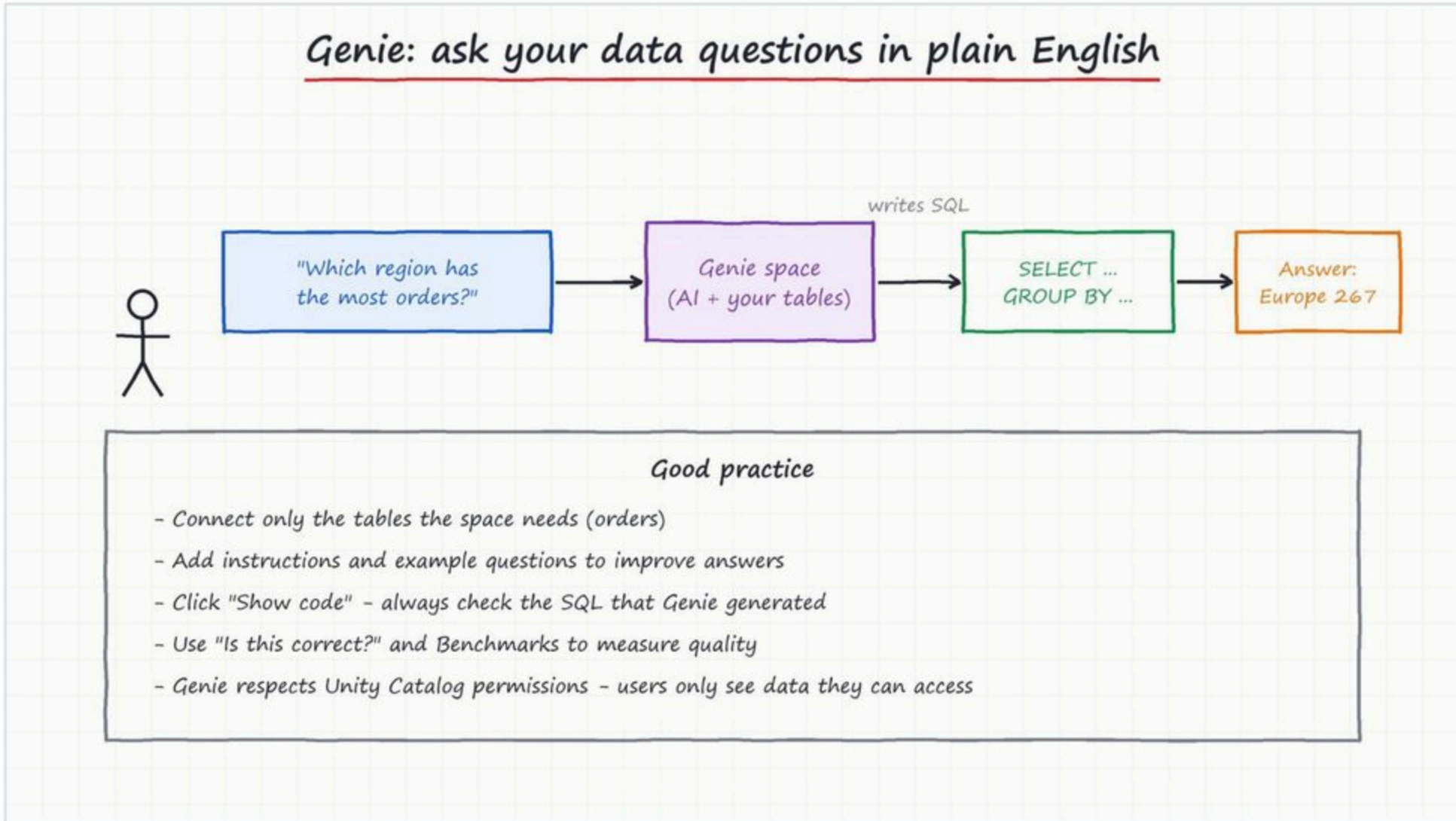
- select * from orders
- 1,000 rows in 0.66 s
- Same table for notebooks, SQL, dashboards, Genie

LAB 5

Genie in Databricks

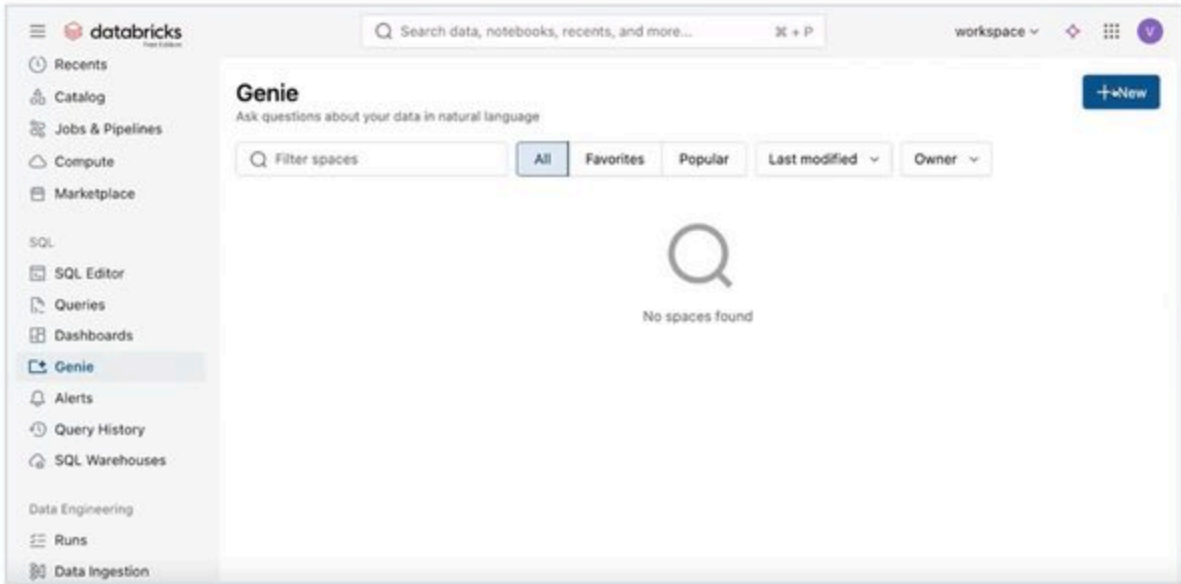
Ask your data questions in natural language

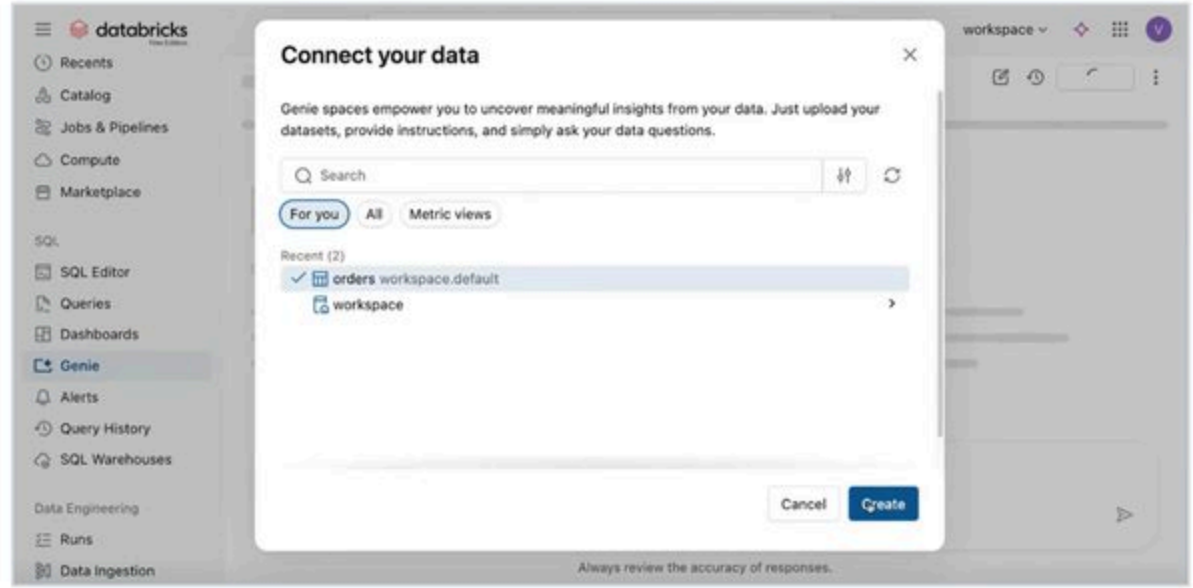
What is Genie?



LAB 5

Create a Genie space

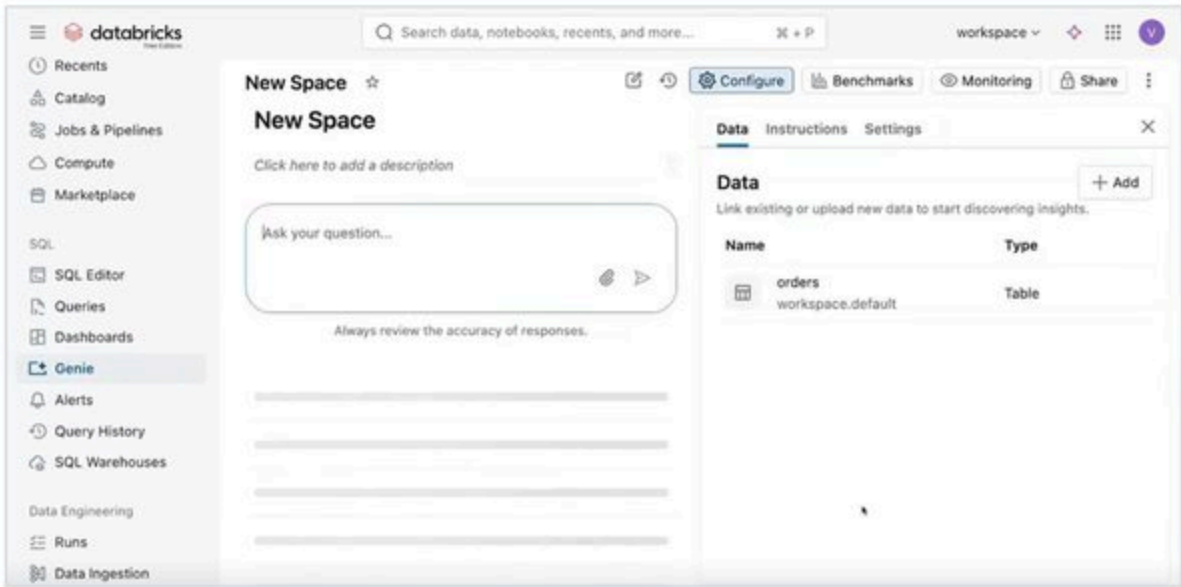


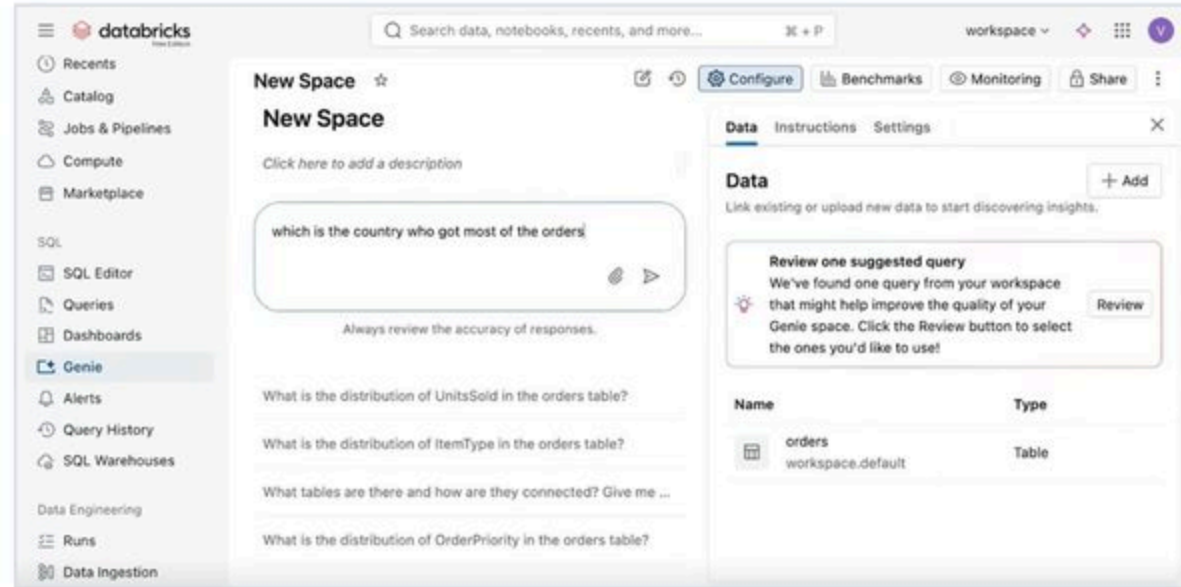


- Genie → + New
- Connect your data: choose orders
- Create

LAB 5

The Genie space



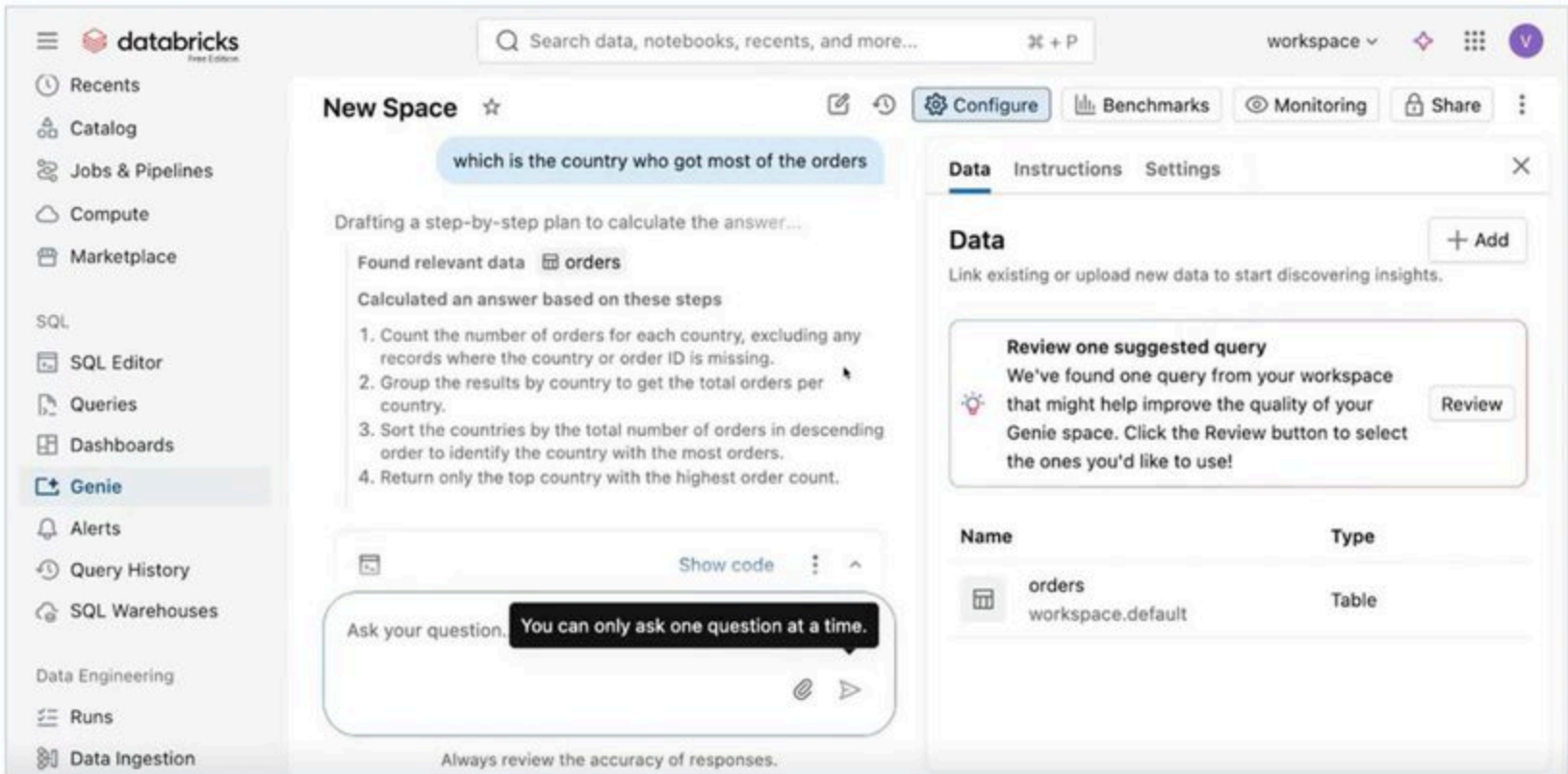


- Data: orders (Table)
- Suggested questions

- Configure: instructions, settings
- Type your question

LAB 5

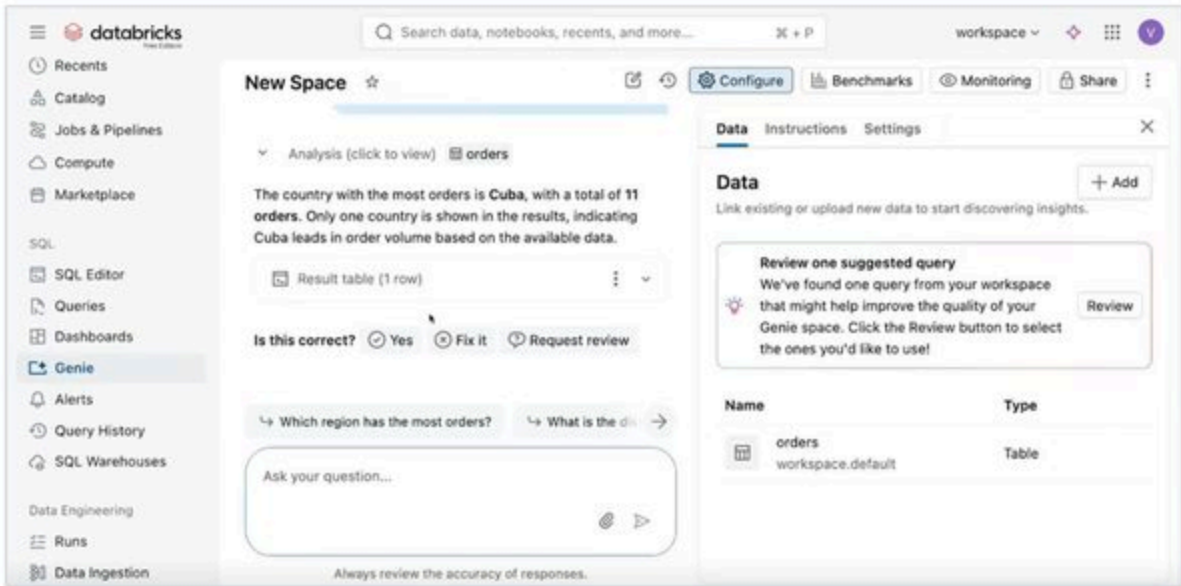
Genie thinks step by step

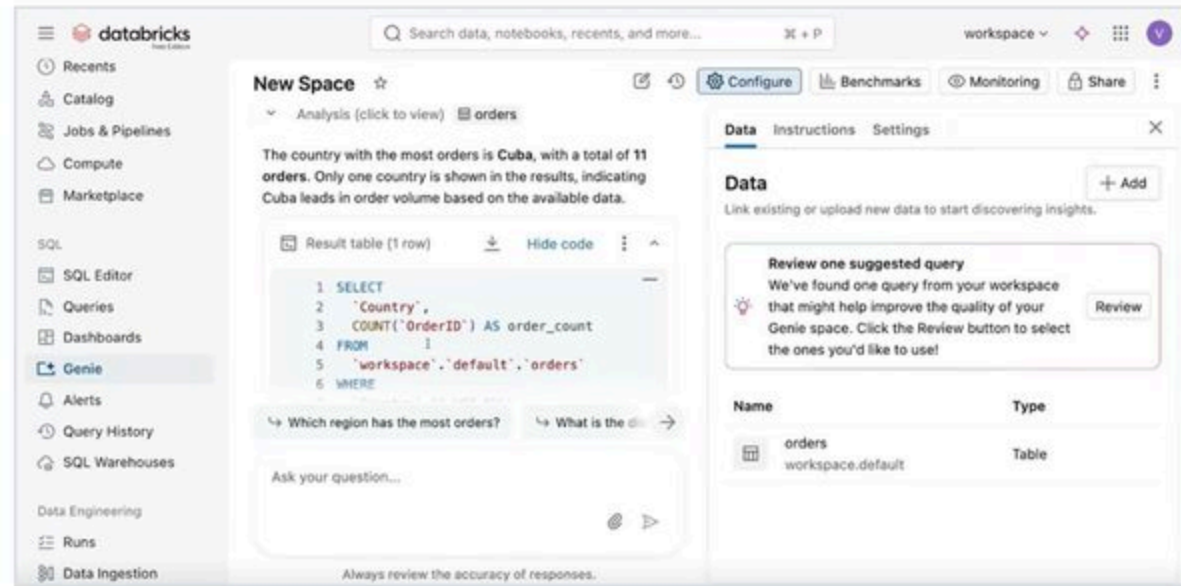


- Drafts a step-by-step plan
- Finds the relevant data: orders
- Counts orders per country, sorts, takes the top

LAB 5

Answer + generated SQL





- Answer: Cuba, 11 orders
- Show code → the SQL Genie wrote

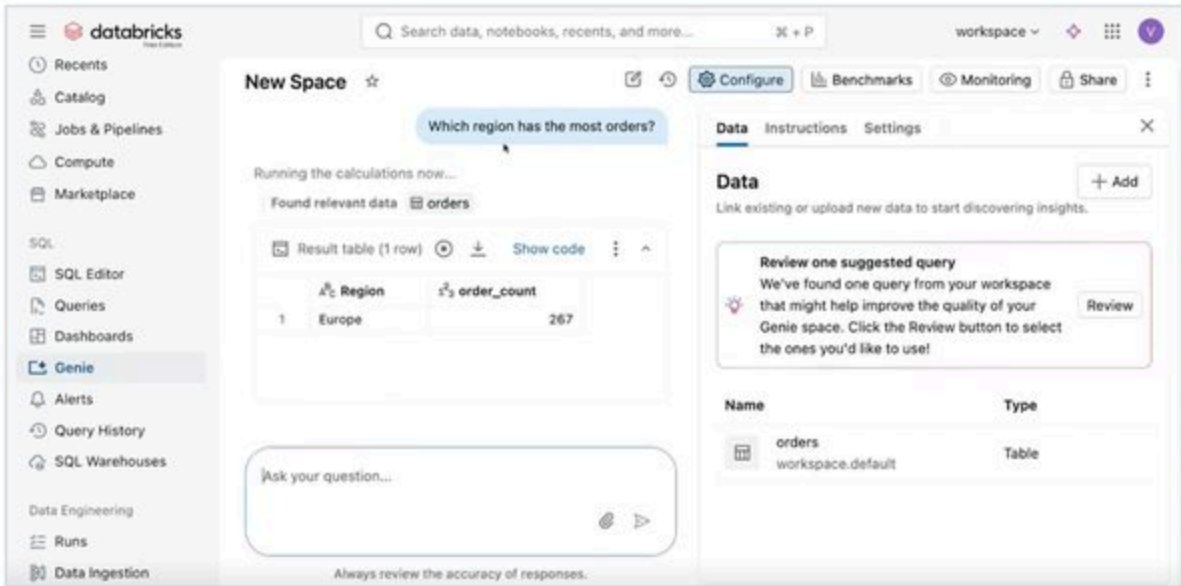
- COUNT(OrderID) … GROUP BY … ORDER BY … LIMIT
- Is this correct? Yes / Fix it

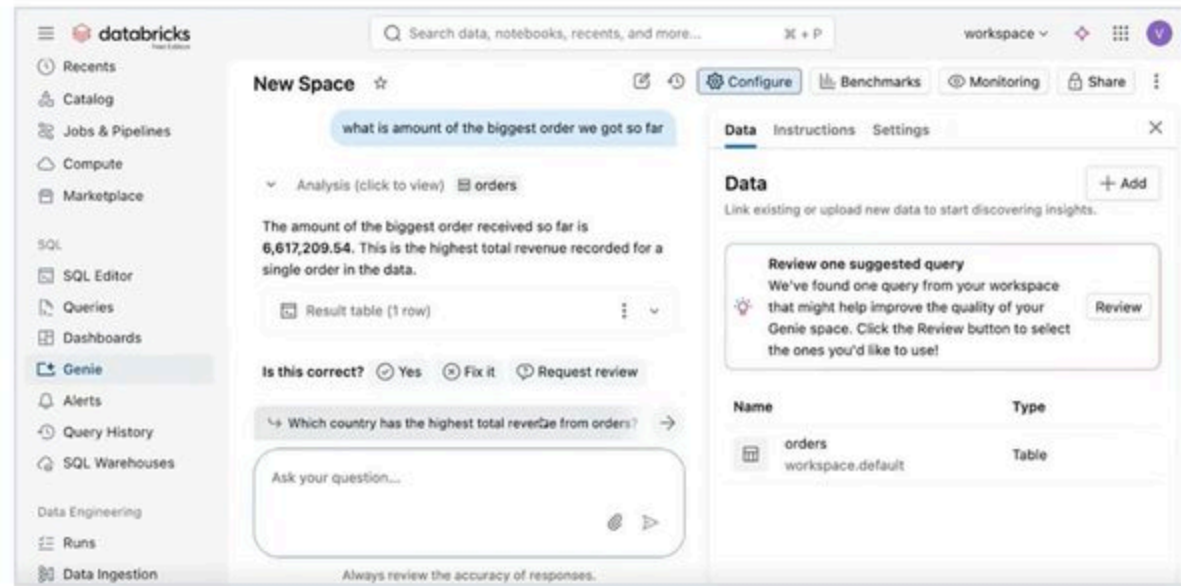
⚠ Genie can be wrong. Read the SQL before you trust the number.

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

LAB 5

More questions

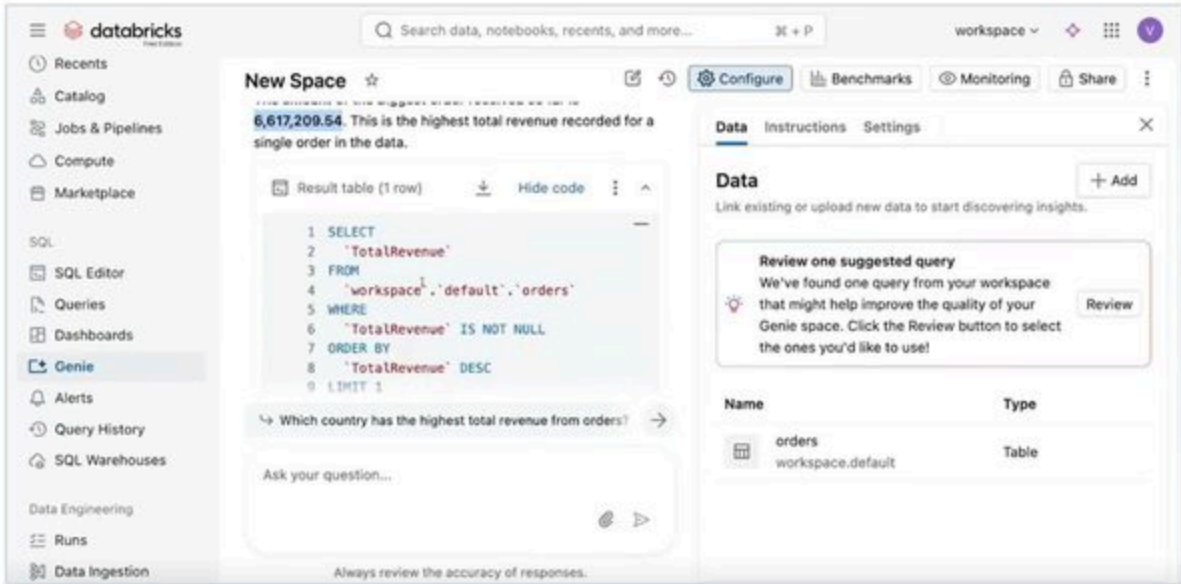


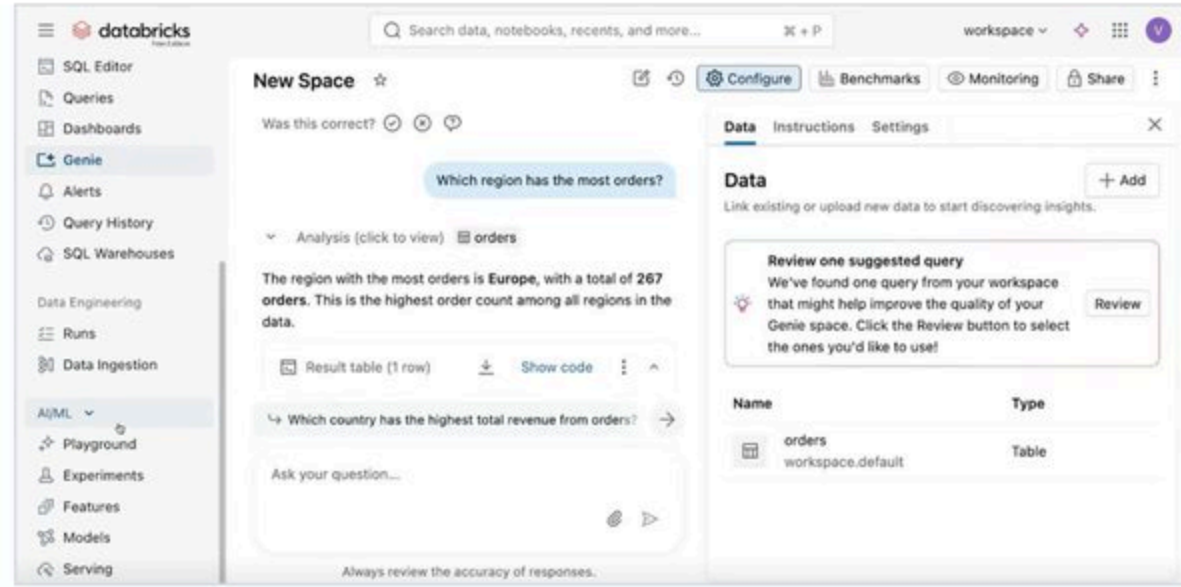


- Which region has the most orders? → Europe, 267
- Biggest order? → TotalRevenue 6,617,209.54
- Follow-up suggestions appear

LAB 5

Check the SQL and give feedback





- SELECT TotalRevenue … ORDER BY TotalRevenue DESC LIMIT 1
- Was this correct?

- Benchmarks and Monitoring
- Always review the accuracy of responses

WRAP-UP

troubleshooting, Q&A, homework

Errors we met • FAQ • Quiz • Homework • Next session

Troubleshooting – errors from today

Error	Cause → fix
Columns _c0, _c1 …	No header option → header=True
All columns are string	No type detection → inferSchema=True or a schema
TABLE_OR_VIEW_NOT_FOUND (notebook)	Temp-view cell not run → run cells in order
TABLE_OR_VIEW_NOT_FOUND (SQL editor)	Temp view is session-only → saveAsTable
PATH_NOT_FOUND	Wrong file name / case → copy the path
SyntaxError: unmatched ')'	Extra bracket or quote → fix syntax
AttributeError … no attribute	Wrong method name → check spelling / Diagnose error
Not enough CPU cores (quota)	Fewer workers or request a quota increase

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

Frequently asked questions (1/2)

Question	Short answer
Is Databricks free?	Free Edition is free for learning; Azure Databricks is pay-per-use (DBU + VMs).
Do I need to learn Scala?	No – PySpark and SQL cover most data engineering jobs.
Spark or pandas?	pandas = one machine; Spark = a cluster. Use Spark for big data.
What is a DBU?	Databricks Unit – the billing unit per hour of compute.
Databricks or ADF?	Both: ADF orchestrates and copies, Databricks transforms.
What format is a Databricks table?	Delta Lake (Parquet files + transaction log) by default.

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

Frequently asked questions (2/2)

Question	Short answer
Volume or table?	Volume for files, table for queryable data.
Managed vs external?	Managed: Databricks controls storage. External: your own ADLS path.
Can Databricks read our ADLS files?	Yes – via an external location / storage credential (later session).
Can ADF run a notebook?	Yes – the Databricks Notebook activity in a pipeline.
Is Genie always correct?	No – check the SQL, add instructions and benchmarks.
Is Free Edition enough for interviews?	Yes for practice; learn cluster settings from Azure docs too.

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

Quick quiz

Question	Answer
Which V means "data arrives very fast"?	Velocity
Spark is written in …	Scala (runs on the JVM)
PySpark is …	The Python API for Spark
Option that uses the first line as column names?	header=True
Option that detects integer / double types?	inferSchema=True
Why did the SQL editor not find "orders"?	It was only a temporary view
Command that creates a permanent table?	df.write.saveAsTable("orders")
Three-level name of the table?	workspace.default.orders

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

Homework / practice

- Create your own Free Edition workspace and rebuild Labs 1–5
- Total revenue per Region and per ItemType with %sql (group by)
- Read the file with an explicit schema instead of inferSchema
- Save a table orders_india with only Indian orders (mode overwrite)
- Ask Genie 3 new questions and check the SQL each time
- Bonus: upload a JSON file and read it with spark.read.json

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: linkedin.com/in/uwayo

Key takeaways

- Big data = too big / fast / varied for one machine → distributed processing
- Spark = engine, PySpark = Python API, Databricks = managed platform
- Volumes hold files; tables hold data (Delta) – catalog.schema.object
- header=True + inferSchema=True (or a schema) for CSV
- Temp views are session-only; saveAsTable shares data
- Genie answers in English – always check the SQL

Thank you

Questions? • Rebuild the labs in Free Edition • Bring your Q&A

Prepared by Ir. UWAYO Jacques · Follow me on LinkedIn: **linkedin.com/in/uwayo**